Measurements of Transverse Spin Dependent $\pi^+\pi^-$ Azimuthal Correlation Asymmetry and Unpolarized $\pi^+\pi^-$ Cross-Section in pp Collisions at $\sqrt{s} = 200$ GeV at STAR

Babu Pokhrel
Advisor: Prof. Bernd Surrow
08/30/2023

Supported in part by:
U.S. DEPARTMENT OF ENERGY | Office of Science

Spin Structure of the Proton

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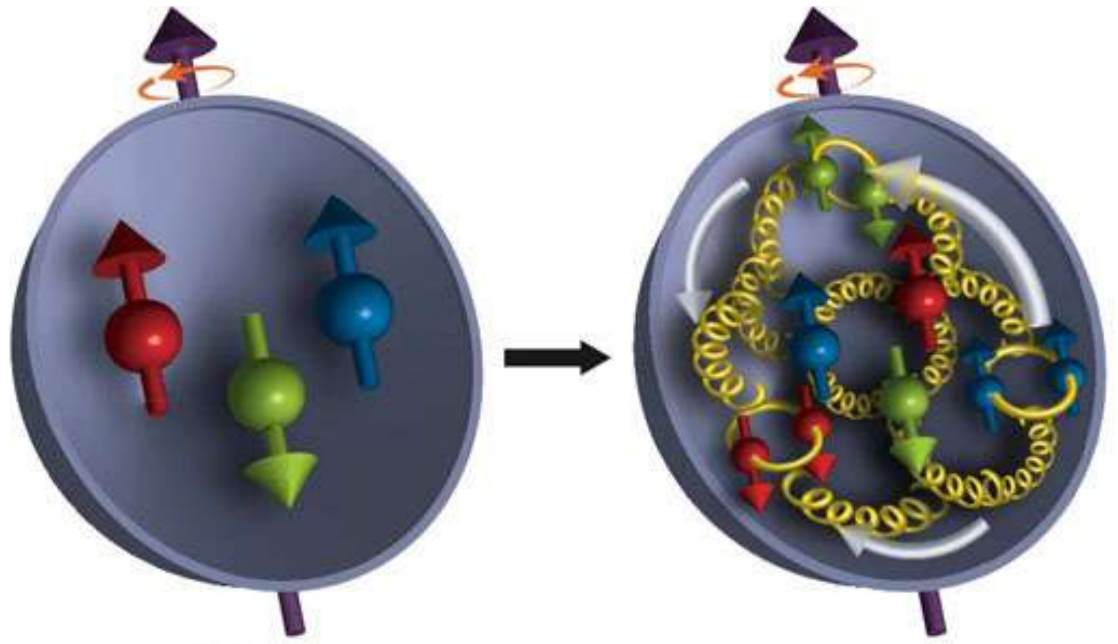
uud $\quad uud + g + sea$



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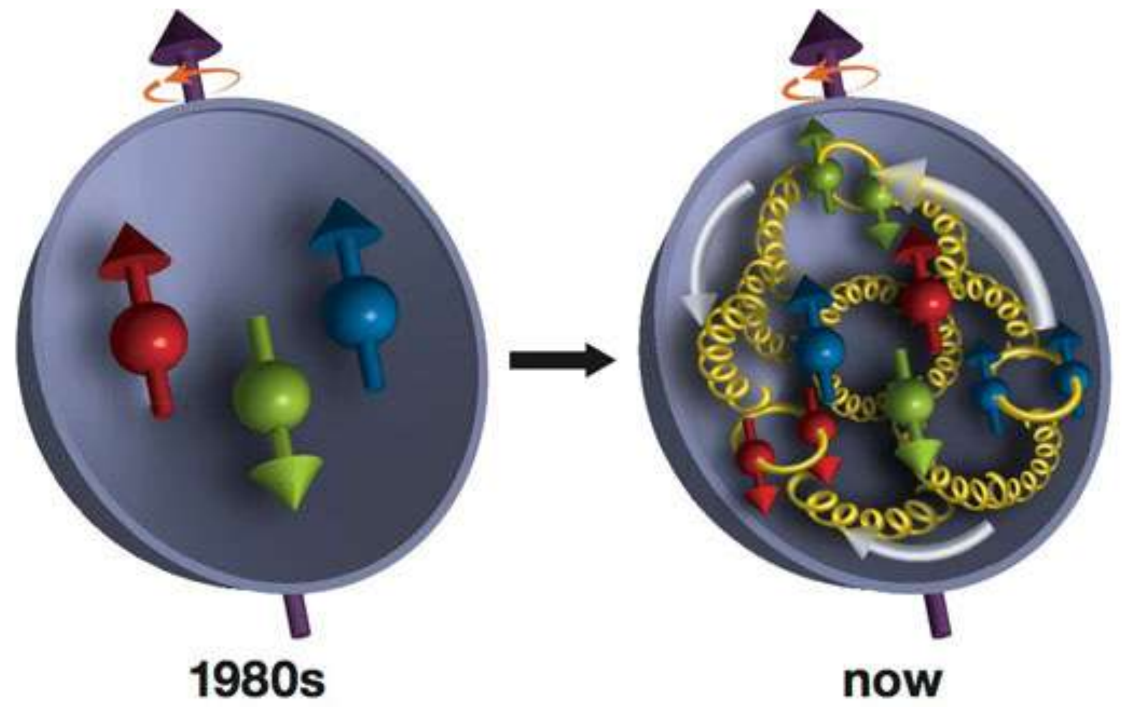
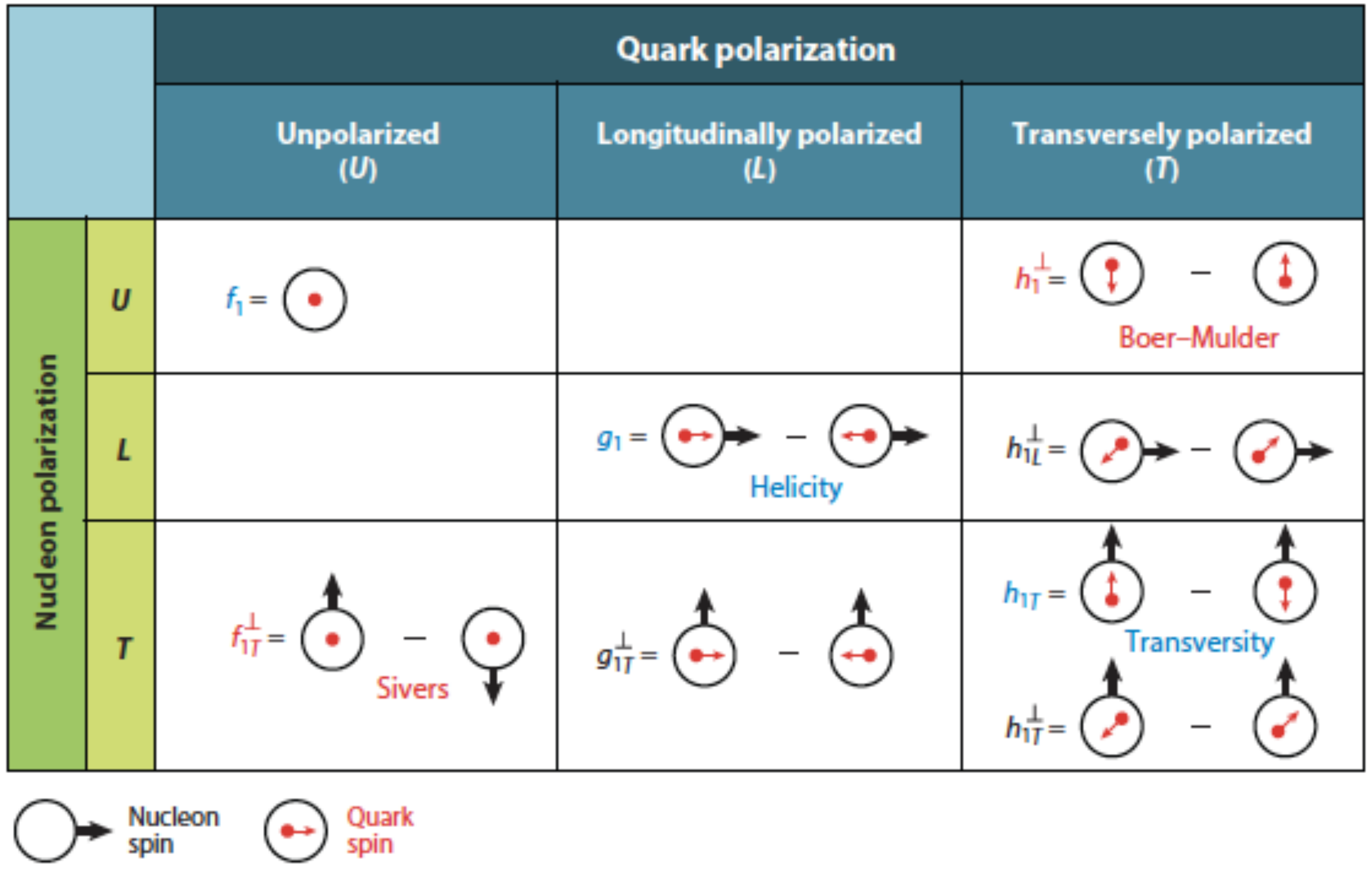


- Proton spin structure in terms of parton distribution functions (PDFs).
- Eight leading twist PDFs.



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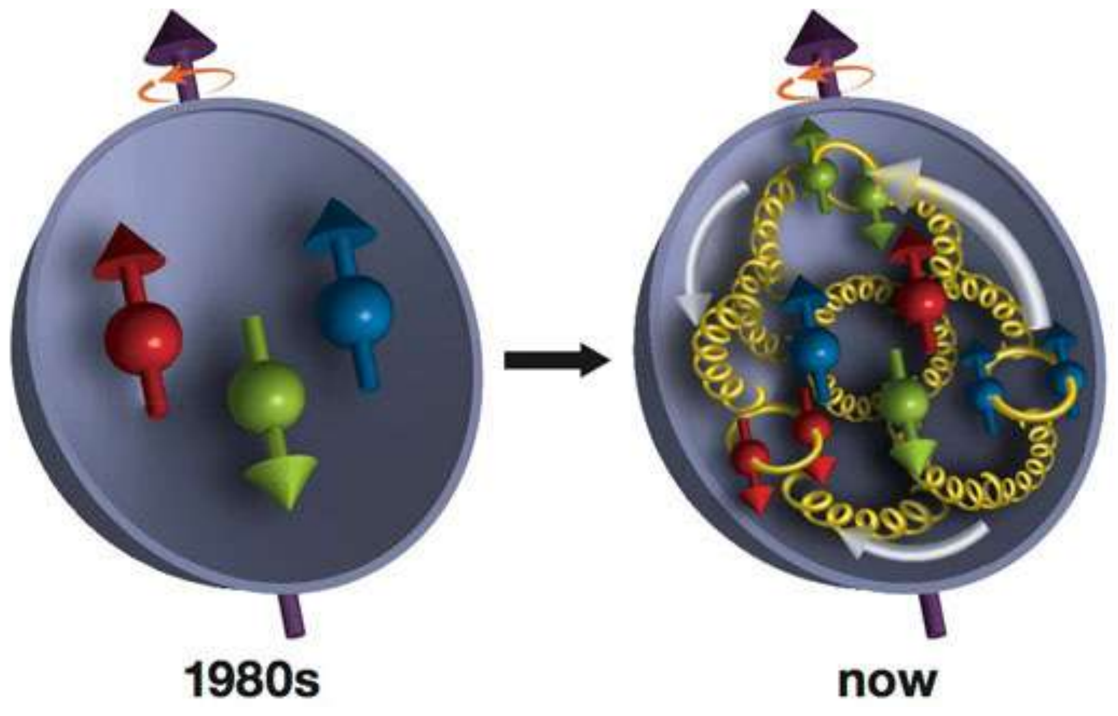

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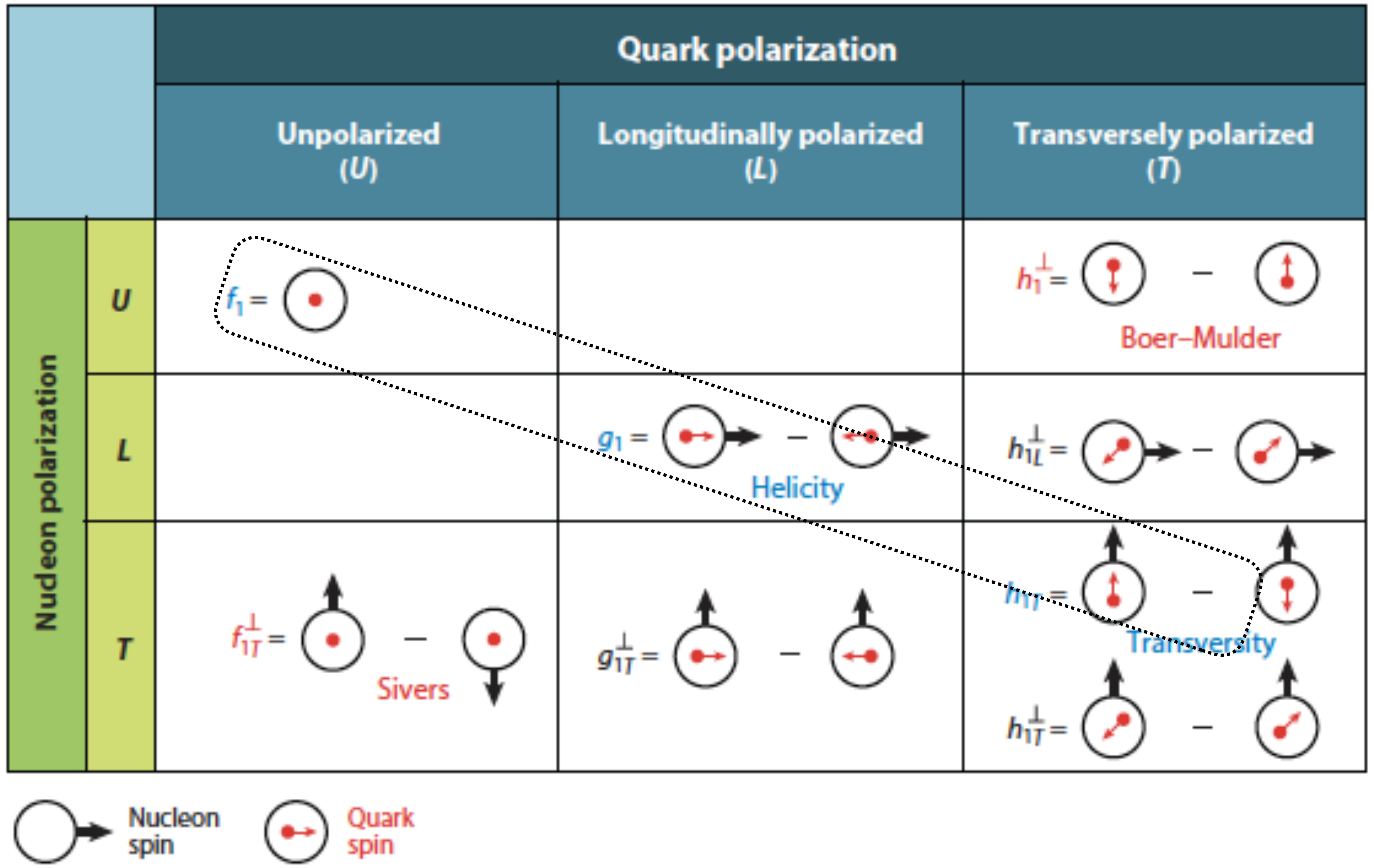
- Three leading twist collinear PDFs:
  - $f_1(x)$ = Unpolarized PDF
  - $g_1(x)$ = Helicity PDF
  - $h_1^q(x)$ = Transversity PDF



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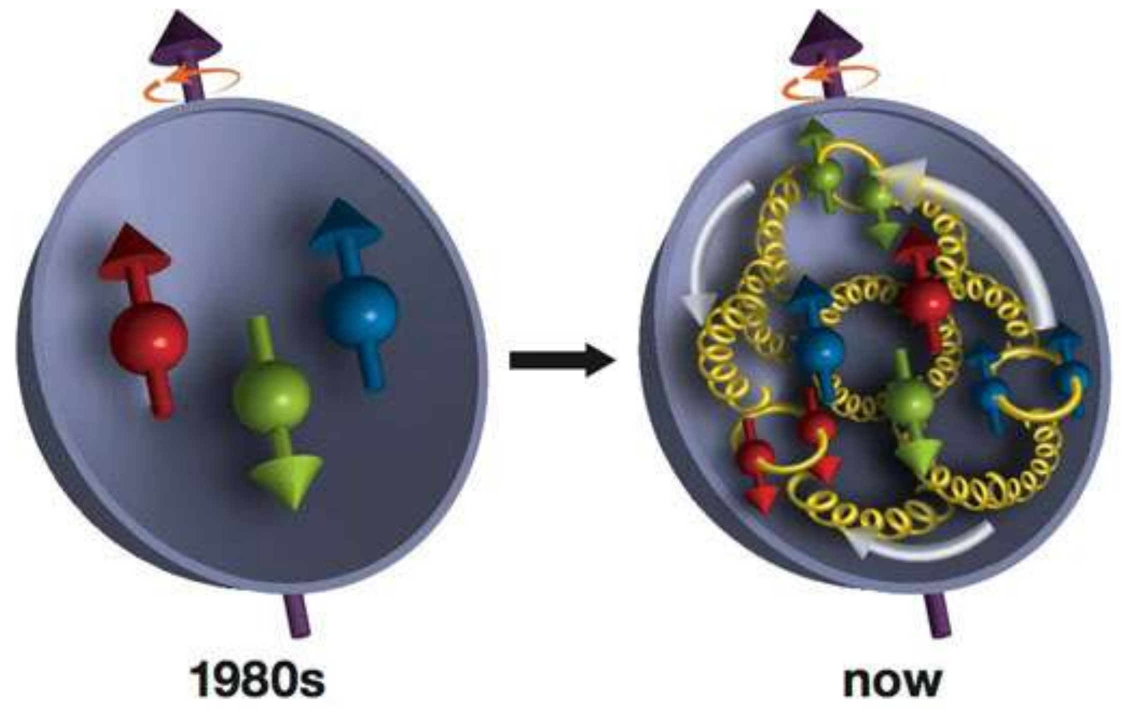

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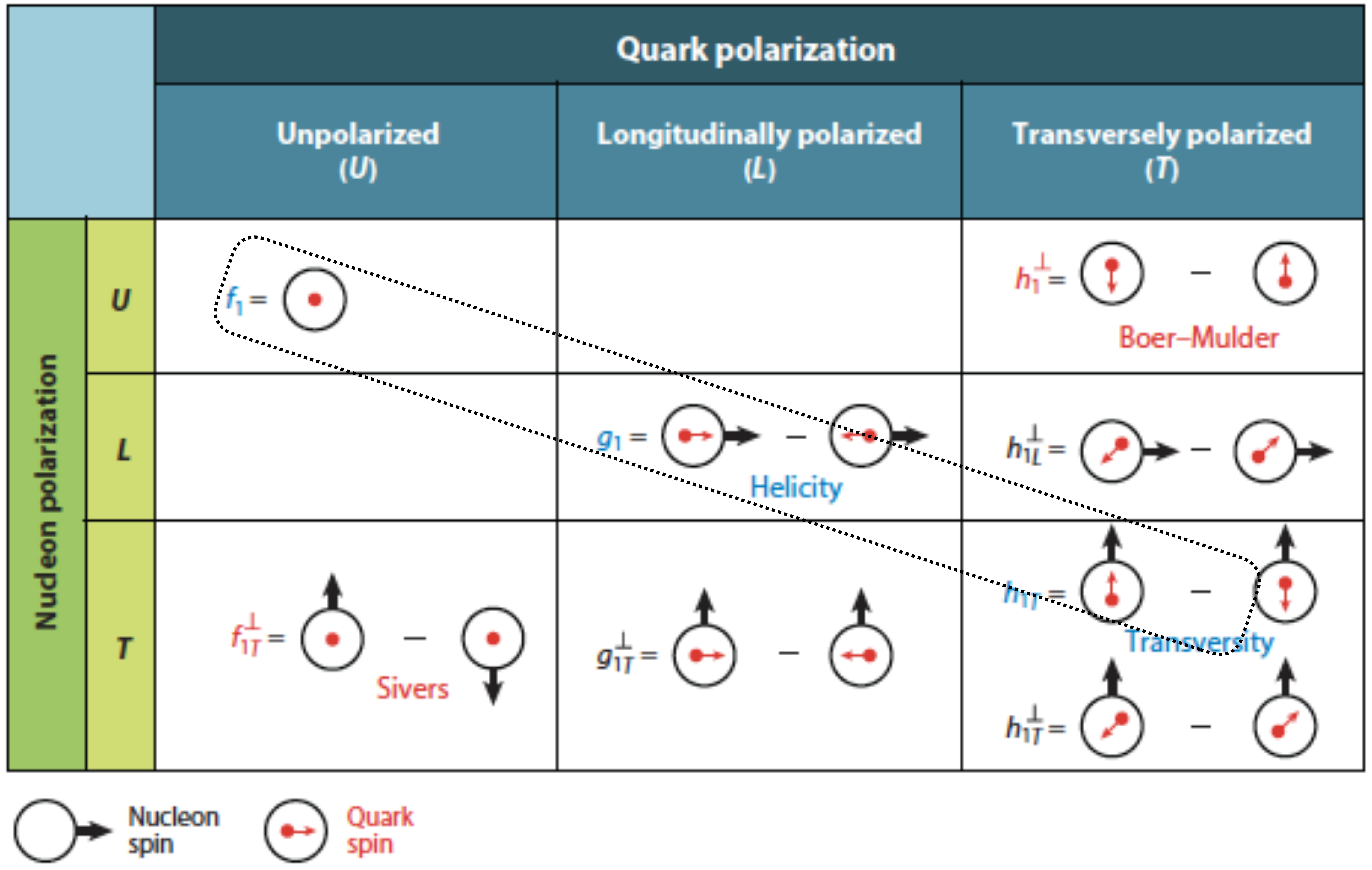

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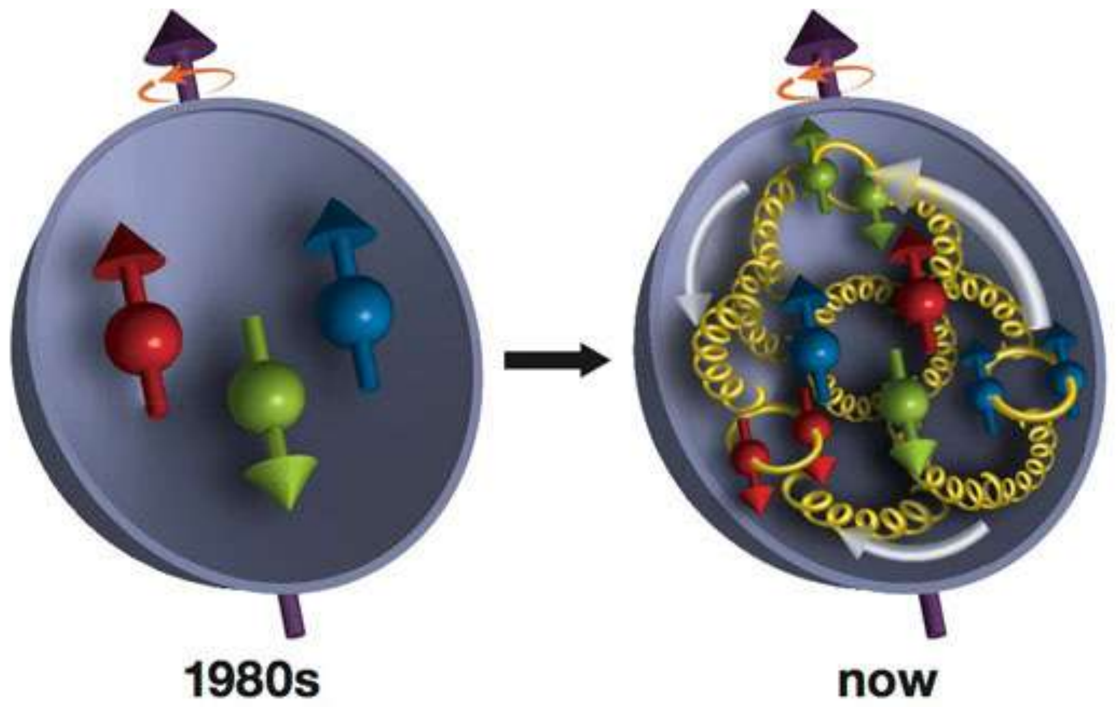

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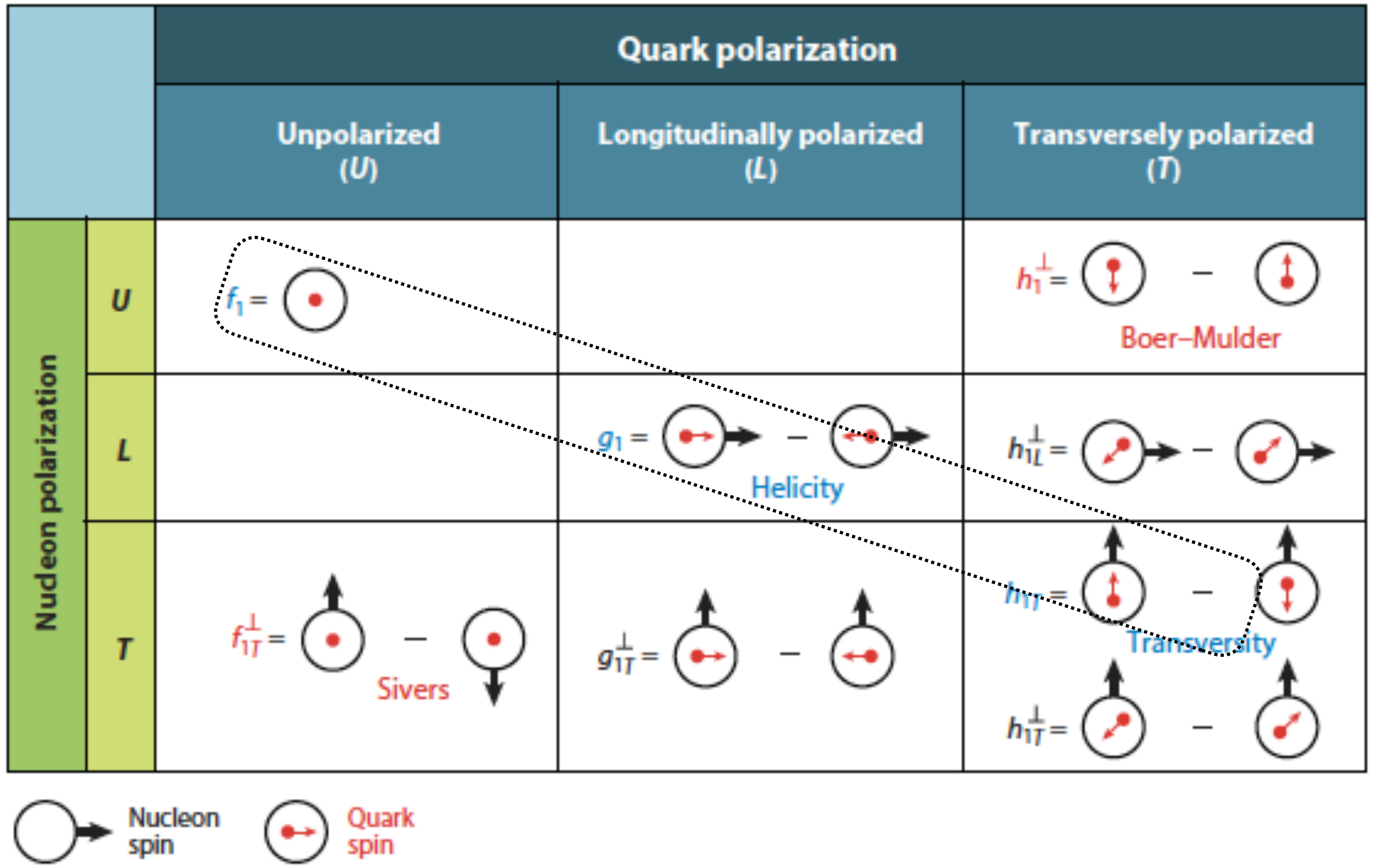

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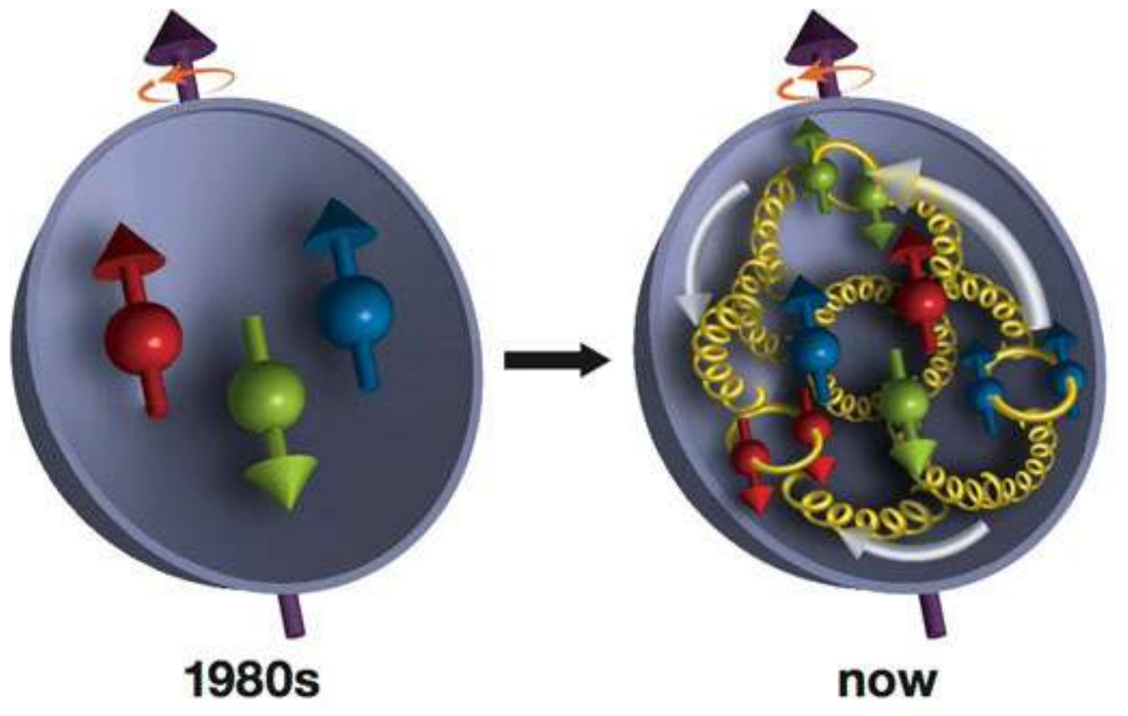

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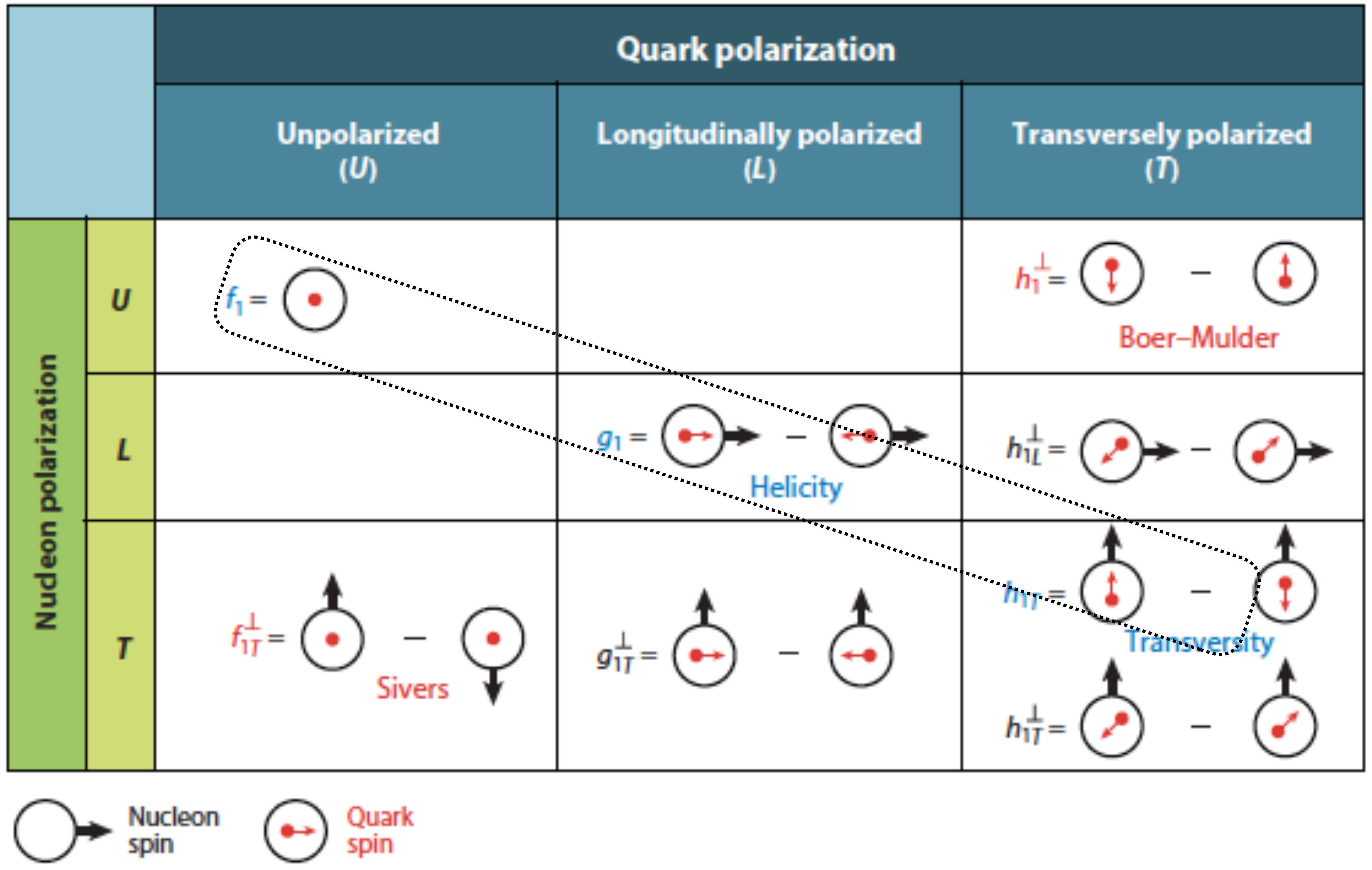
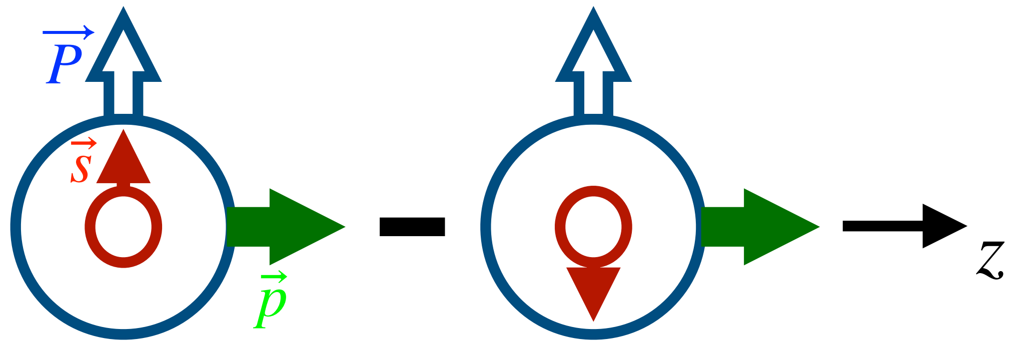


Motivation: Measurement of observables to constrain $h_1^q(x)$ in the collinear framework in pp collision.

Transversity

Transversity, $h_1^q(x)$

Transversity: $h_1^q(x) \approx$



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$\vec{s}$ = Quark polarization

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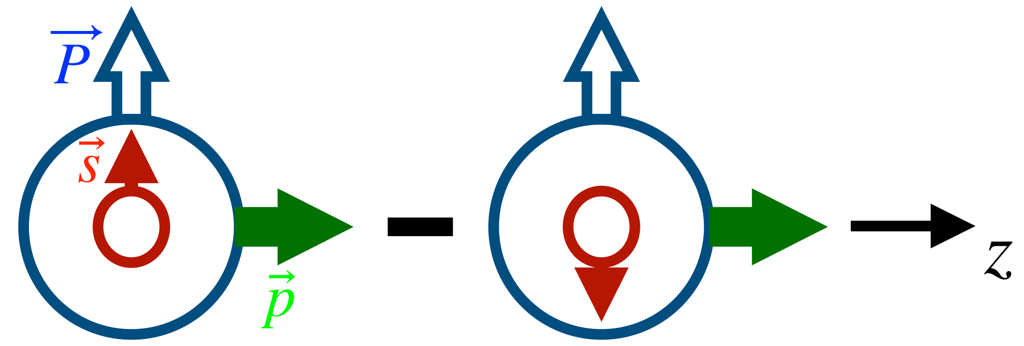
Transversity: $h_1^q(x) \approx$ 

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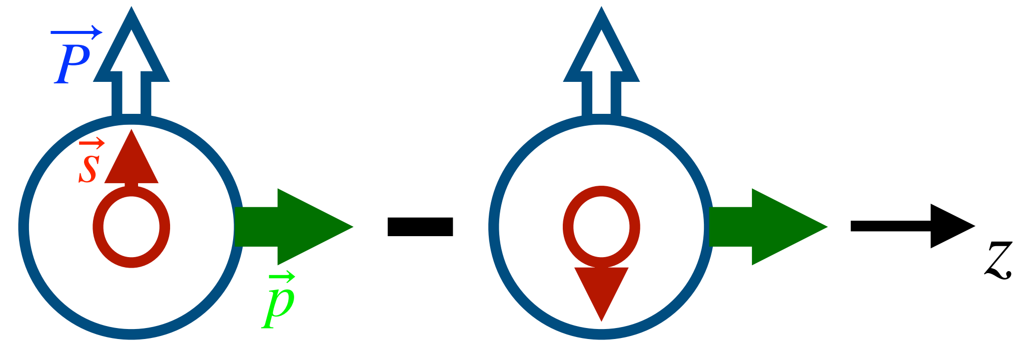

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- Its extraction requires coupling to another chiral-odd object, such as Interference Fragmentation Function (IFF) in dihadron production channel.
- Nucleon tensor charge, g_T, from the experimentally measured transversity allows to compare it with the first principle lattice QCD calculation.
  $g_T = \delta u - \delta d$,
  $$\delta q = \int dx \left(h_1^q(x) - h_1^{\bar{q}}(x) \right)$$

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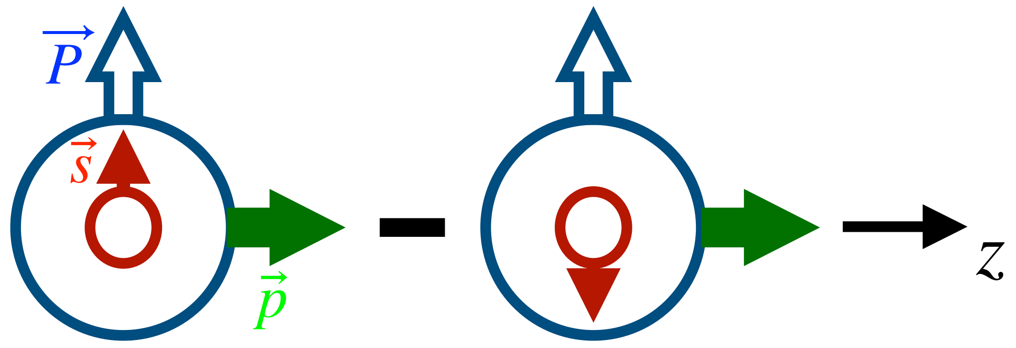

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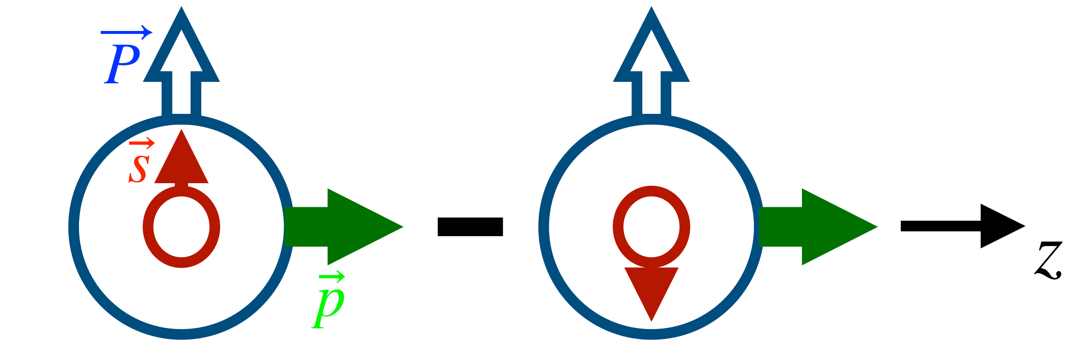
Transversity: $h_1^q(x) \approx$ 

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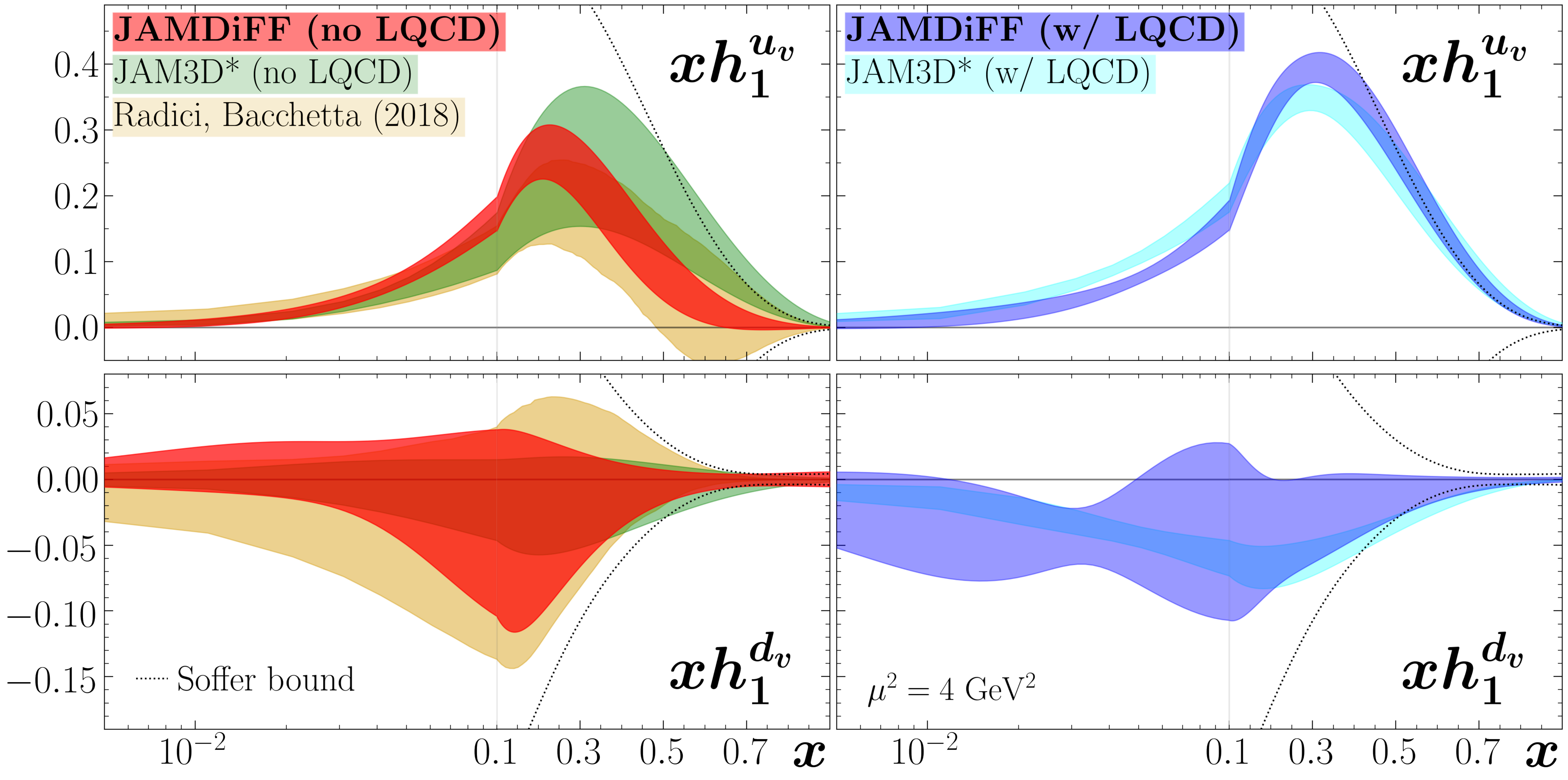


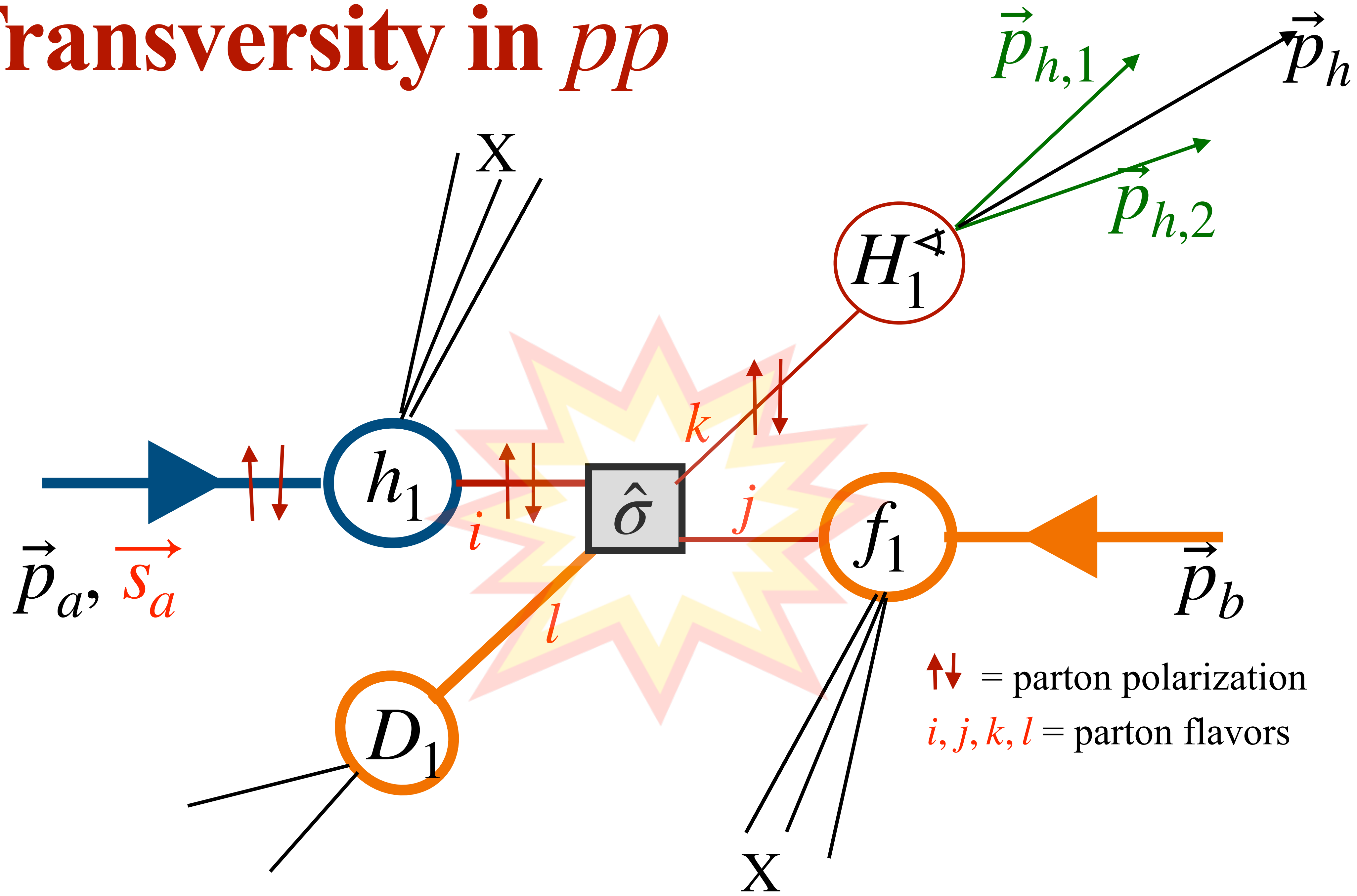
C. Cocuzza et. al., arXiv:2306.12998v1

Observables for Transversity in *pp*

Bachhetta & Radici
Phys.Rev.D 70 (2004) 094032

Observables for Transversity in *pp*

Dihadron Channel: $p^{\uparrow} + p \rightarrow h^{+}h^{-} + X$



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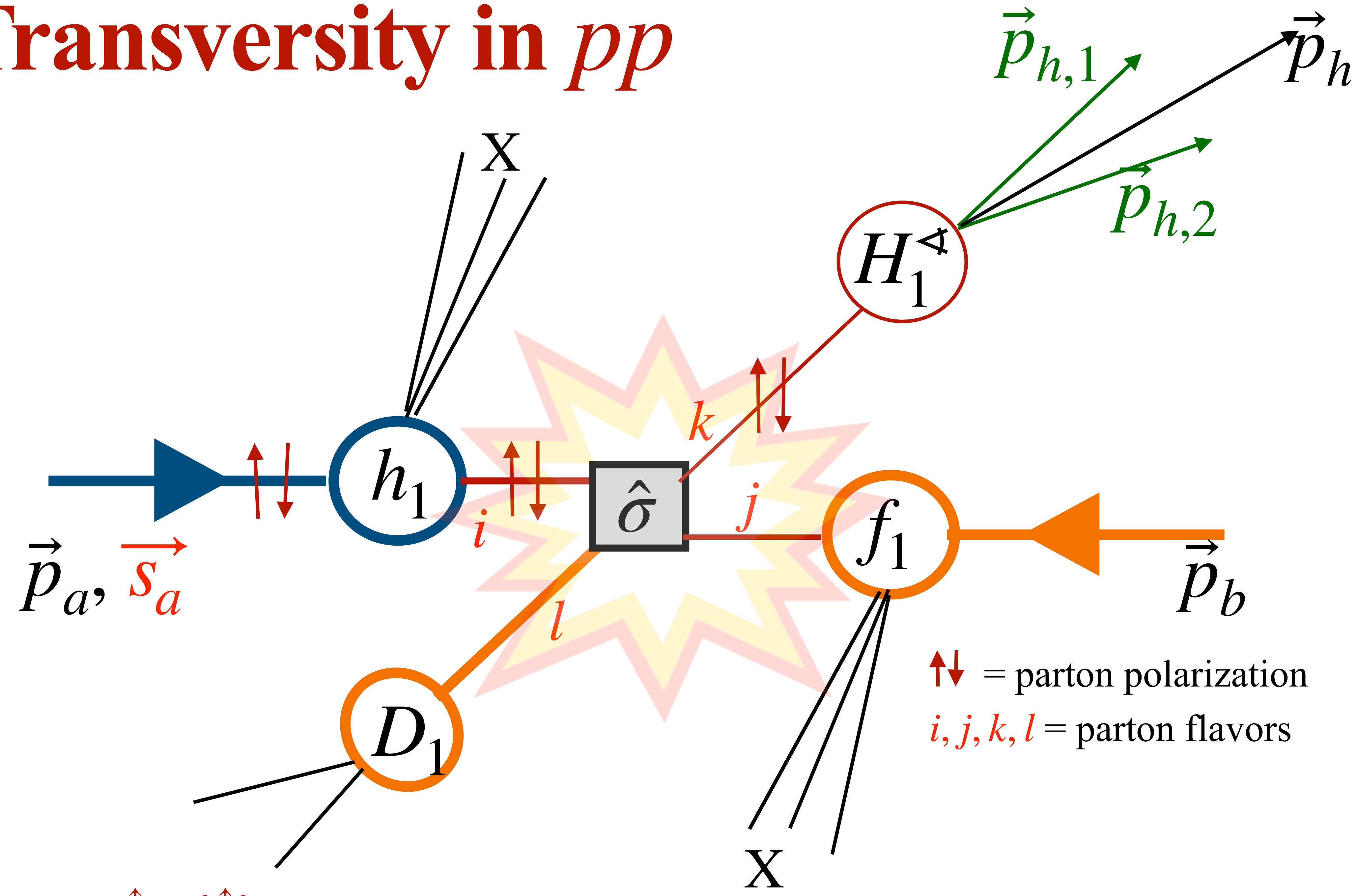

Polarized Cross Section:

$$d\sigma_{UT}^{p_a^{\uparrow}p_b\rightarrow(h_1,h_2)X} \propto \sin(\phi_S - \phi_R)\sum_{i,j,k,l}\int dx_a \int dx_b \int dz\; h_1^{i/p_a}(x_a)\, f_1^{j/p_b}(x_b)\frac{d\Delta\hat{\sigma}^{i^{\uparrow}j\rightarrow k^{\uparrow}l}}{d\hat{t}}\; H_1^{\sphericalangle h_1h_2/k}(z, M_h^2)$$

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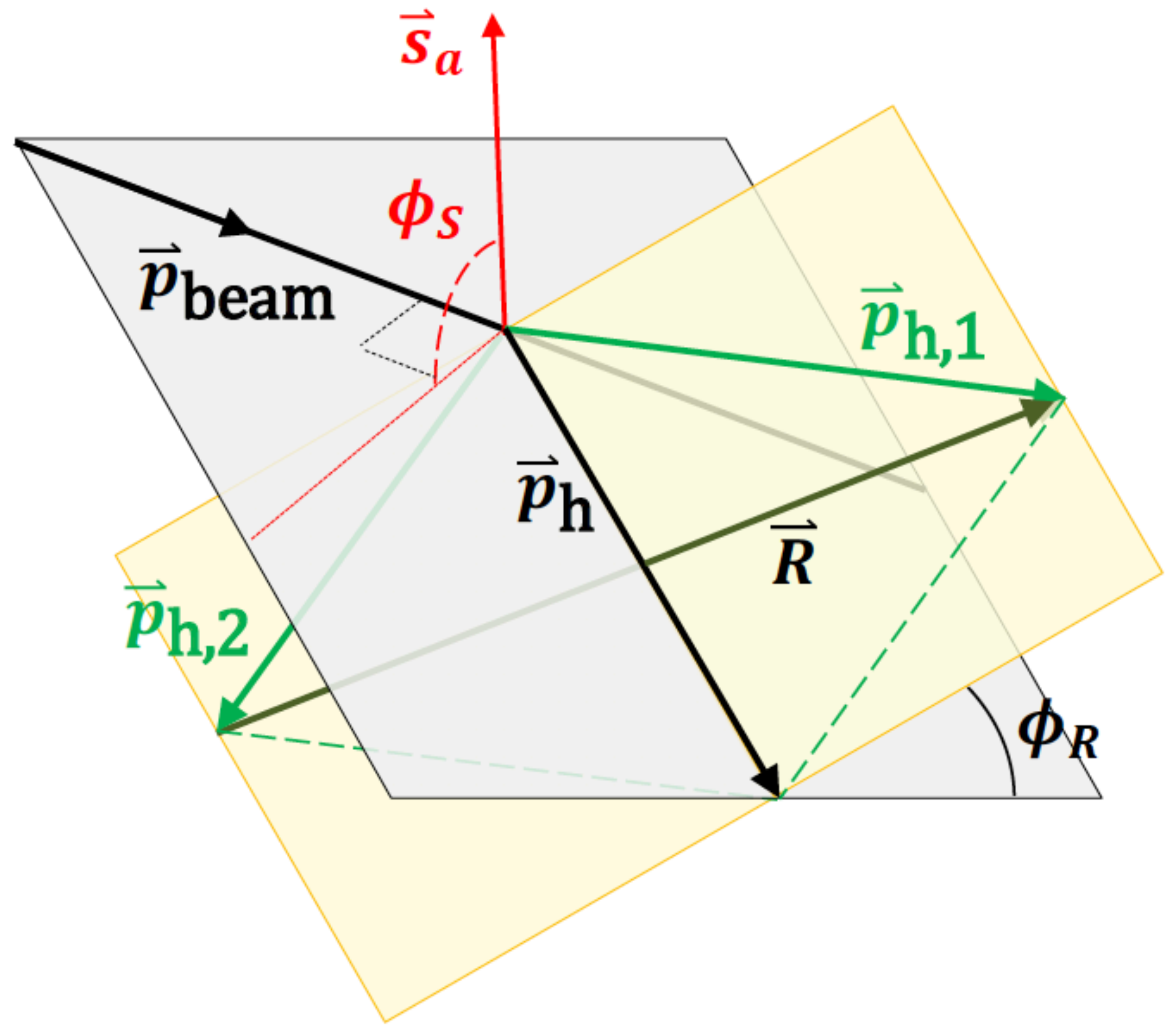

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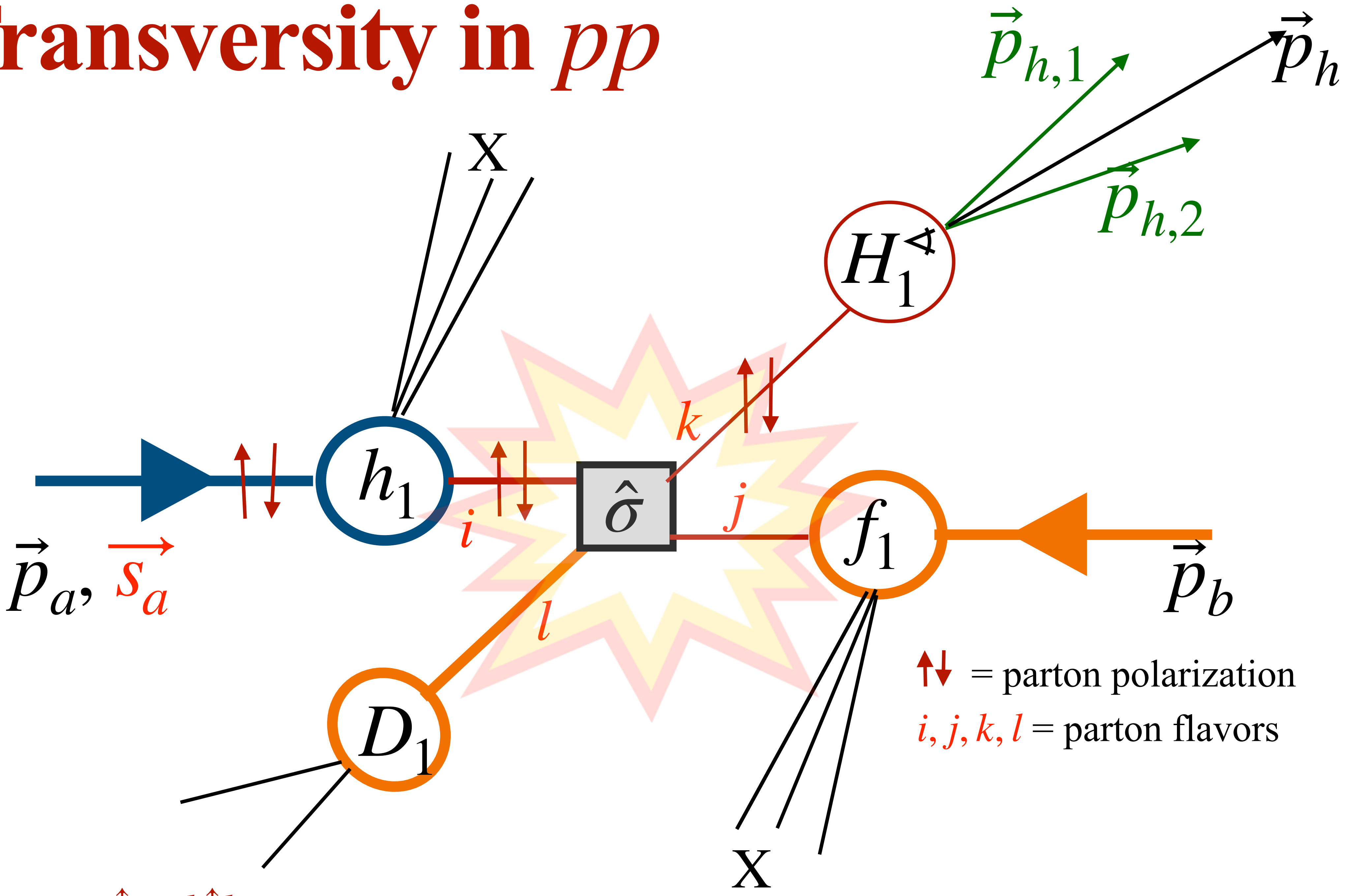

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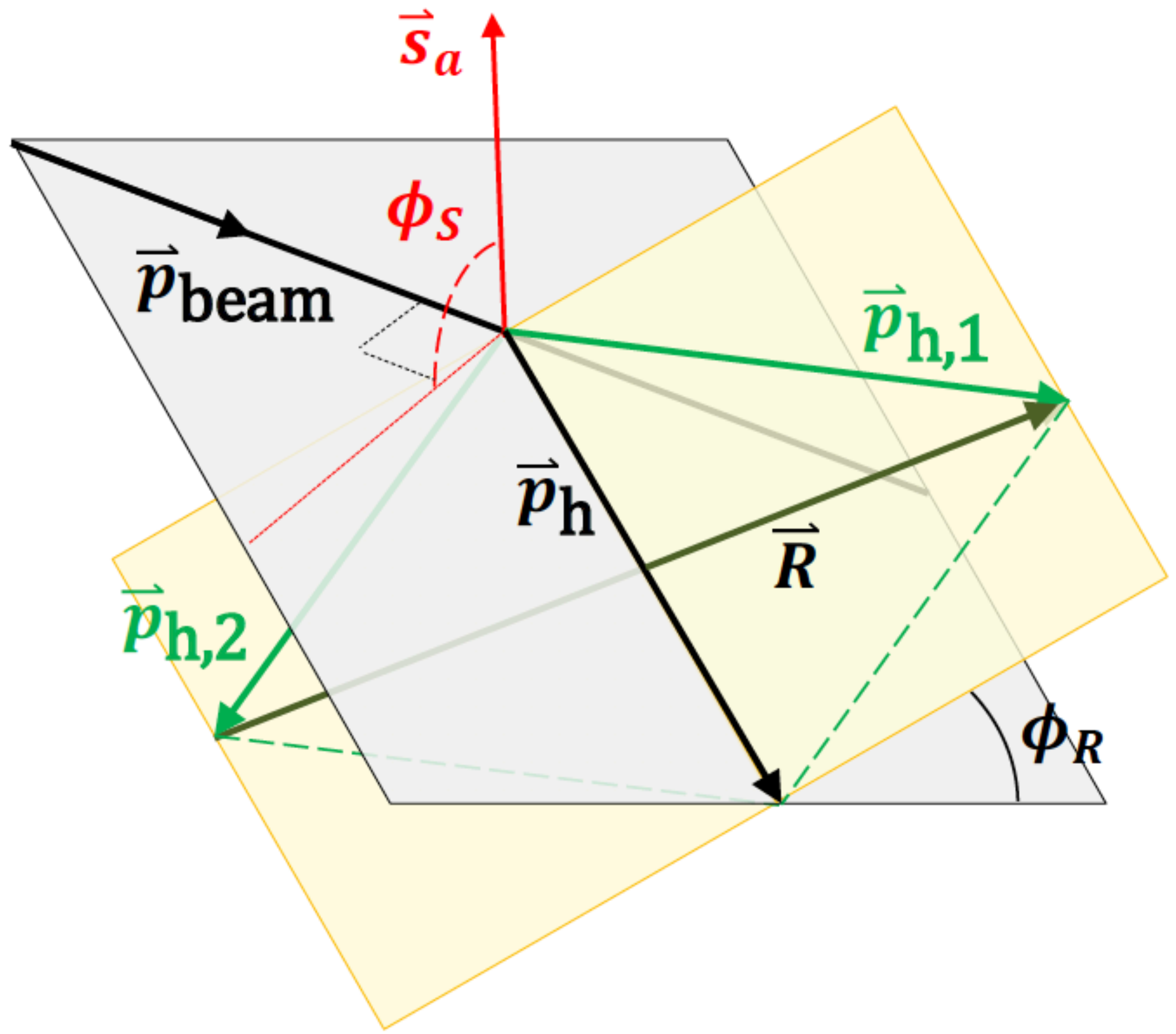

Unpolarized Cross Section:

$\vec{p}_a^{\uparrow} \leftrightarrow p_a,\ h_1 \leftrightarrow f_1,\ \ H_1^{\sphericalangle} \leftrightarrow D_1$

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Observables for Transversity $h_1(x)$ in pp

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- **Dihadron Azimuthal Correlation Asymmetry,** A_{UT} **, in** $p^{\uparrow} + p \rightarrow h^{+}h^{-} + X$

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- $D_1^{h_1h_2}$ is least known, specifically for gluon fragmentation.

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  - $d\sigma_{UU}$ is crucial for the $D_1^{h_1 h_2}$, which provides equal access to quarks and gluons.

- $d\sigma_{UU}$ **and** A_{UT} **allows model-independent extraction of** $h_1^{q}(x)$**.**

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STAR $\pi^+\pi^- A_{UT}$: Proof-of-Principle

$$p^{\uparrow} + p \rightarrow h^+h^- + X$$

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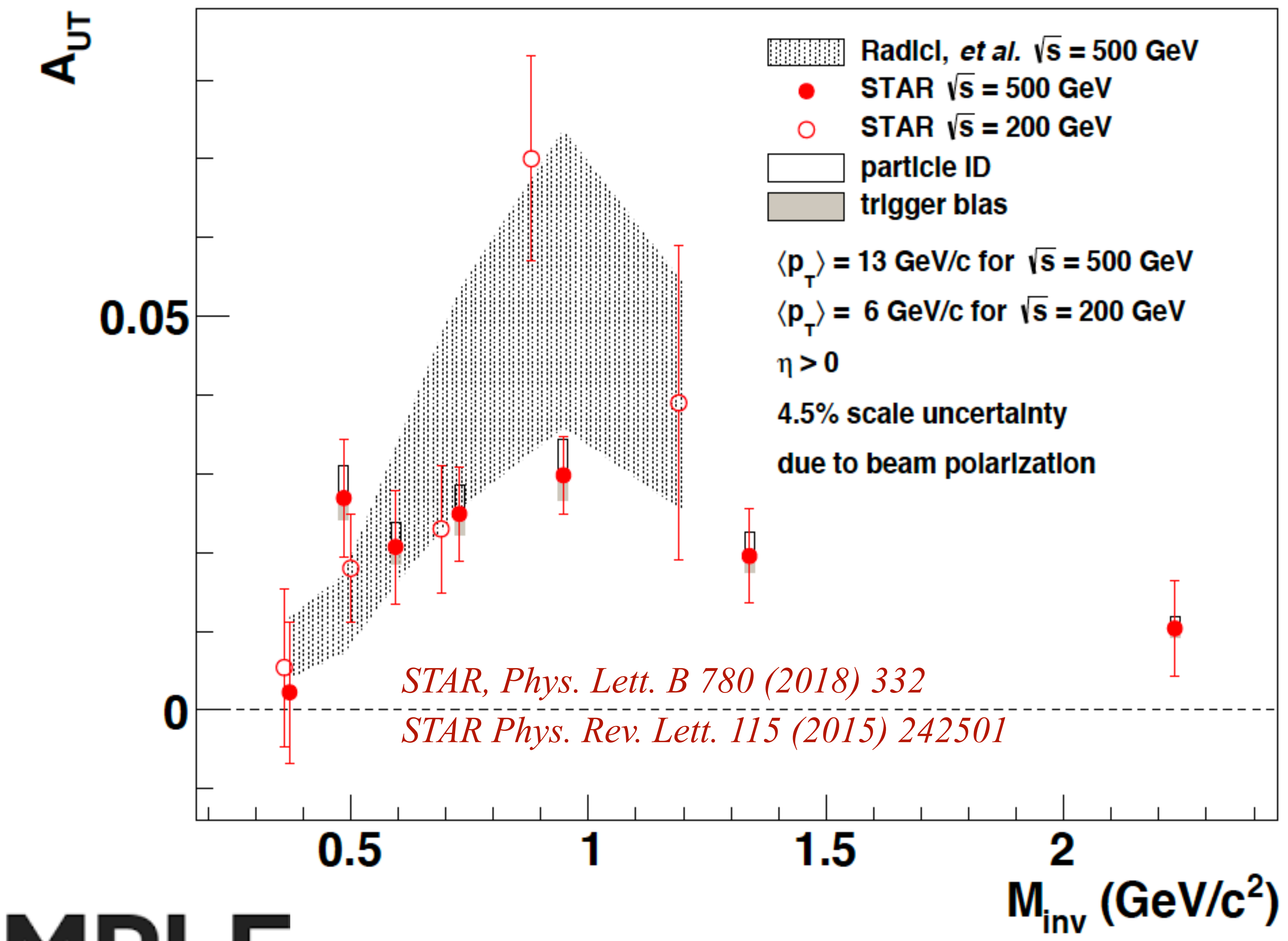

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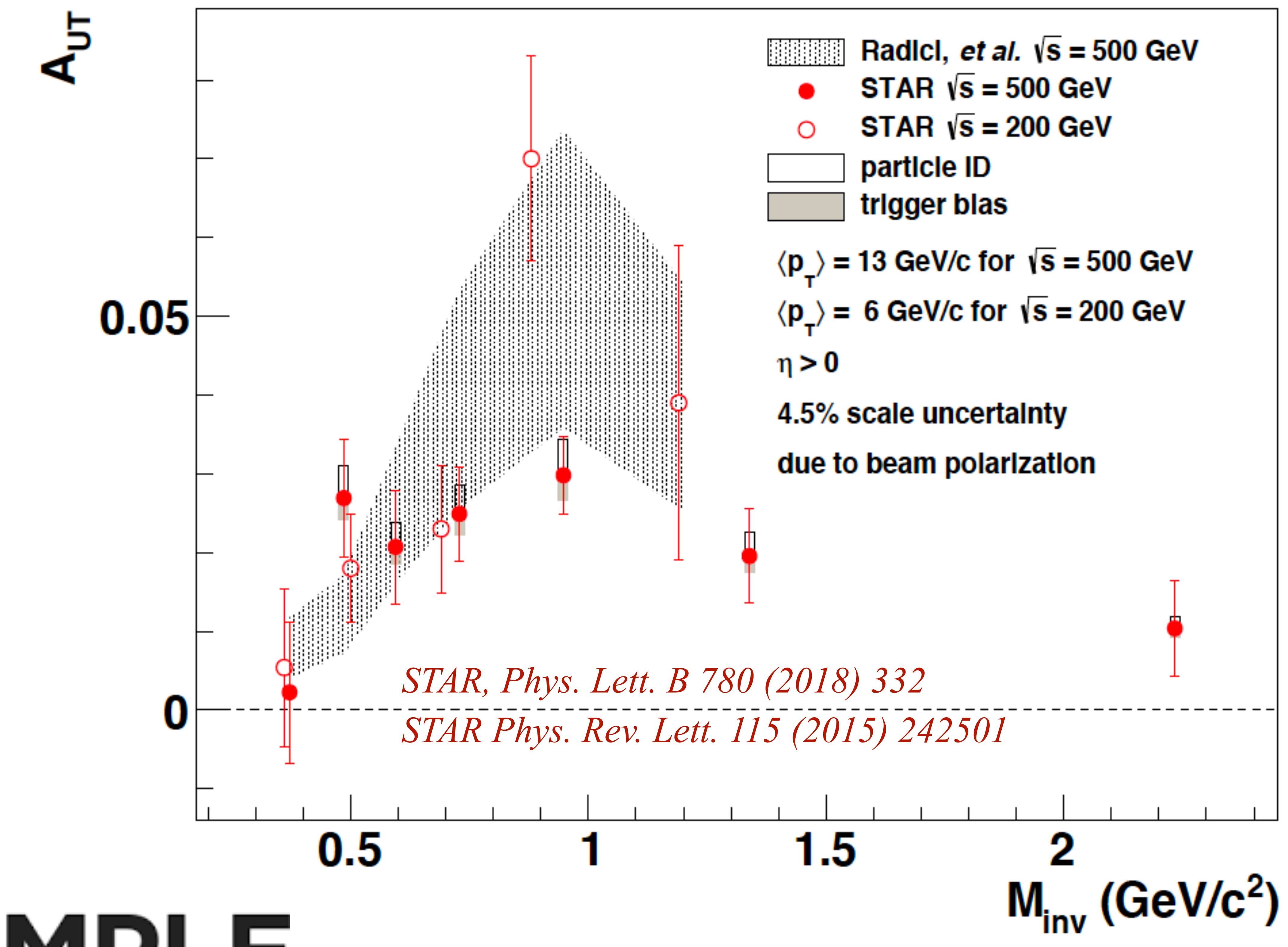
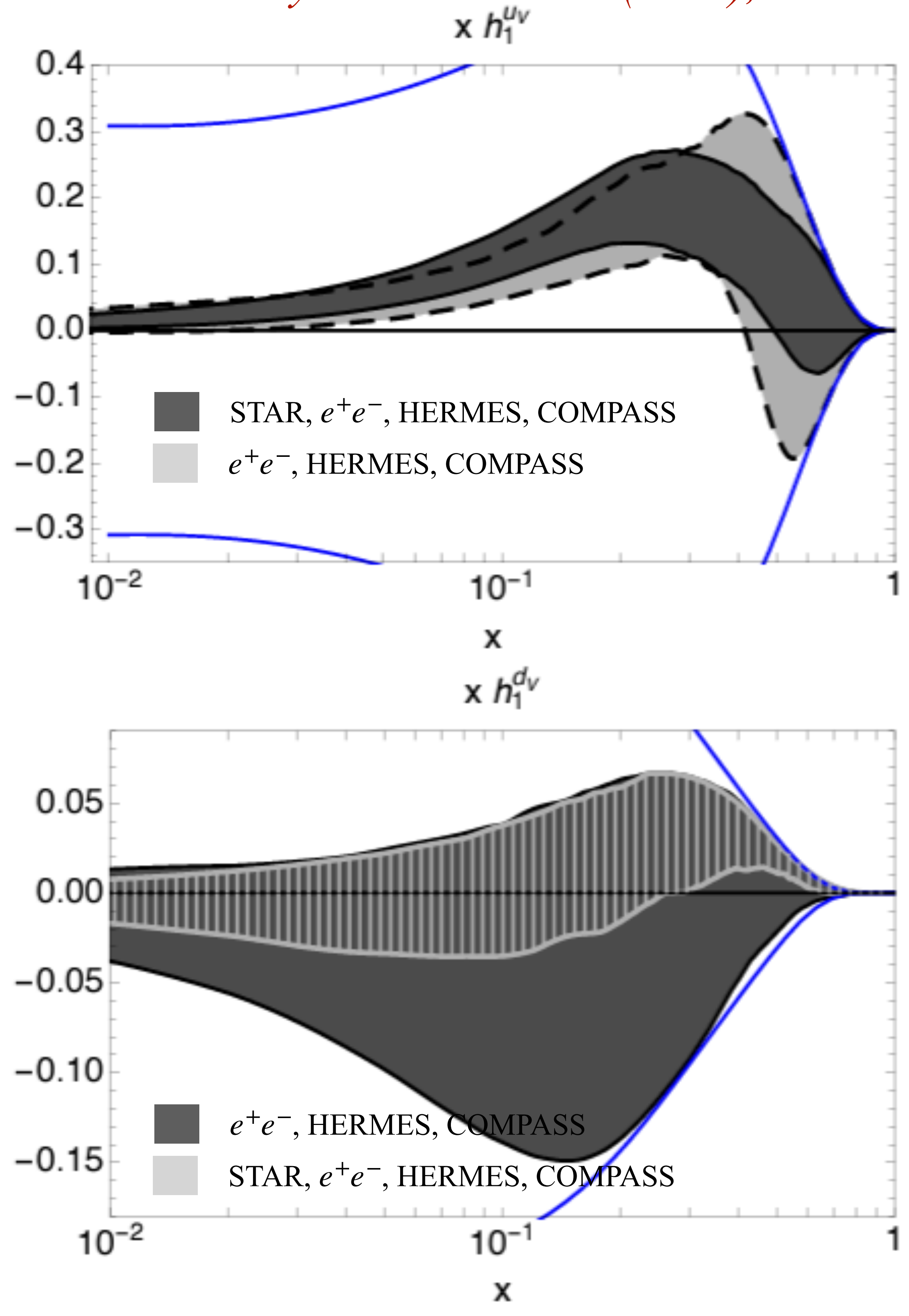


STAR, Phys. Lett. B 780 (2018) 332

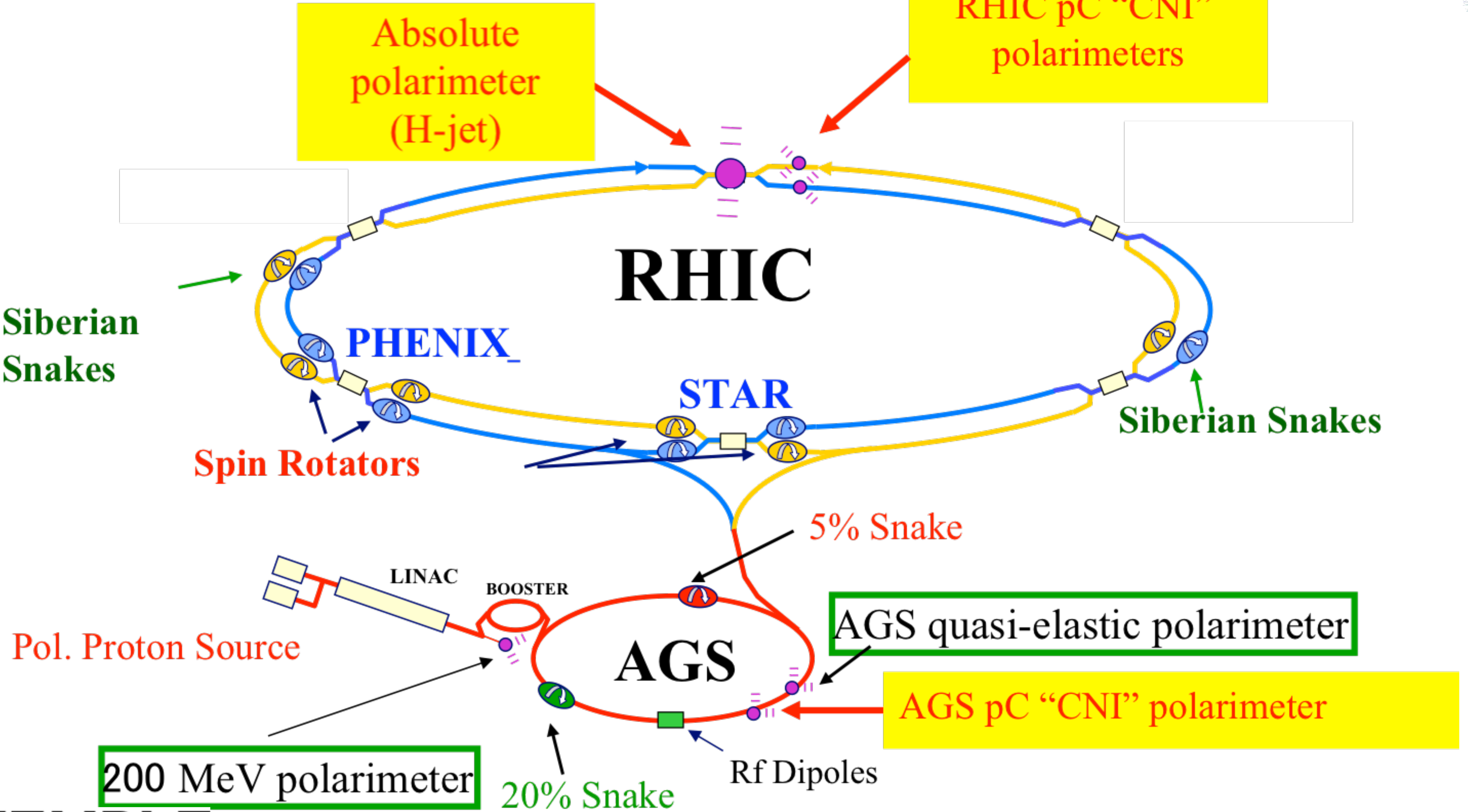
STAR Phys. Rev. Lett. 115 (2015) 242501

Radici et. al. Phys. Rev. Lett. 120 (2018), 19 192001

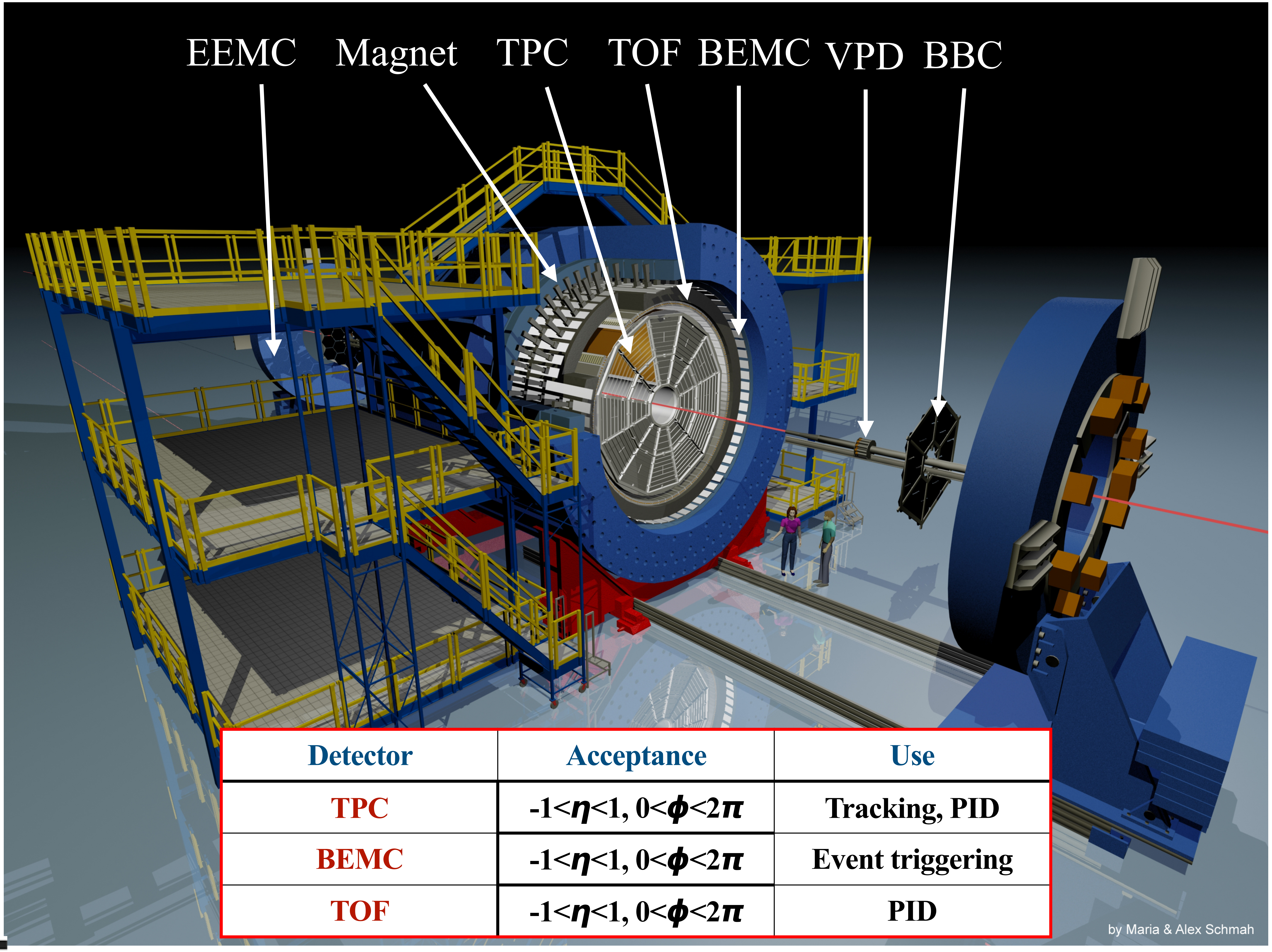


Significant impact on $h_1^q(x)$ from the STAR data at $\sqrt{s} = 200$ GeV

Relativistic Heavy Ion Collider (RHIC)



STAR Detector at RHIC



Detector	Acceptance	Use
TPC	$-1<\eta<1$, $0<\phi<2\pi$	Tracking, PID
BEMC	$-1<\eta<1$, $0<\phi<2\pi$	Event triggering
TOF	$-1<\eta<1$, $0<\phi<2\pi$	PID

STAR Polarized pp Collision Datasets for the IFF Analyses

Collision	proton-proton						
Polarization	transverse						
Year	2006	2011	2012	2015	2017	2022	2024
$\sqrt{s}$ (GeV)	200	500	200	200	510	508	200
L_{int} (pb^{-1})	~ 1.8	~ 25	~ 22	~ 52	~ 350	~ 400	~ 190
$< P_{beam} >$ (%)	~ 60	~ 53	~ 57	~ 57	~58		

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- STAR Preliminaries @ $\sqrt{s}$ = 200 GeV
  - Unpolarized $\pi^+\pi^-$ Cross Section (2012)
  - IFF Asymmetry (2015)

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- STAR IFF Preliminary

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STAR, Phys. Rev. Lett. 115 (2015) 242501

- STAR IFF Preliminary

@ $\sqrt{s}$ = 510 GeV

- Planned IFF and Cross Section Measurements

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TEMPLE UNIVERSITY

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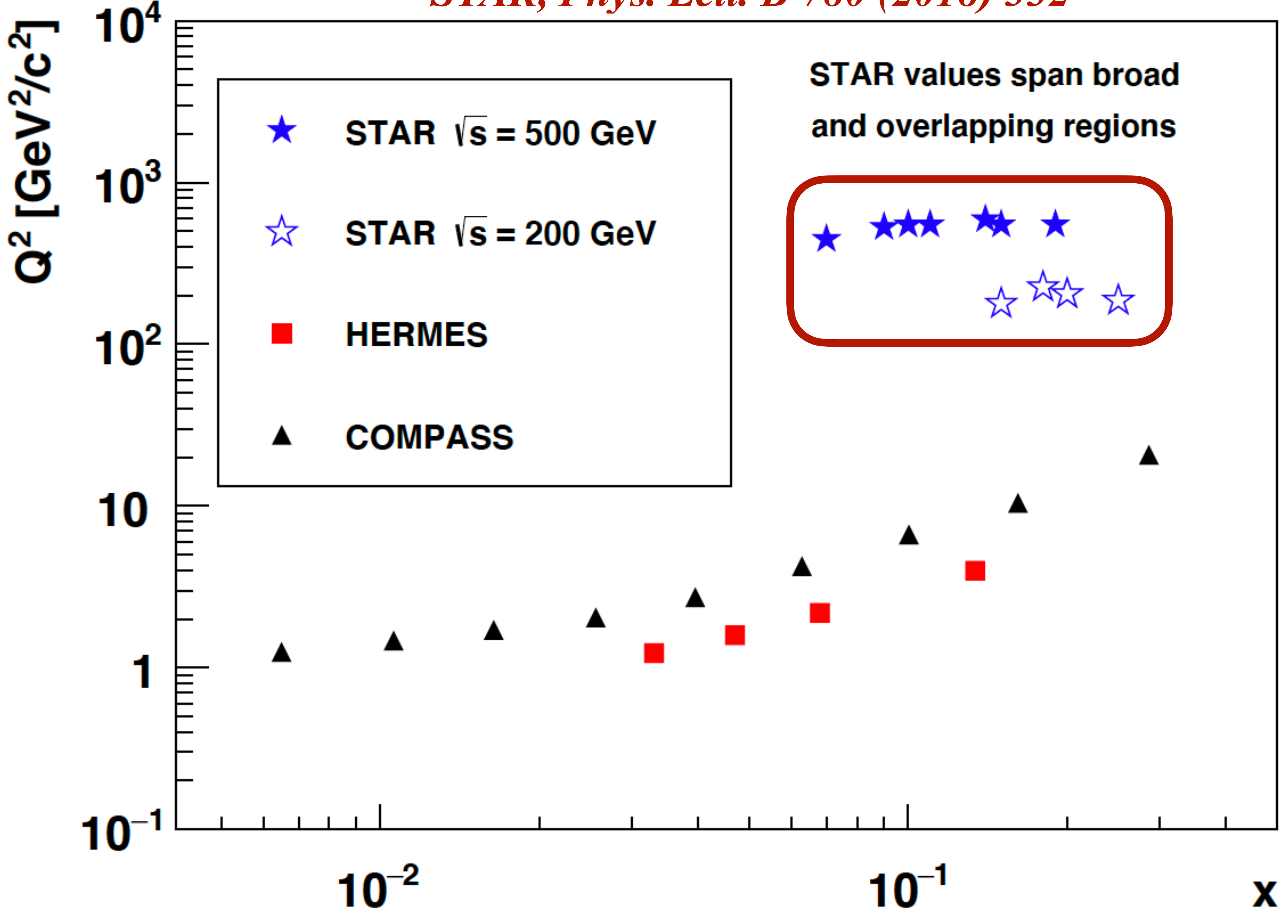


STAR Kinematic Coverage:

- Covers much higher Q^2 than HERMES and COMPASS.
- Intermediate x coverage, which probes the valence quark region.

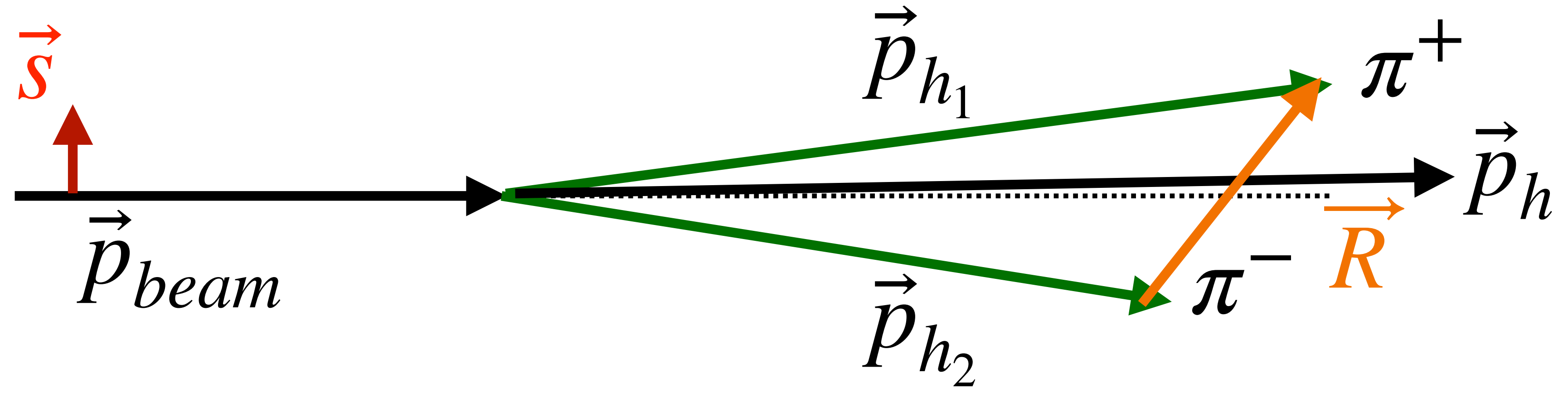
$\pi^+\pi^-$ Formation and Azimuthal Angles

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^\uparrow + p \to \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

$\pi^+\pi^-$ Formation and Azimuthal Angles

- Polarized parton fragments to $\pi^+\pi^-$



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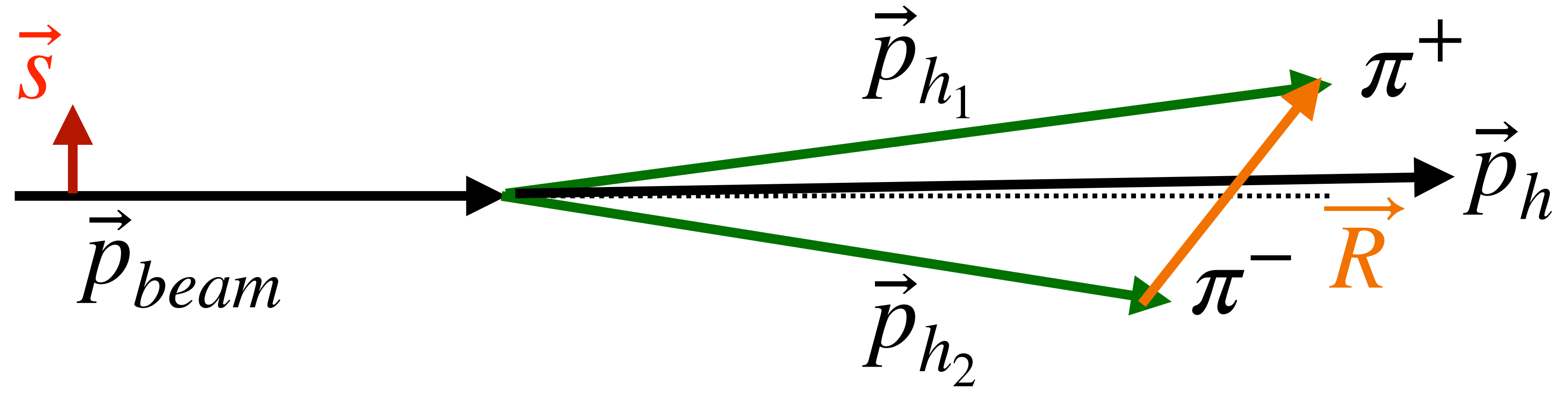
$\pi^+\pi^-$ Formation and Azimuthal Angles

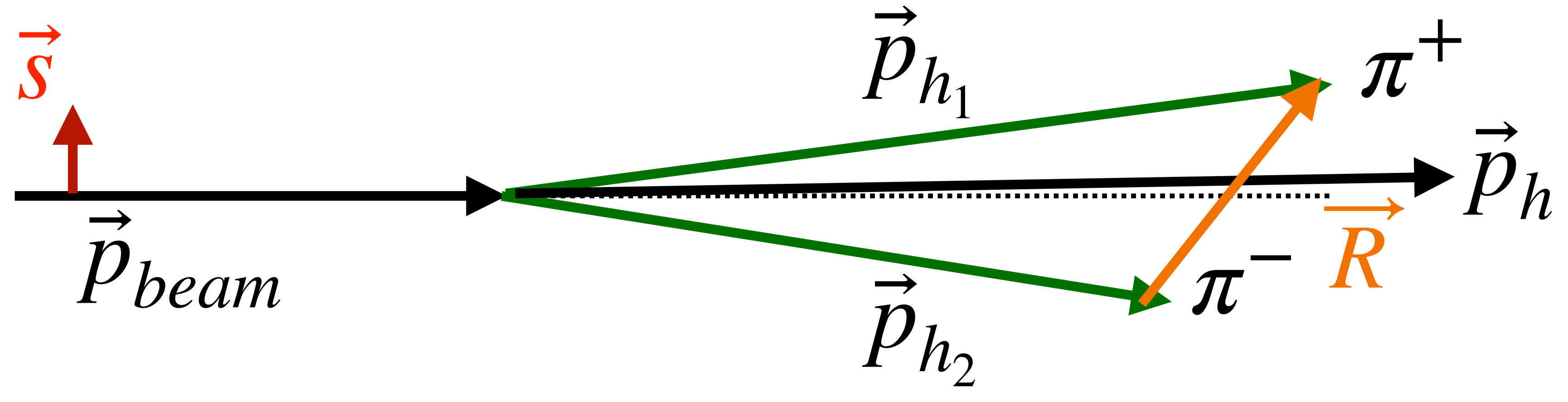


- Polarized parton fragments to $\pi^+\pi^-$
- **Two crucial vectors**: $\vec{p}_h = \vec{p}_{h_1} + \vec{p}_{h_2}$, $\overrightarrow{R} = \frac{1}{2}(\vec{p}_{h_1} - \vec{p}_{h_2})$

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

$\pi^+\pi^-$ Formation and Azimuthal Angles



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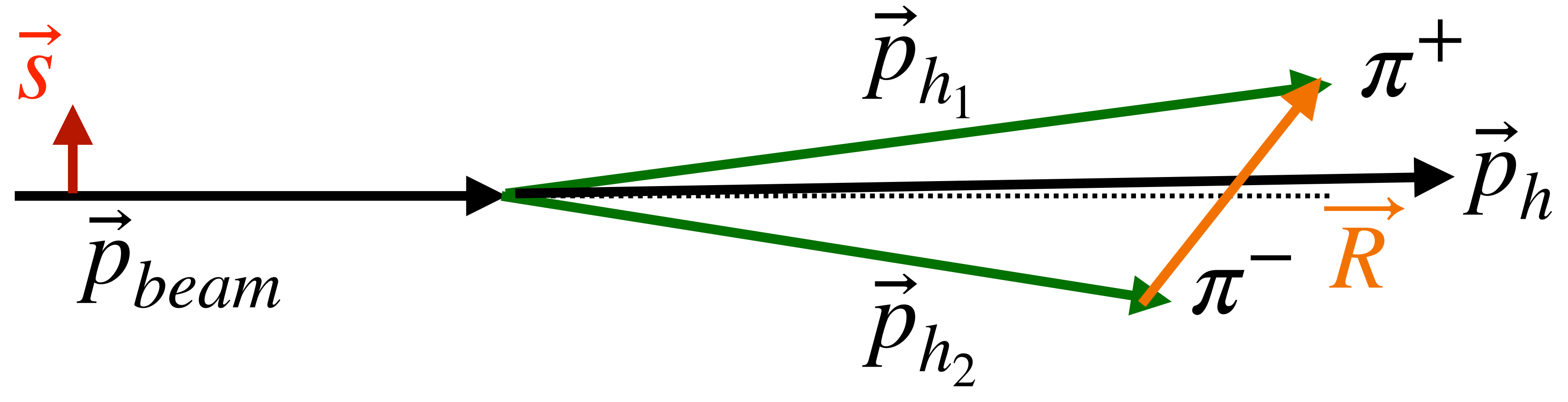

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- Oppositely charged pion pairs, $\pi^+\pi^-$
- Direction of $\vec{R}$ always points from π^- to π^+ (or the other way); otherwise A_{UT} gets diluted.

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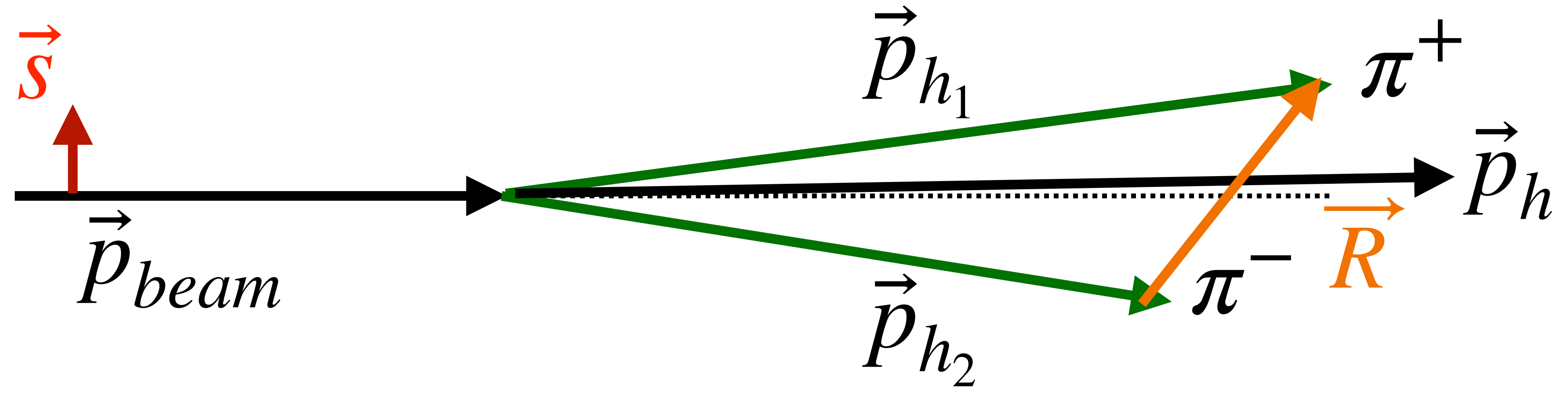

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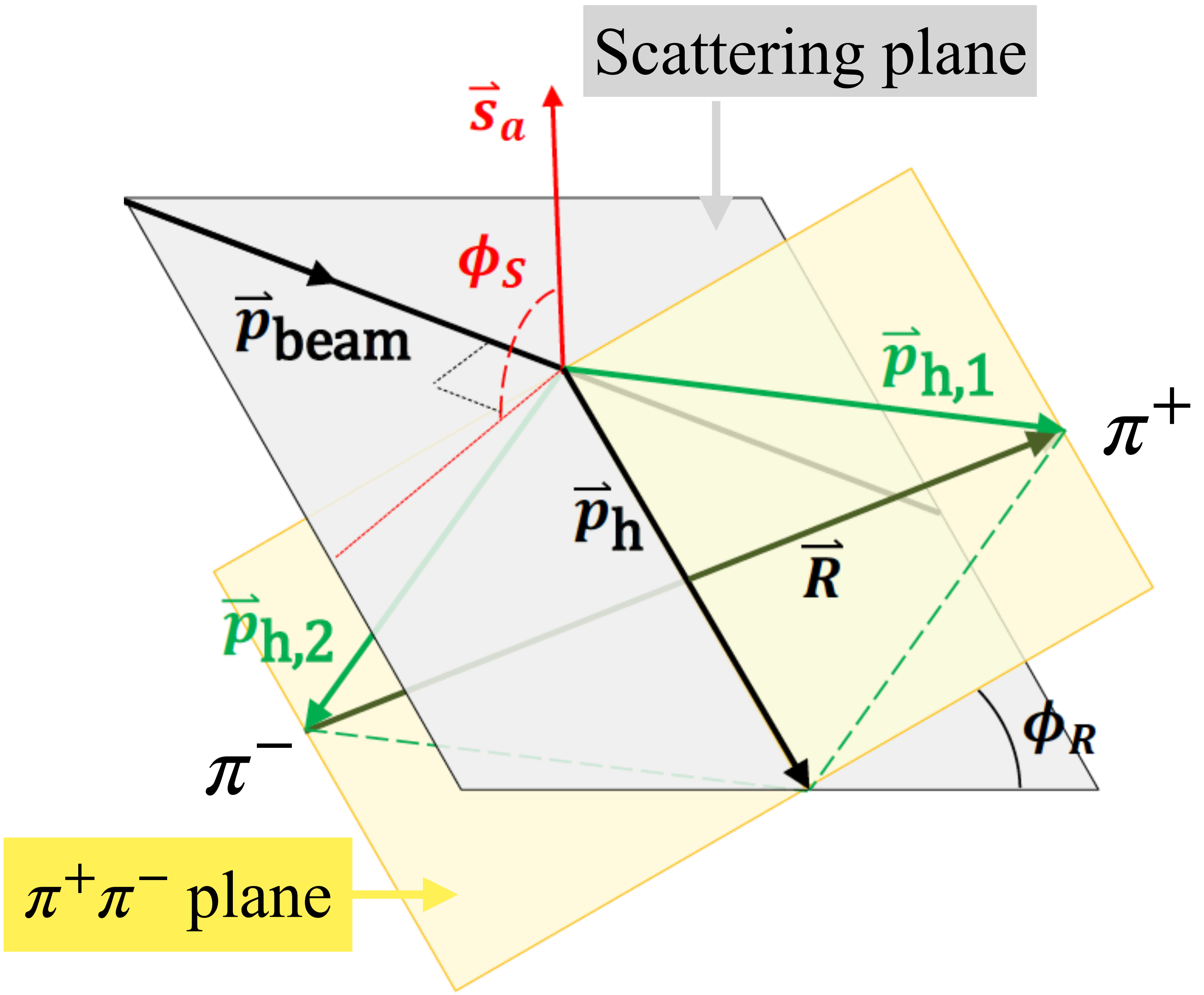
- Direction of $\vec{R}$ always points from π^- to π^+ (or the other way); otherwise A_{UT} gets diluted.
- $\pi^+\pi^-$ Azimuthal angle, $\phi_{RS} = \phi_S - \phi_R$



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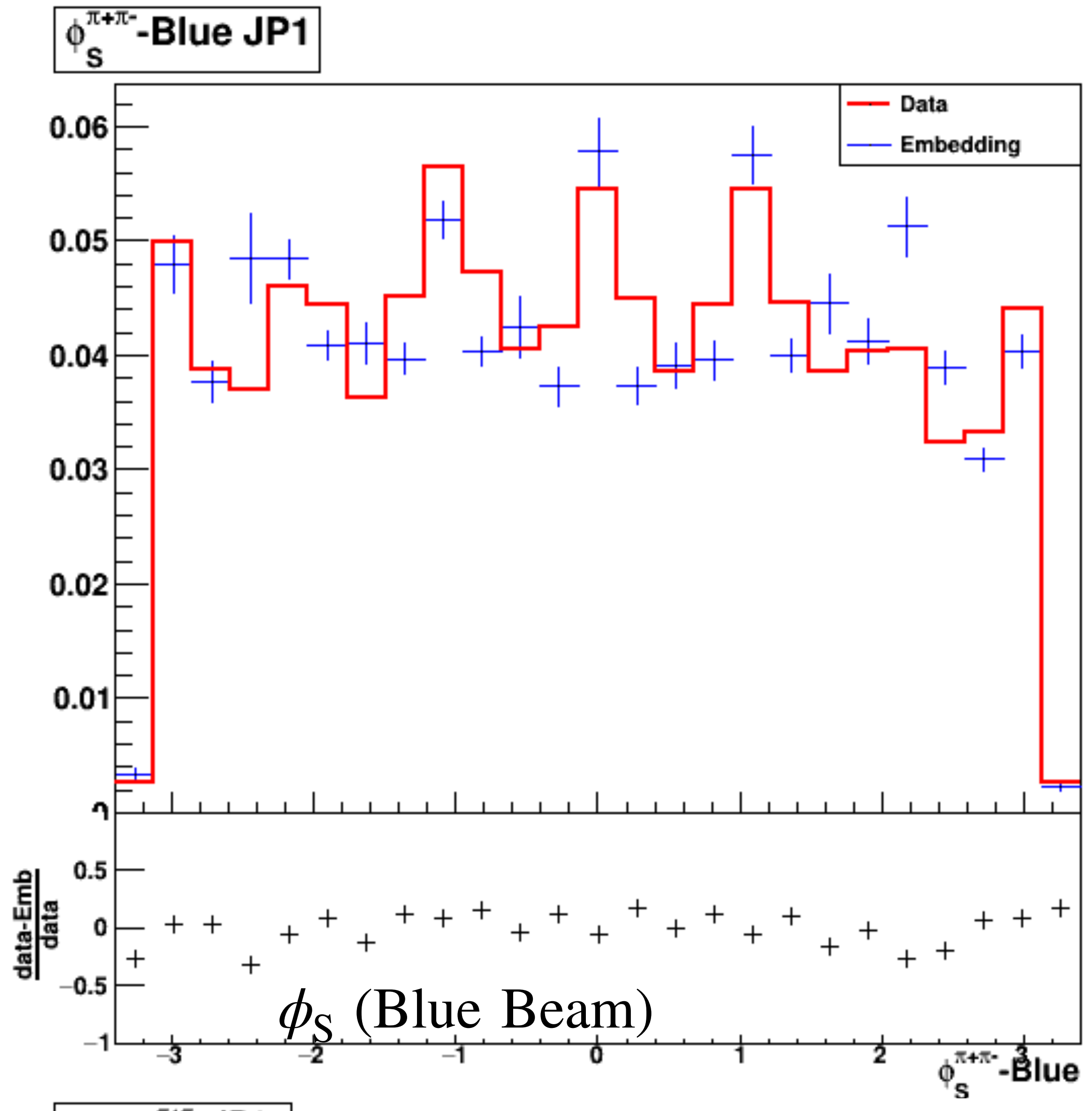
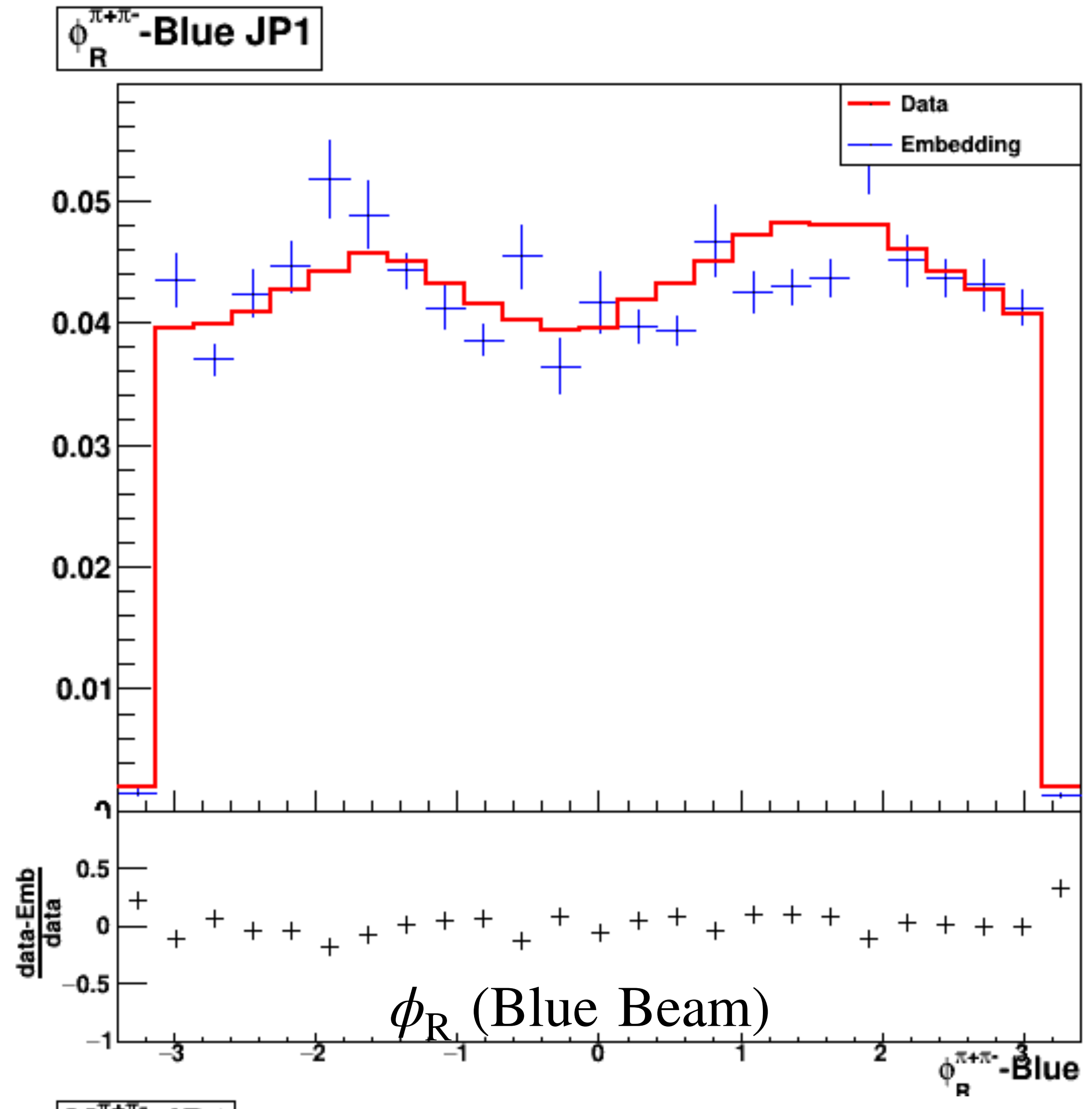
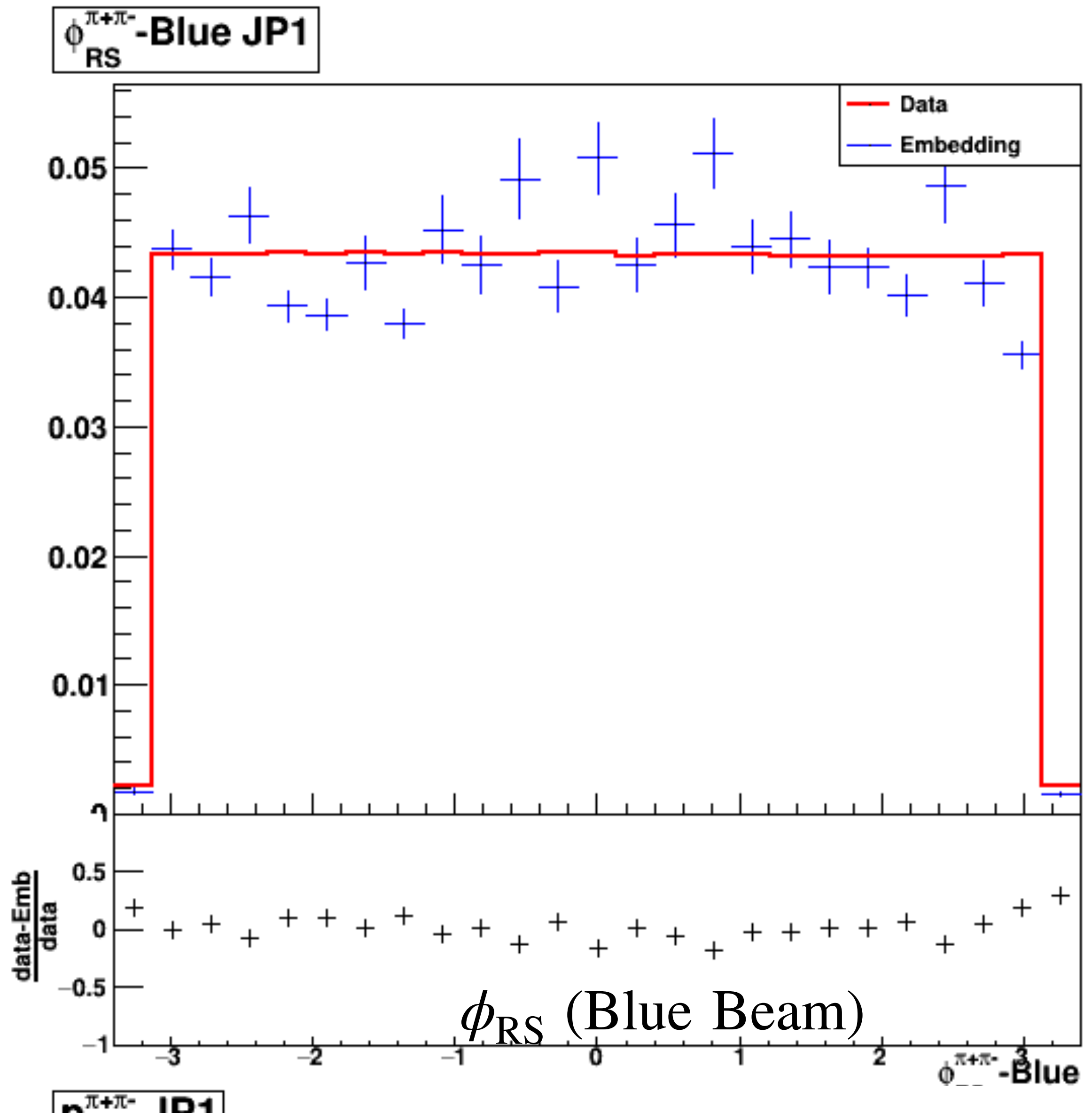
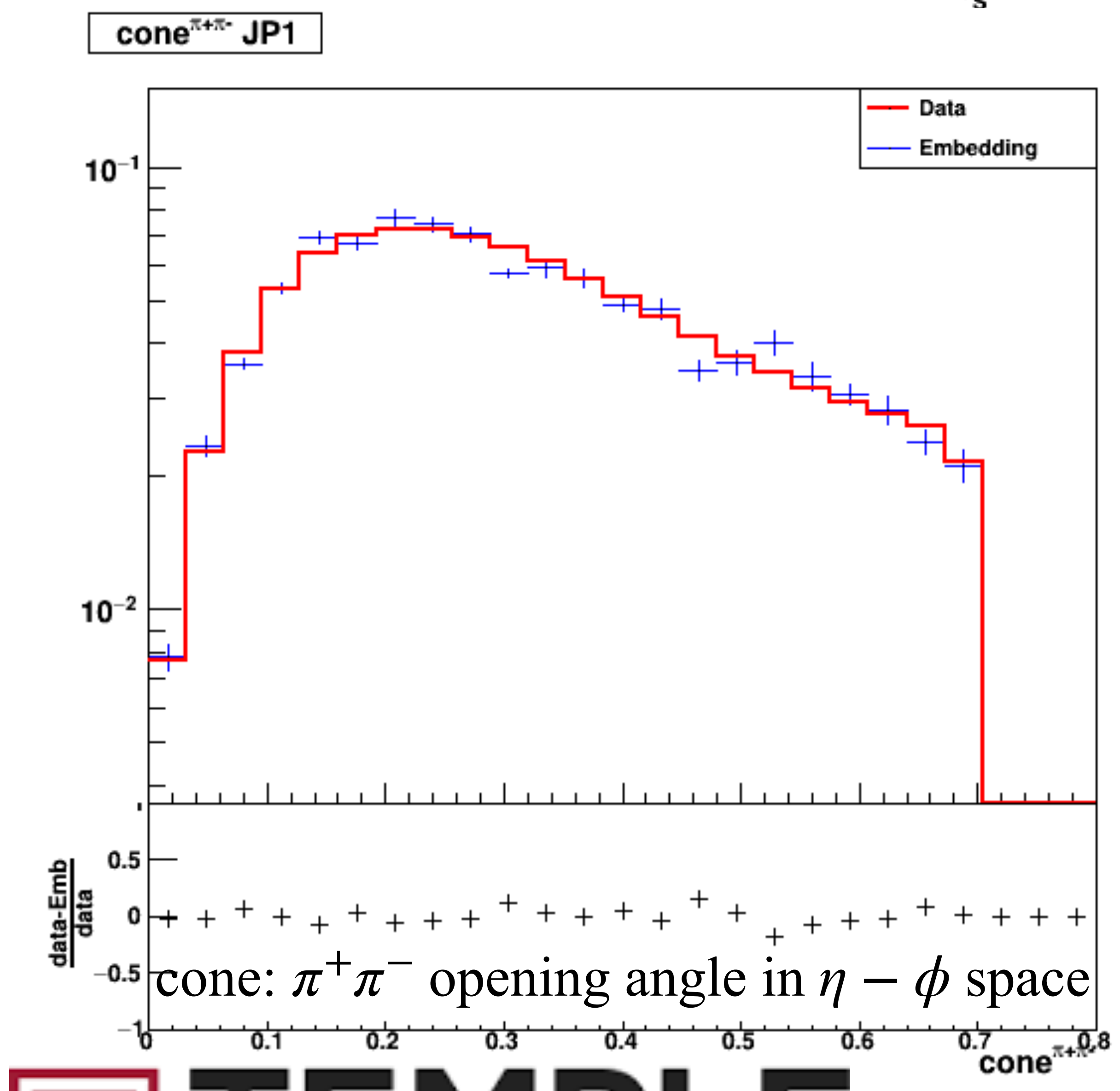
$\pi^+\pi^-$ Kinematics

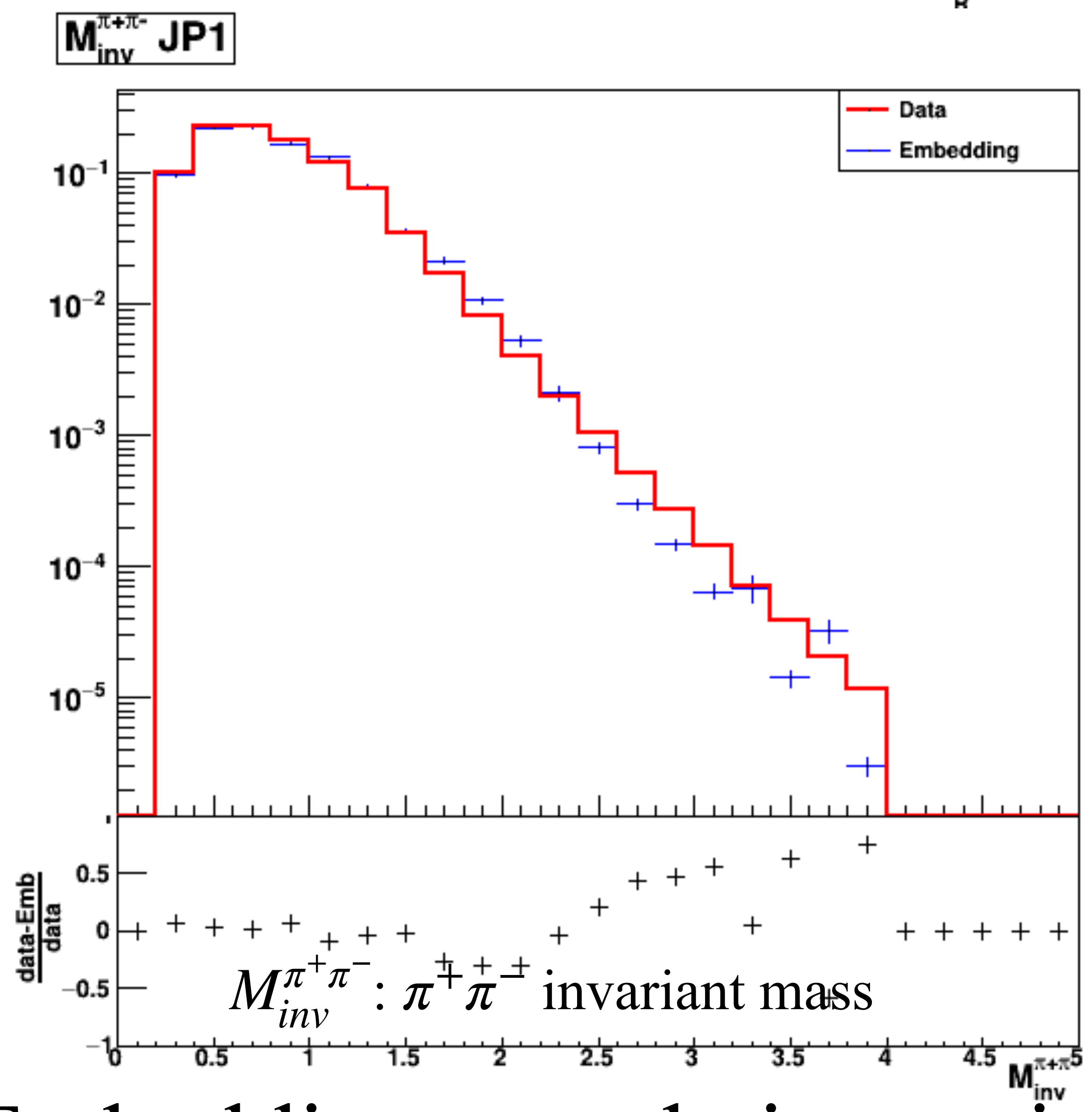


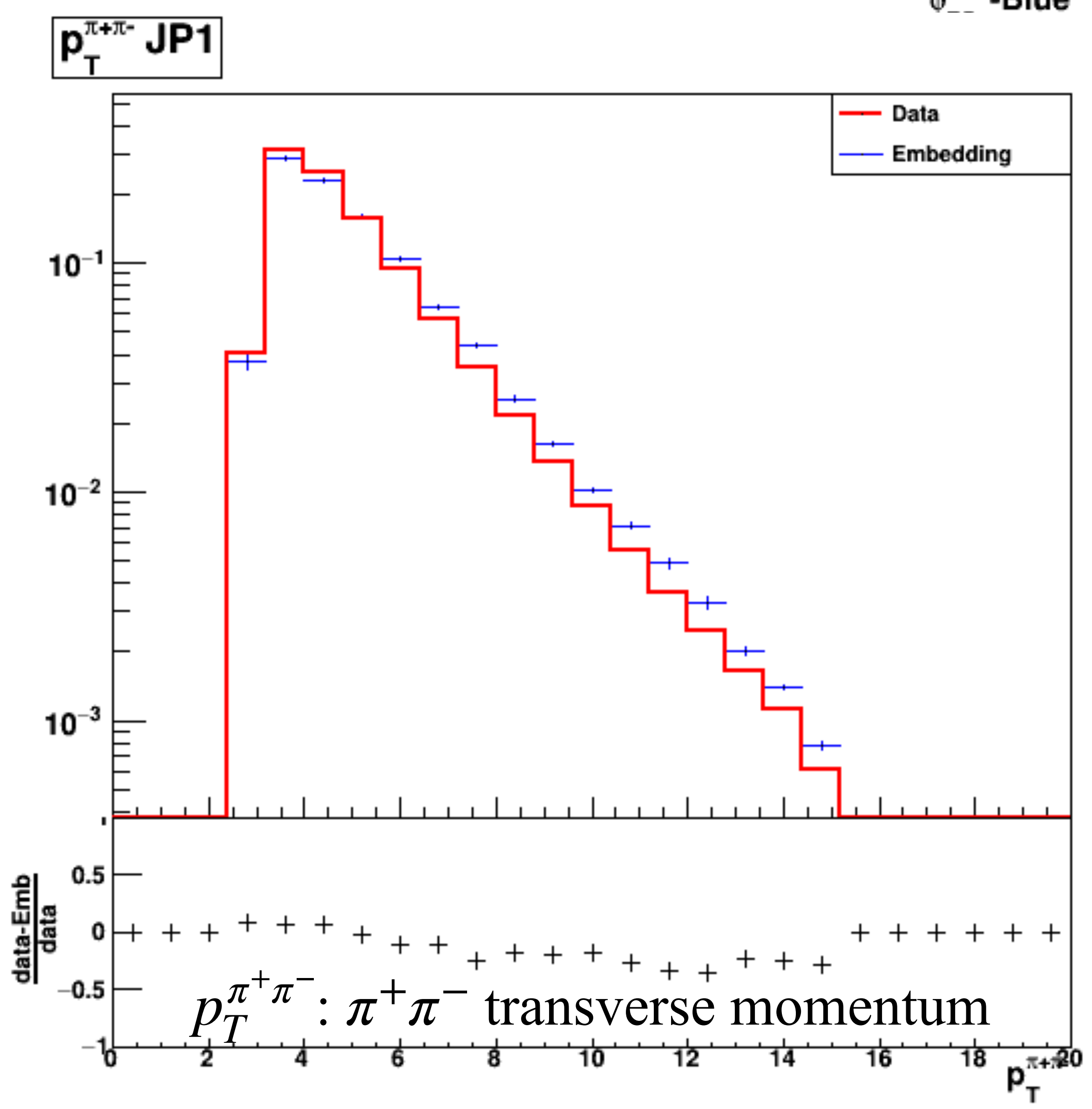


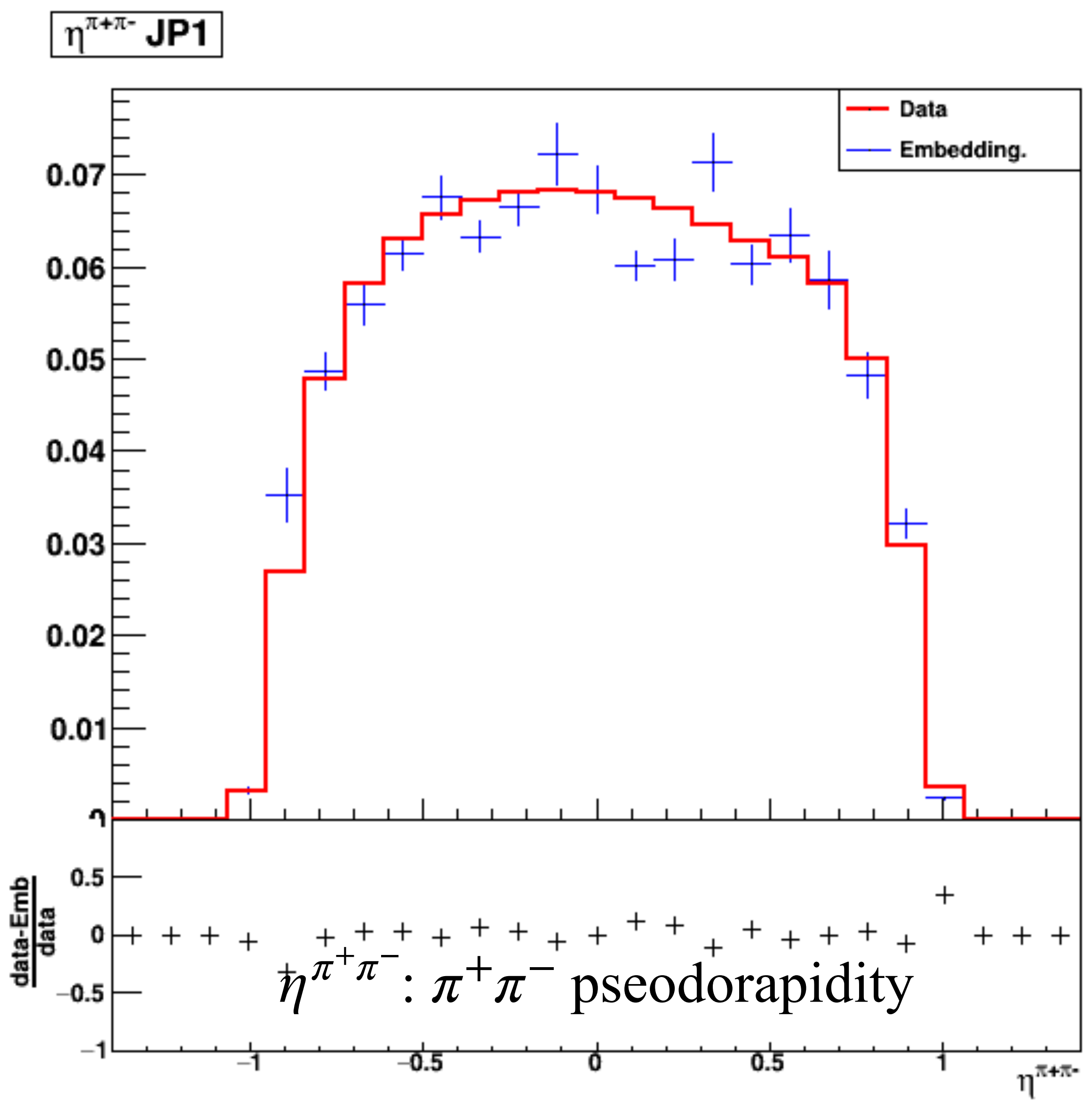


*Embedding: PYTHIA simulated events reconstructed through GEANT3 package.









- Embedding sample is required for the systematic studies.

ϕ_{RS} Binning for A_{UT} Extraction

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

ϕ_{RS} Binning for A_{UT} Extraction

- Cross-ratio formula:

$$A_{UT}\sin(\phi_{RS}) = \frac{1}{P}\frac{\sqrt{N^{\uparrow}(\phi_{RS})N^{\downarrow}(\phi_{RS}+\pi)} - \sqrt{N^{\downarrow}(\phi_{RS})N^{\uparrow}(\phi_{RS}+\pi)}}{\sqrt{N^{\uparrow}(\phi_{RS})N^{\downarrow}(\phi_{RS}+\pi)} + \sqrt{N^{\downarrow}(\phi_{RS})N^{\uparrow}(\phi_{RS}+\pi)}}$$

  - Free from relative luminosity terms (cancels out in symmetric detector system!)

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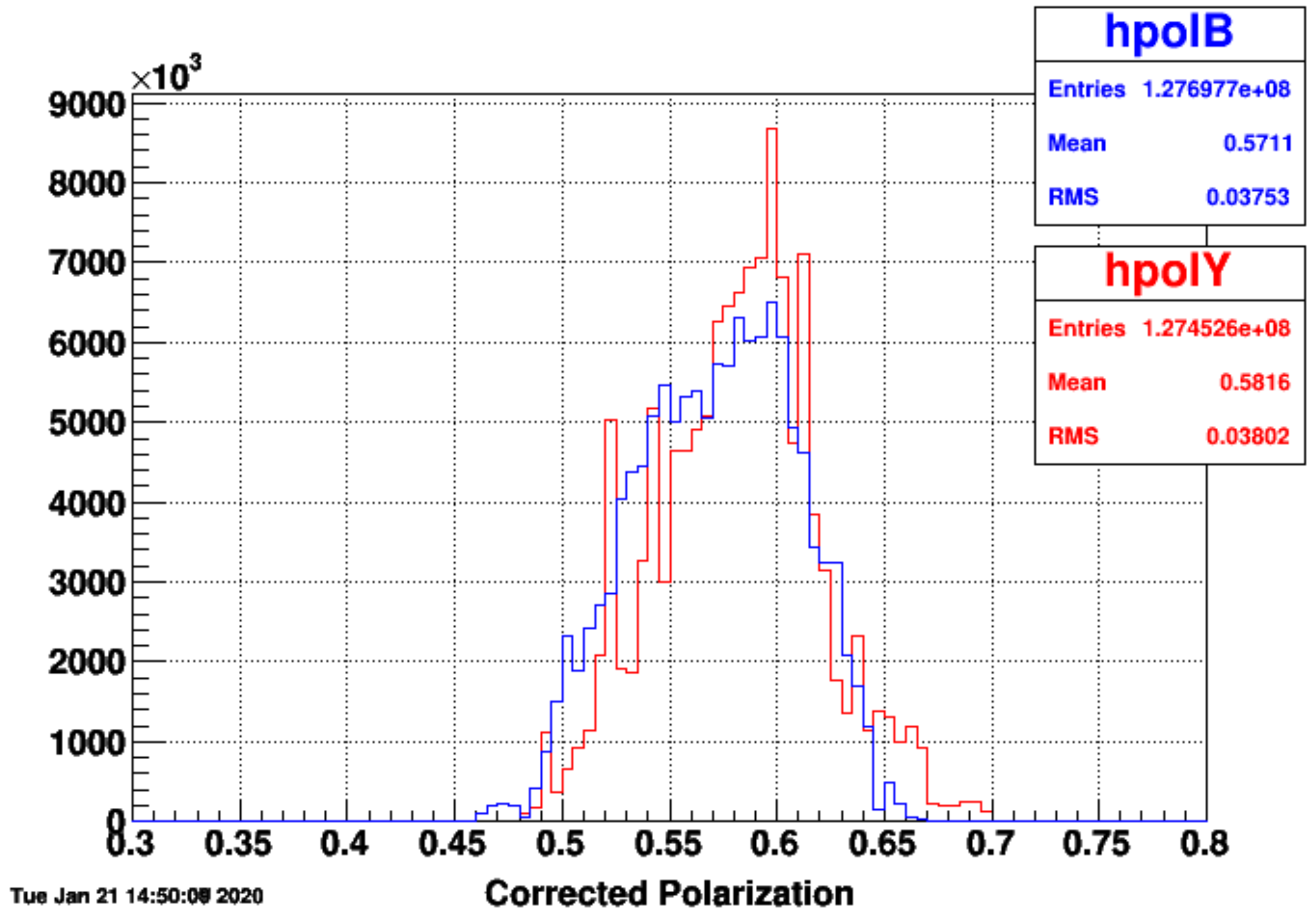

$P \equiv$ Average beam polarization

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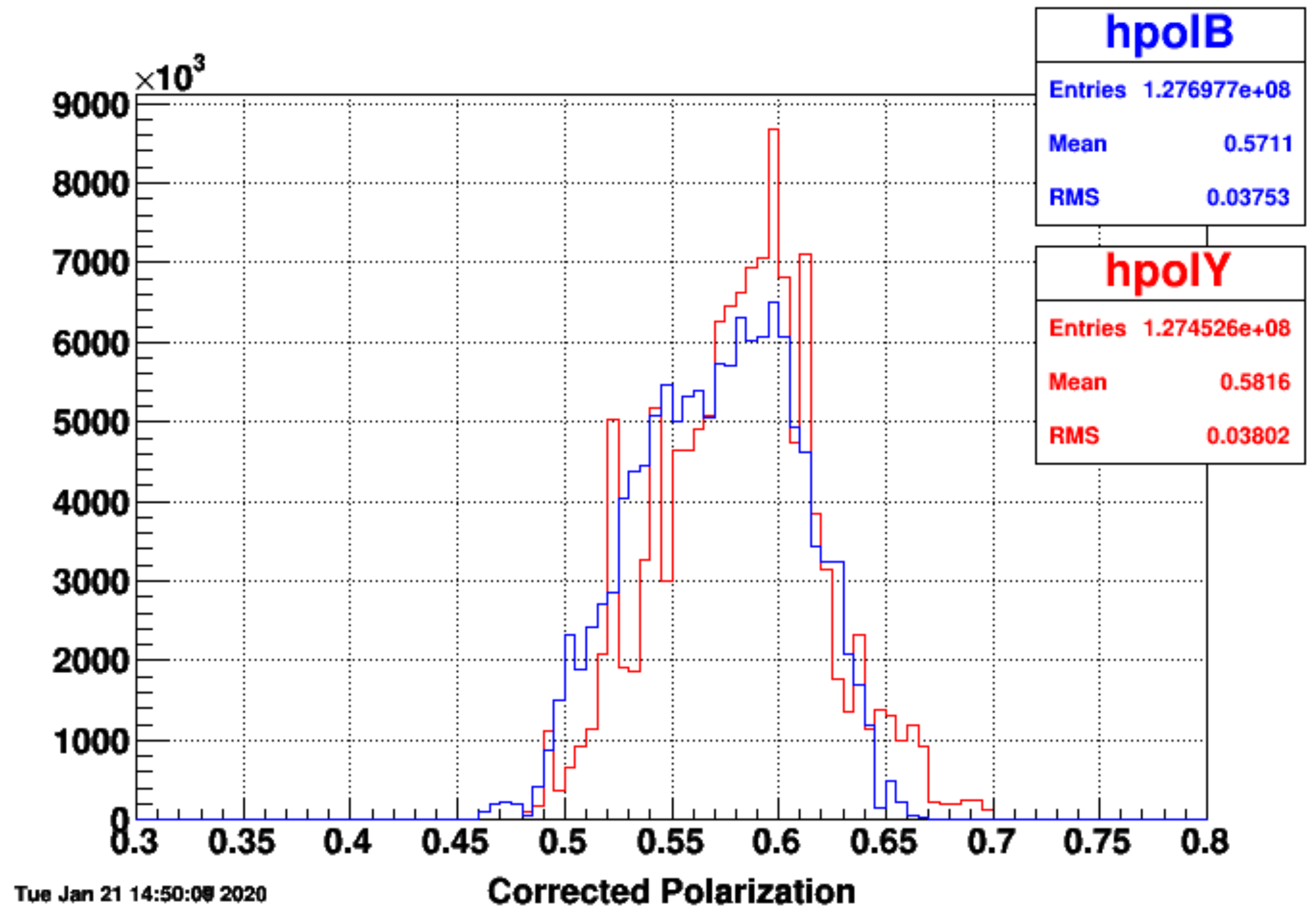
  - Free from relative luminosity terms (cancels out in symmetric detector system!)
- Two transverse polarization states: ↑, ↓
- 16 ϕ_{RS} bins of uniform widths over $[-\pi, \pi]$.
- Symmetry between $[-\pi, 0]$ and $[0, \pi]$ hemispheres.
- Count $\pi^+\pi^-$ yields in each 16 ϕ_{RS} bins for each polarization states: $N^\uparrow(\phi_{RS})$, $N^\downarrow(\phi_{RS})$.



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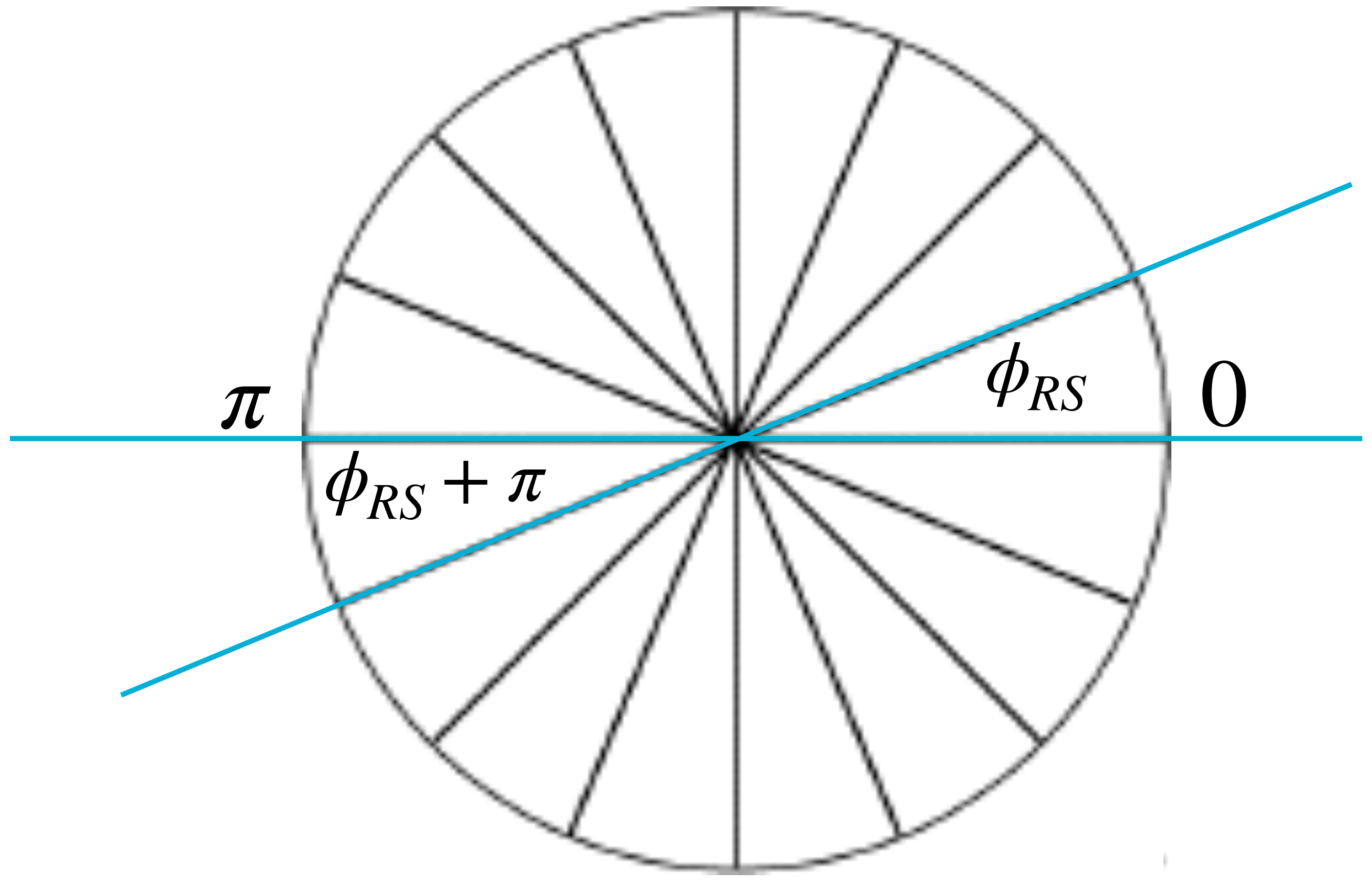

ϕ_{RS} Binning Scheme

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \ \text{at} \sqrt{s} = 200 \text{ GeV}$$

ϕ_{RS} Binning for A_{UT} Extraction

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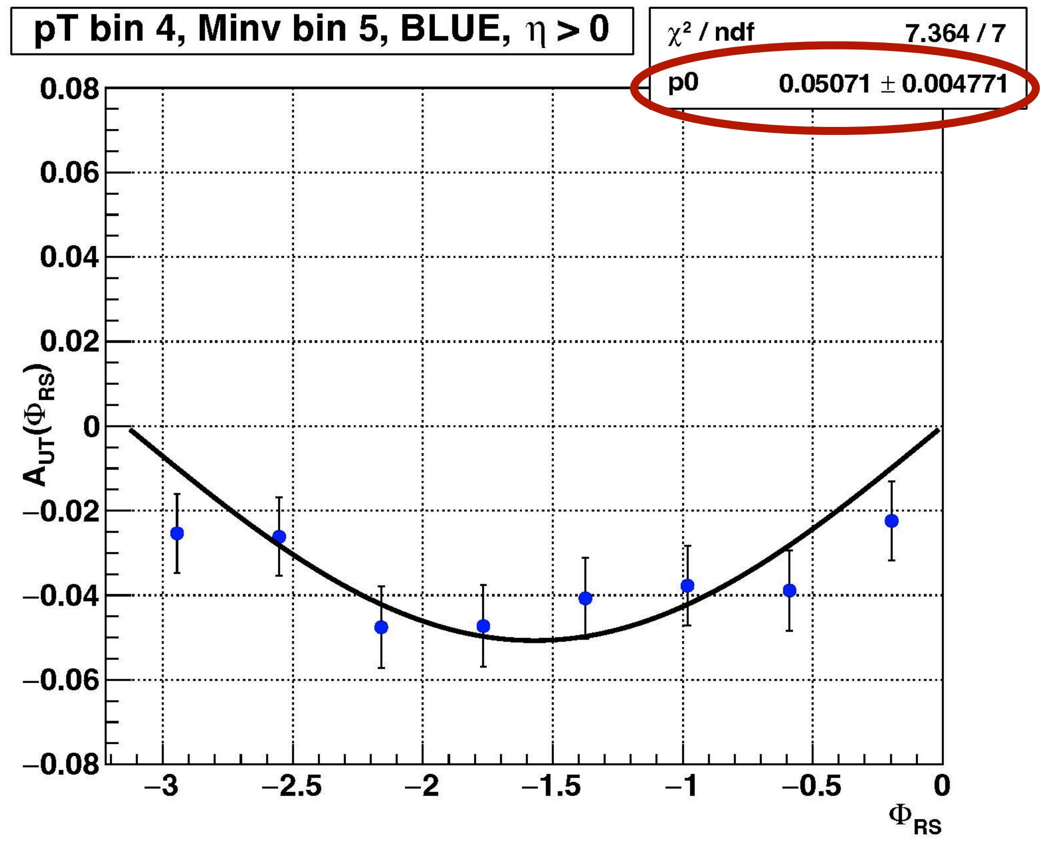
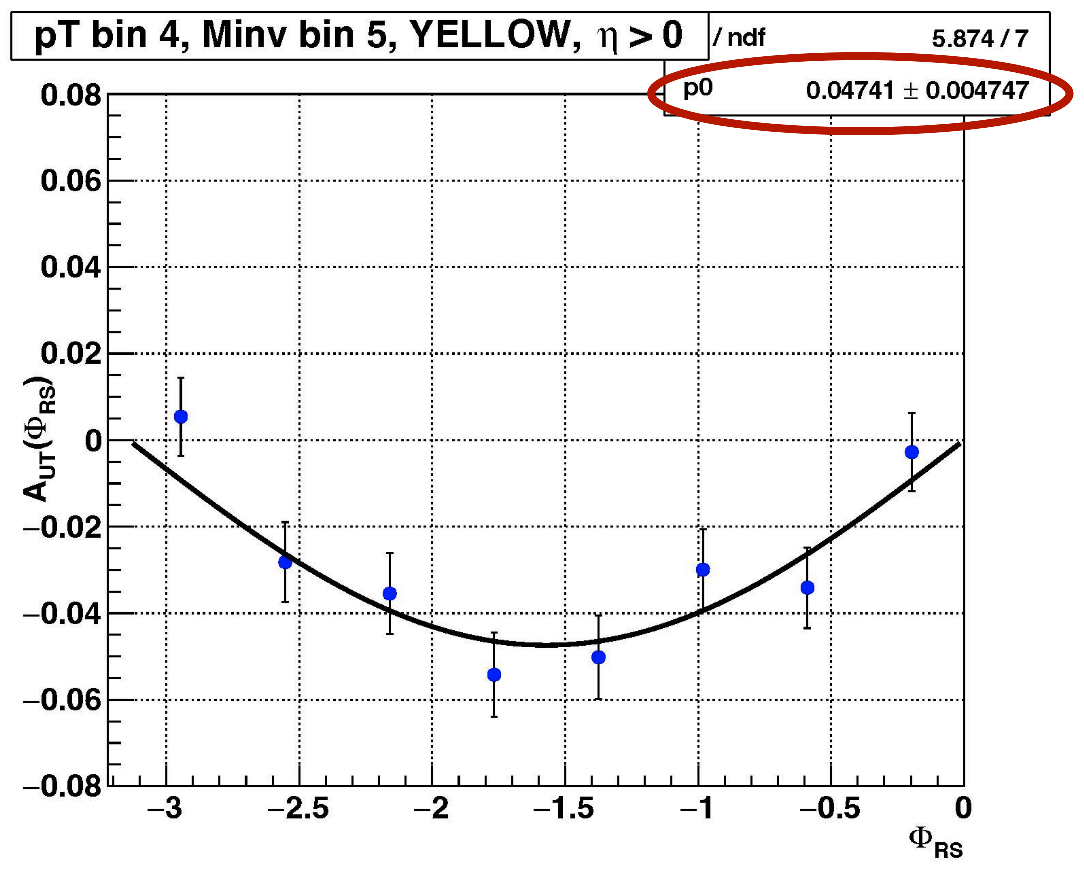
- Symmetry between $[-\pi, 0]$ and $[0, \pi]$ hemispheres.
- Count $\pi^+\pi^-$ yields in each 16 ϕ_{RS} bins for each polarization states: $N^{\uparrow}(\phi_{RS})$, $N^{\downarrow}(\phi_{RS})$.
- Amplitude of the fit in $[-\pi, 0]$ gives the A_{UT} .
- A_{UT} extracted for Blue and Yellow beams separately. Final A_{UT} is the weighted average of both.

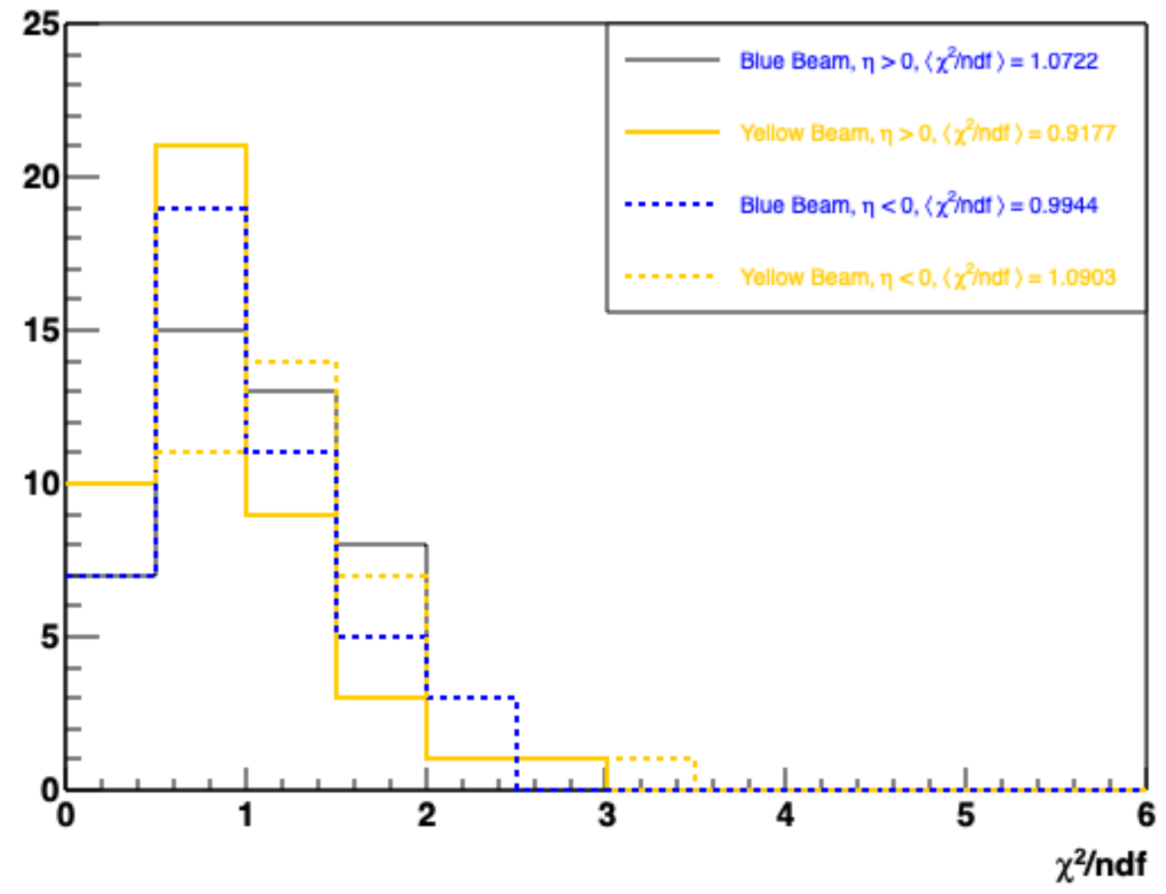


Blue Beam: Sample sine fit in $\eta > 0$



Yellow Beam: Sample sine fit in $\eta > 0$

The fit description is monitored by χ^2/ndf.



Systematic Uncertainties: Particle Identification (PID)

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

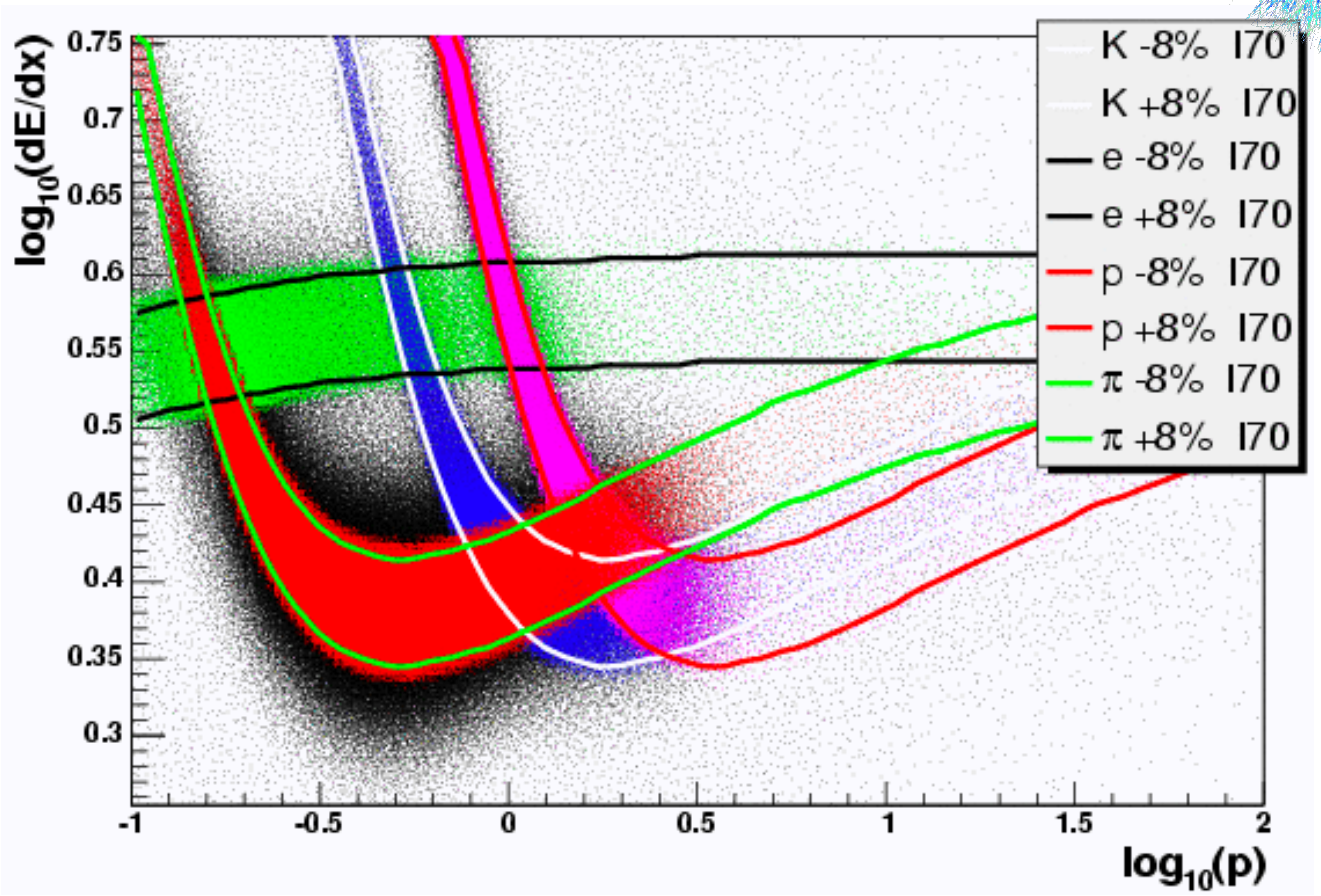
$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Systematic Uncertainties: Particle Identification (PID)

- Pions are selected using $-1 < n\sigma_\pi < 2$ cut, where

$$\mathrm{n}\sigma_\pi = \frac{1}{\sigma_{\exp}} \ln\left(\frac{\mathrm{dE/dx}_{\mathrm{obs}}}{\mathrm{dE/dx}_{\pi,\mathrm{calc}}}\right) \sim \text{Gaussian distribution}$$

- PID is challenging for particle $p > 1$ GeV/c, where the dE/dx separation is not clean.



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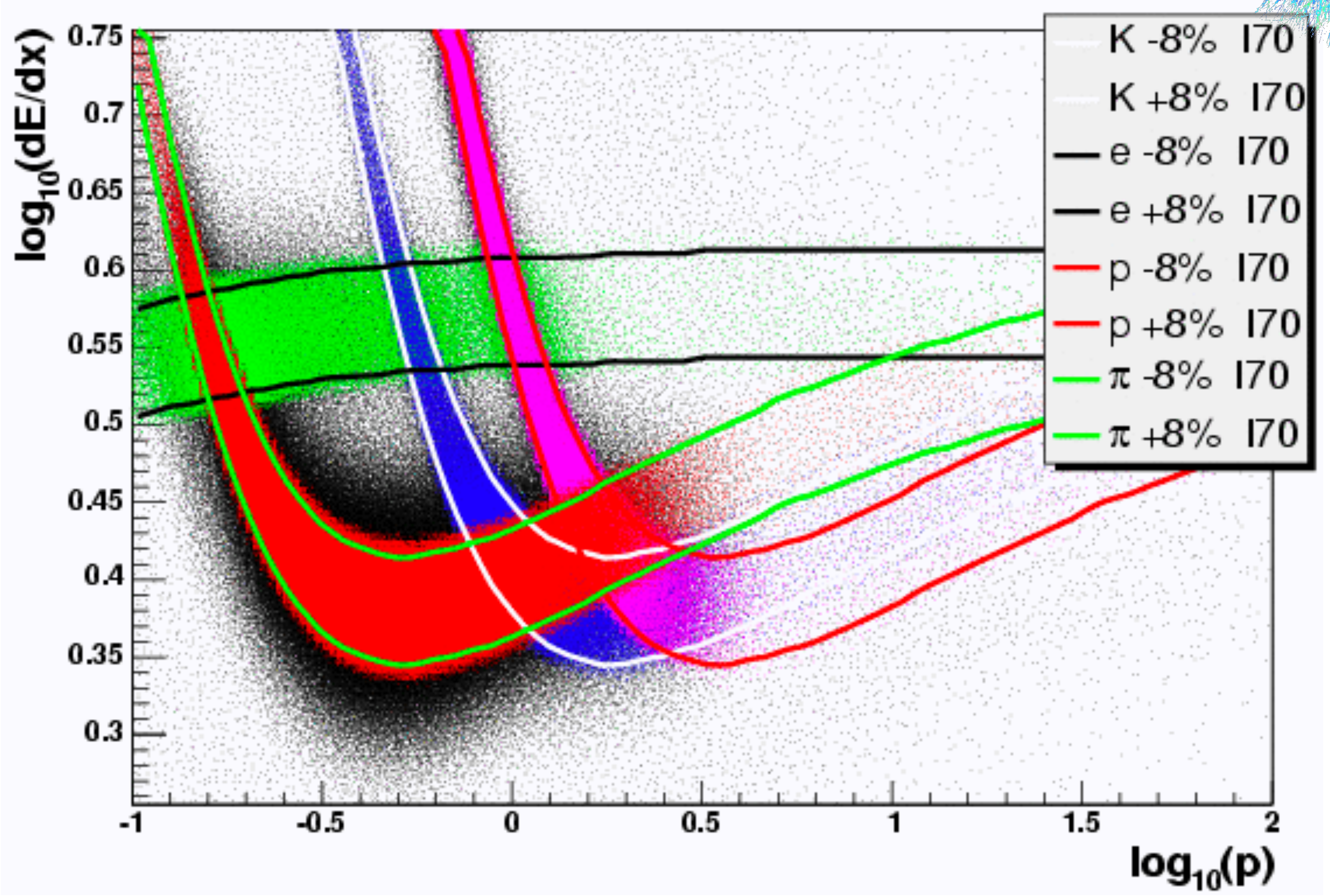
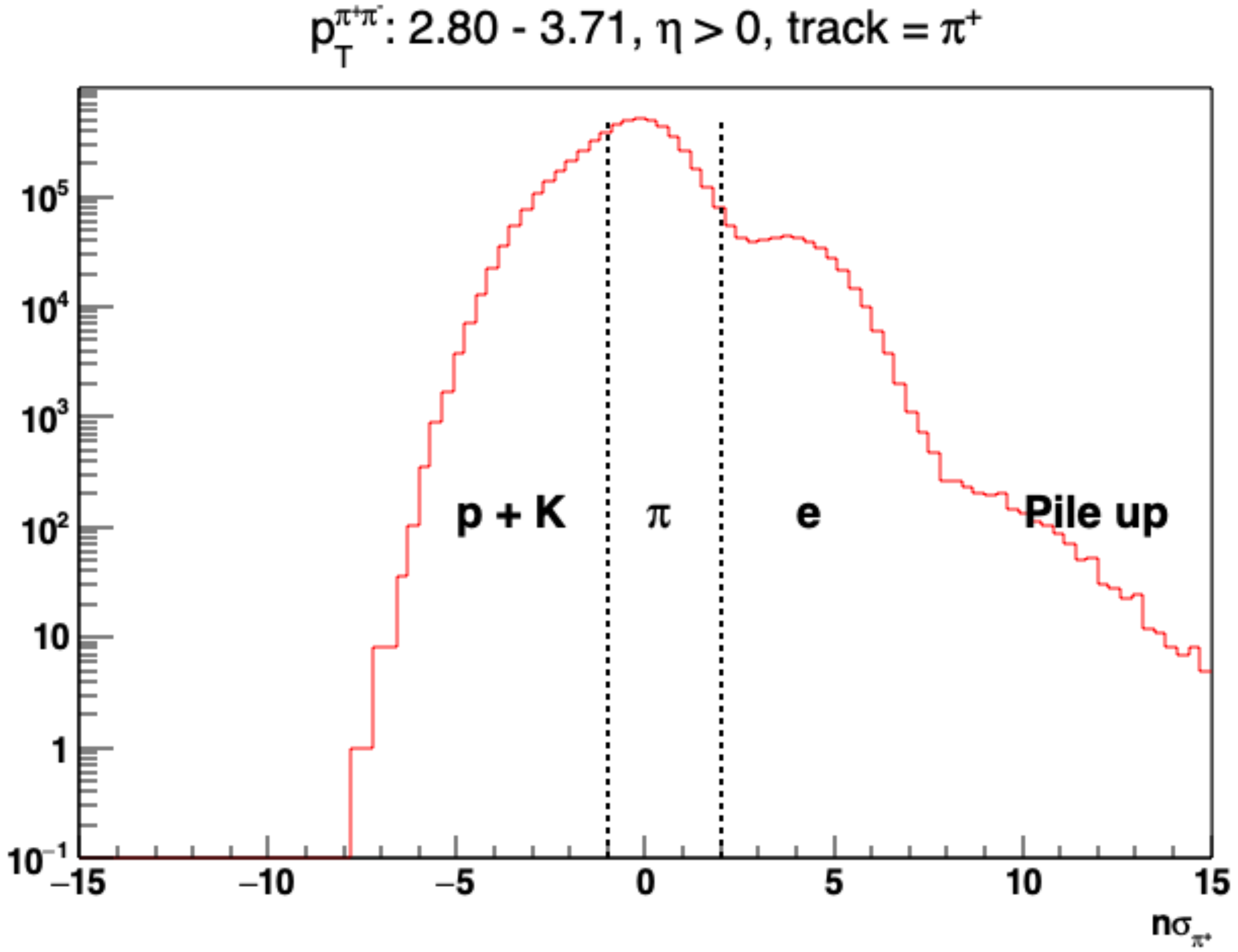


The fraction of proton, kaon, and electron (backgrounds) in the pion signal region estimates the PID systematic uncertainty.



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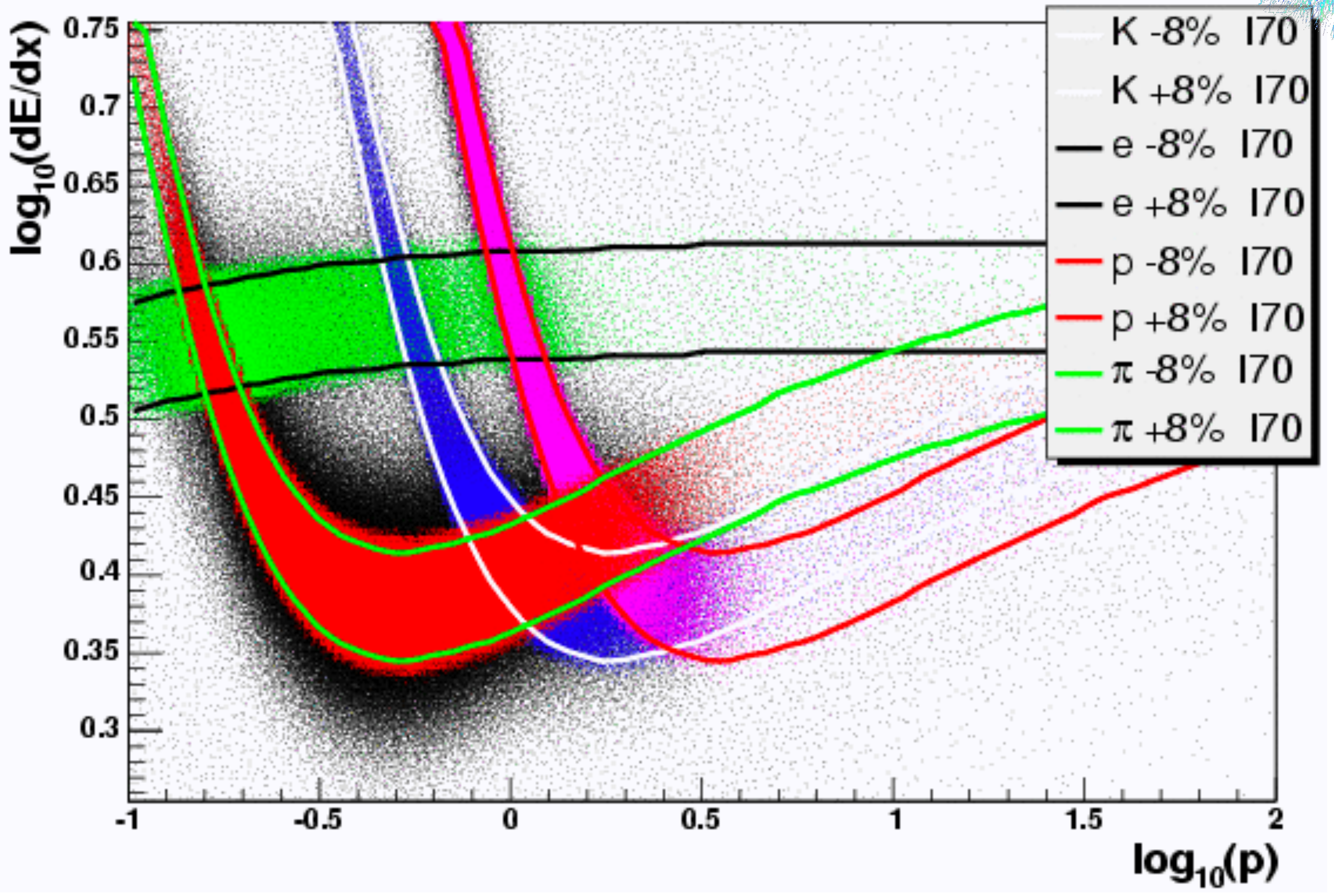

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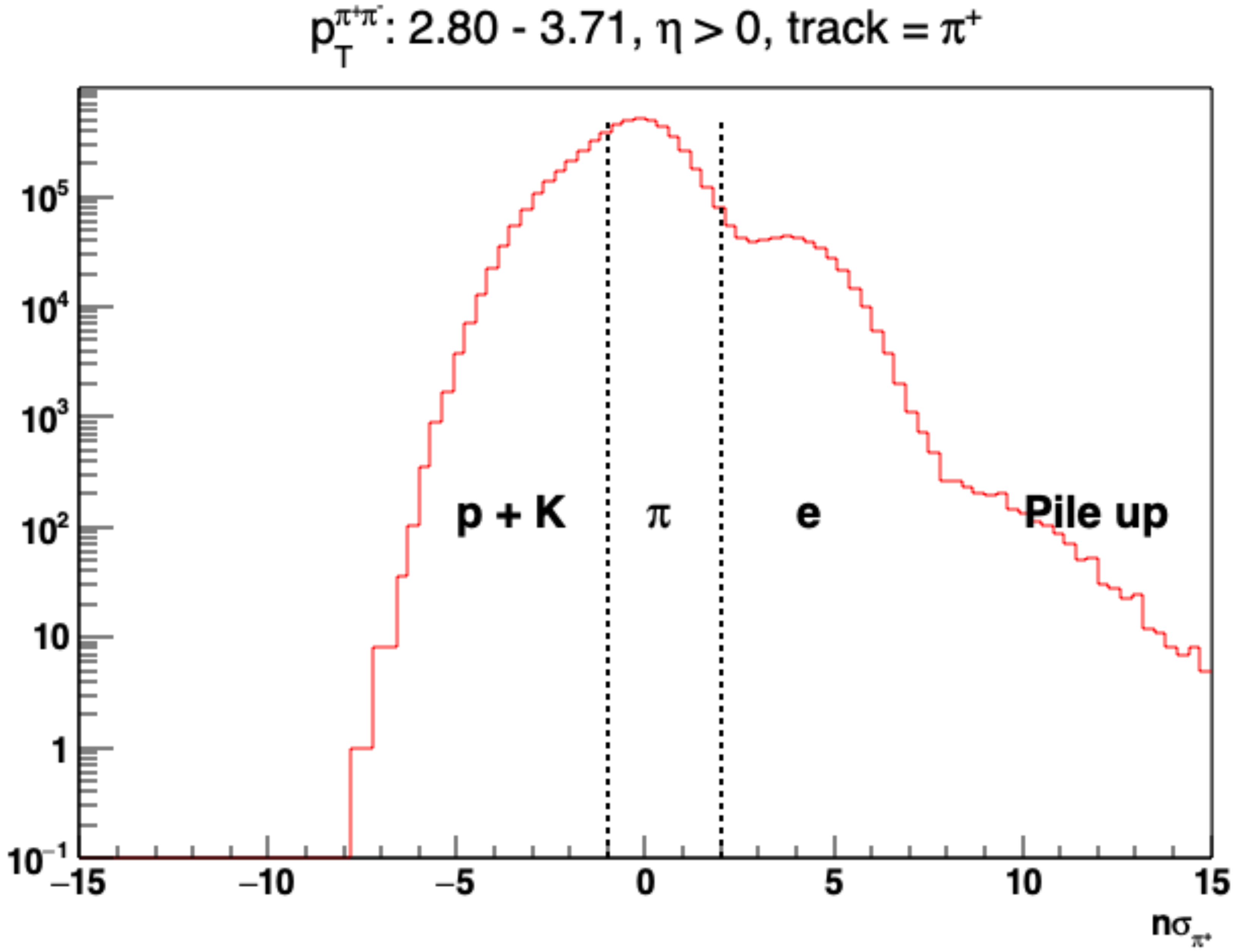

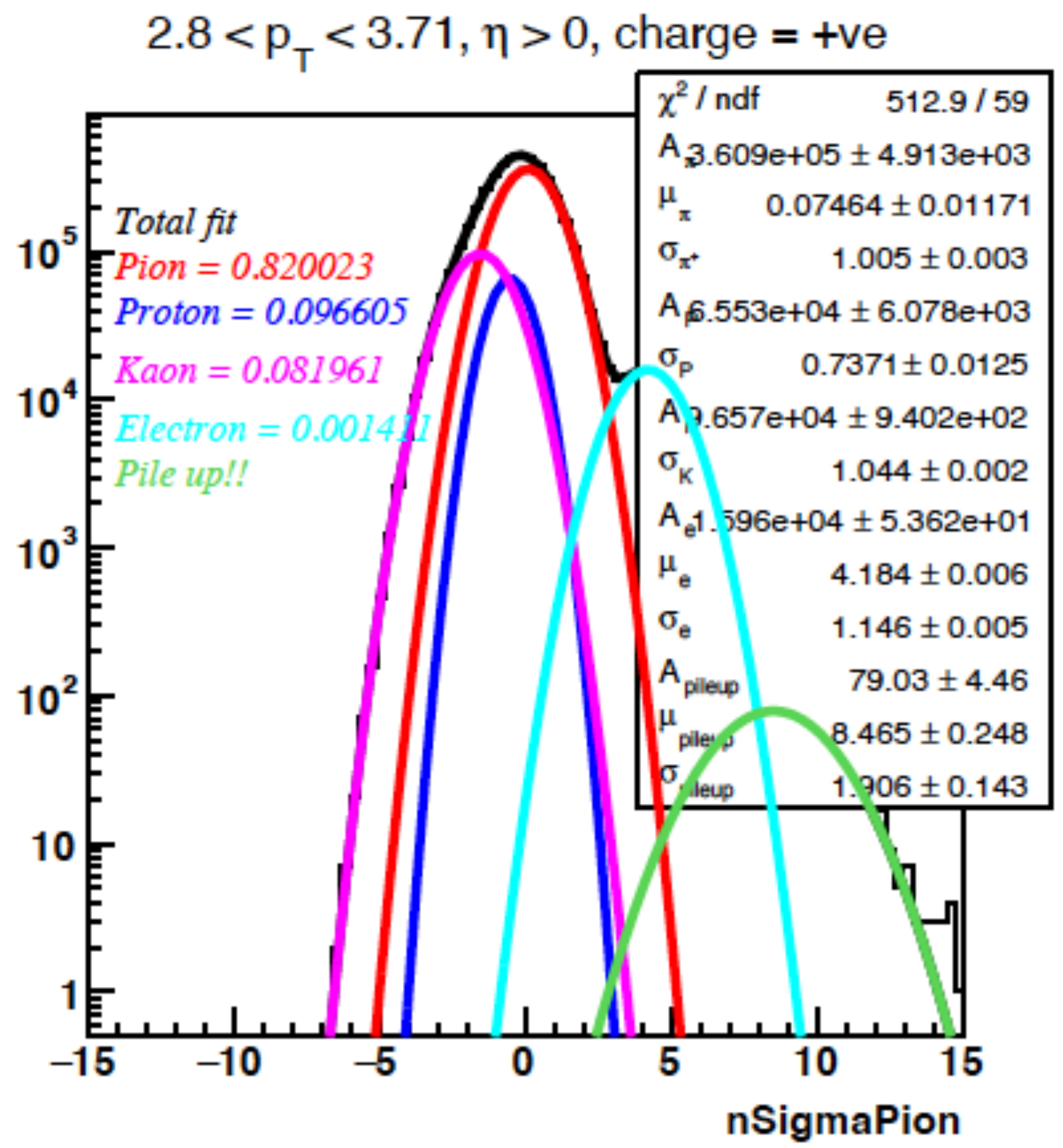

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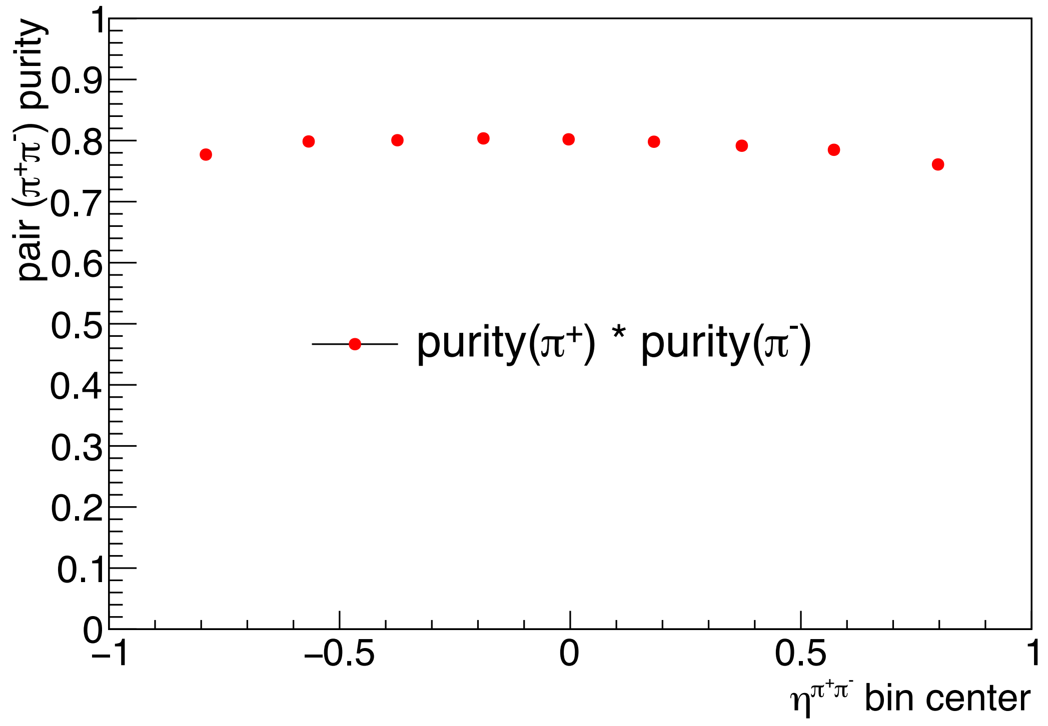
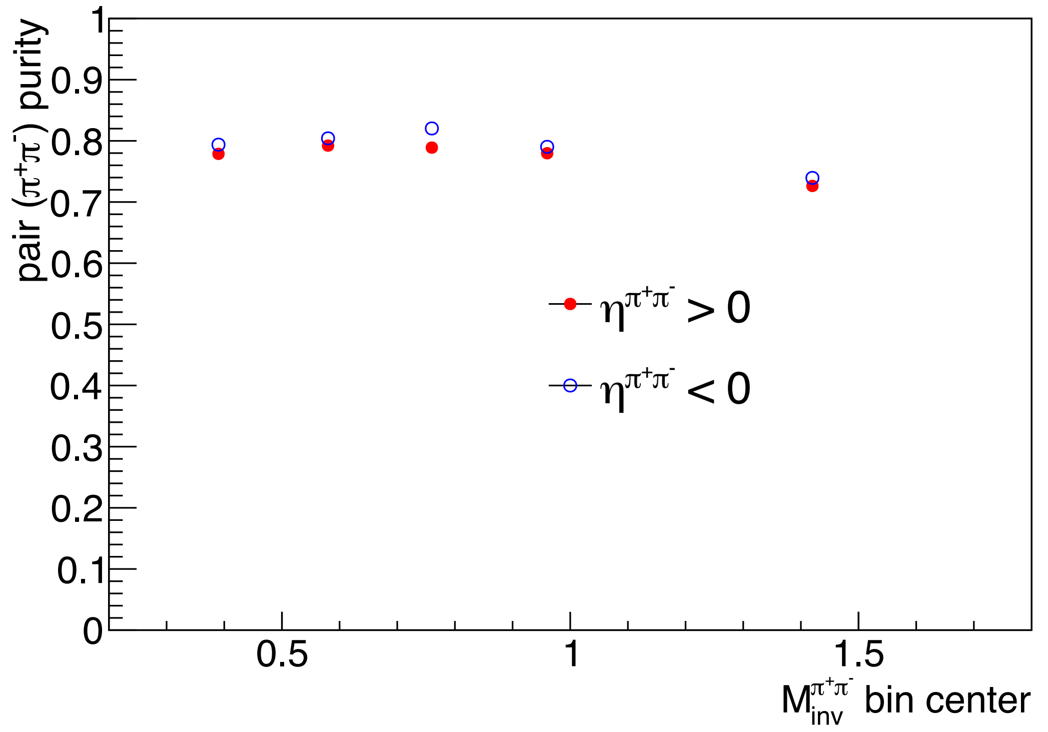
- PID is challenging for particle $p > 1$ GeV/c, where the dE/dx separation is not clean.
- $\pi^+\pi^-$ Purity: ~ 70% - 80%

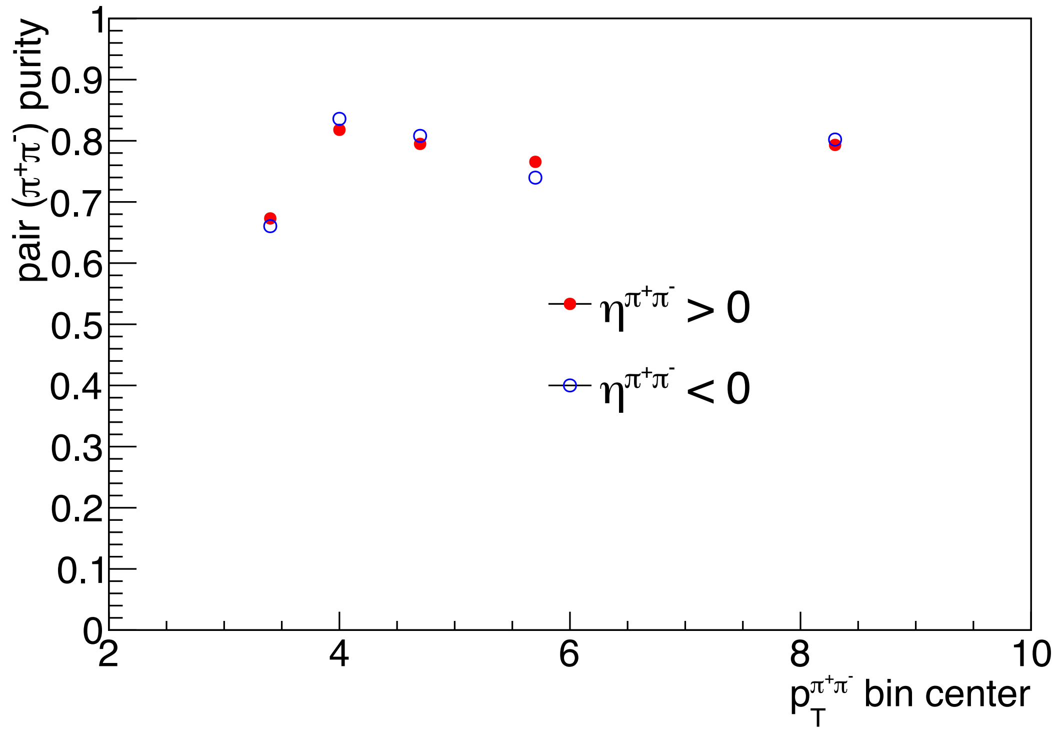
- $\delta_{sys,pid} = (1 - f_{\pi^+\pi^-}) \cdot \text{Max}(A_{UT}, \delta_{stat}^{A_{UT}})$

$\delta_{sys,pid}$ is estimated assuming the background asymmetry is about the size of the signal asymmetry or smaller.

$\pi^+\pi^-$ fraction in asymmetry bins







STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \;\text{ at } \sqrt{s} = 200 \text{ GeV}$$

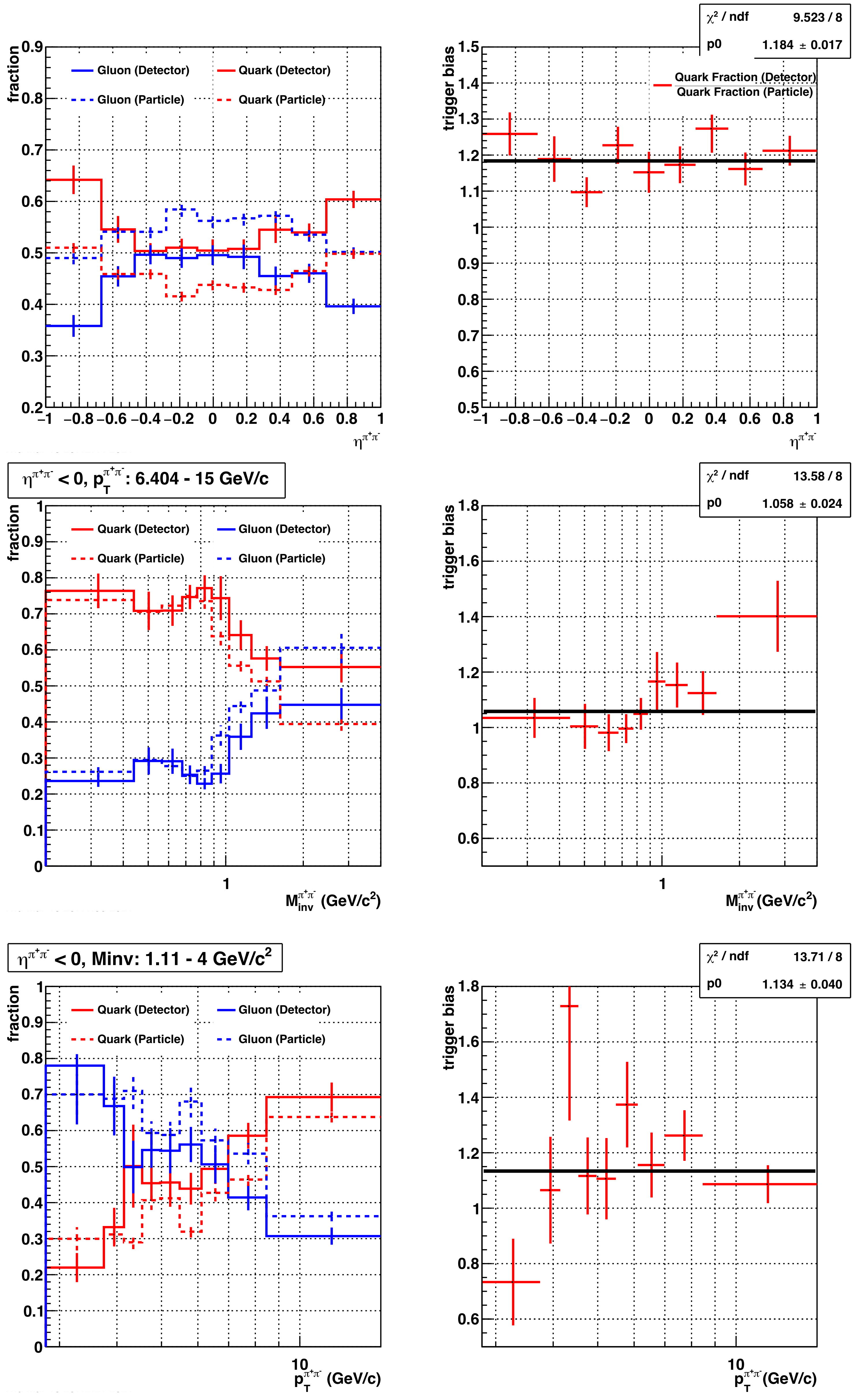
Systematic Uncertainties: Trigger Bias

- Event triggering may cause bias in the measurement favoring quark events over gluons.
- The size of the trigger bias can be estimated from the embedding sample.

$$\text{Trigger Bias Fraction, } f_{bias} = \frac{f_{quark}^{detector}}{f_{quark}^{particle}}$$

- $\delta_{sys,bias} = (1 - f_{bias}) \cdot \mathrm{Max}(A_{UT}, \delta_{stat}^{A_{UT}})$

Total Systematic Uncertainty: $\delta_{sys} = \sqrt{\delta_{sys,pid}^2 + \delta_{sys,bias}^2}$



STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

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STAR Preliminary: $A_{UT}^{\sin(\phi_{RS})}$ **vs** $\eta^{\pi^+\pi^-}$

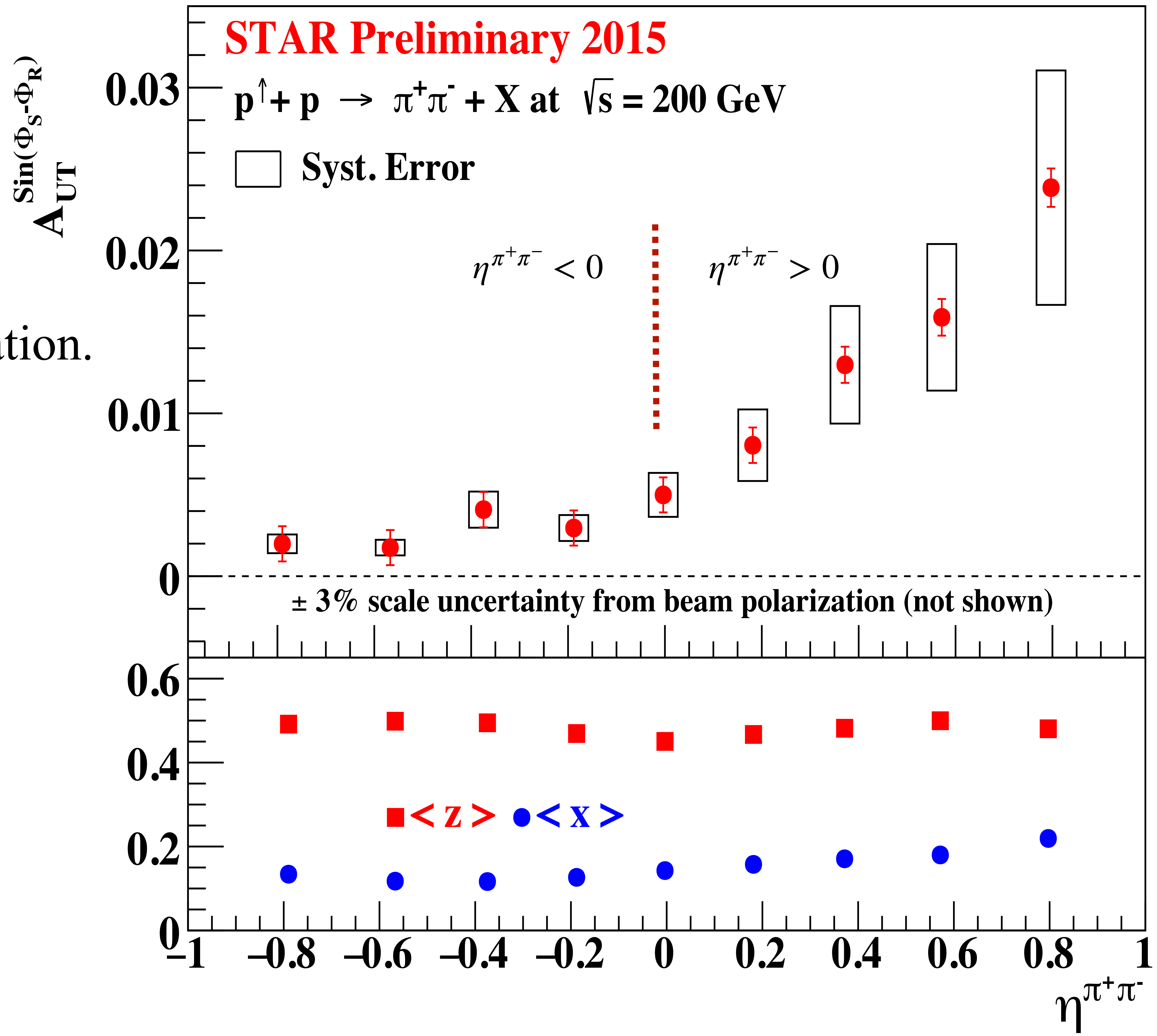
Top Panel:

- A_{UT} increases with the $\eta^{\pi^+\pi^-}$, where $\eta \equiv x$.
- Sizable $h_1^q(x)$ is expected in the $\eta > 0$ region.

Bottom Panel:

- Mean x and z as a function of $\eta^{\pi^+\pi^-}$ from simulation.
- $0.1 < \langle x \rangle < 0.22$, $\langle z \rangle \sim 0.46$

$$z \rightarrow \frac{E_{\pi^+\pi^-}}{E_{quark}}, \; x = \frac{\vec{p}_{quark}}{\vec{p}_{proton}}$$



- Systematic uncertainty includes effects related to PID and trigger bias.
- **Dominant systematic uncertainty from the PID.**

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STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

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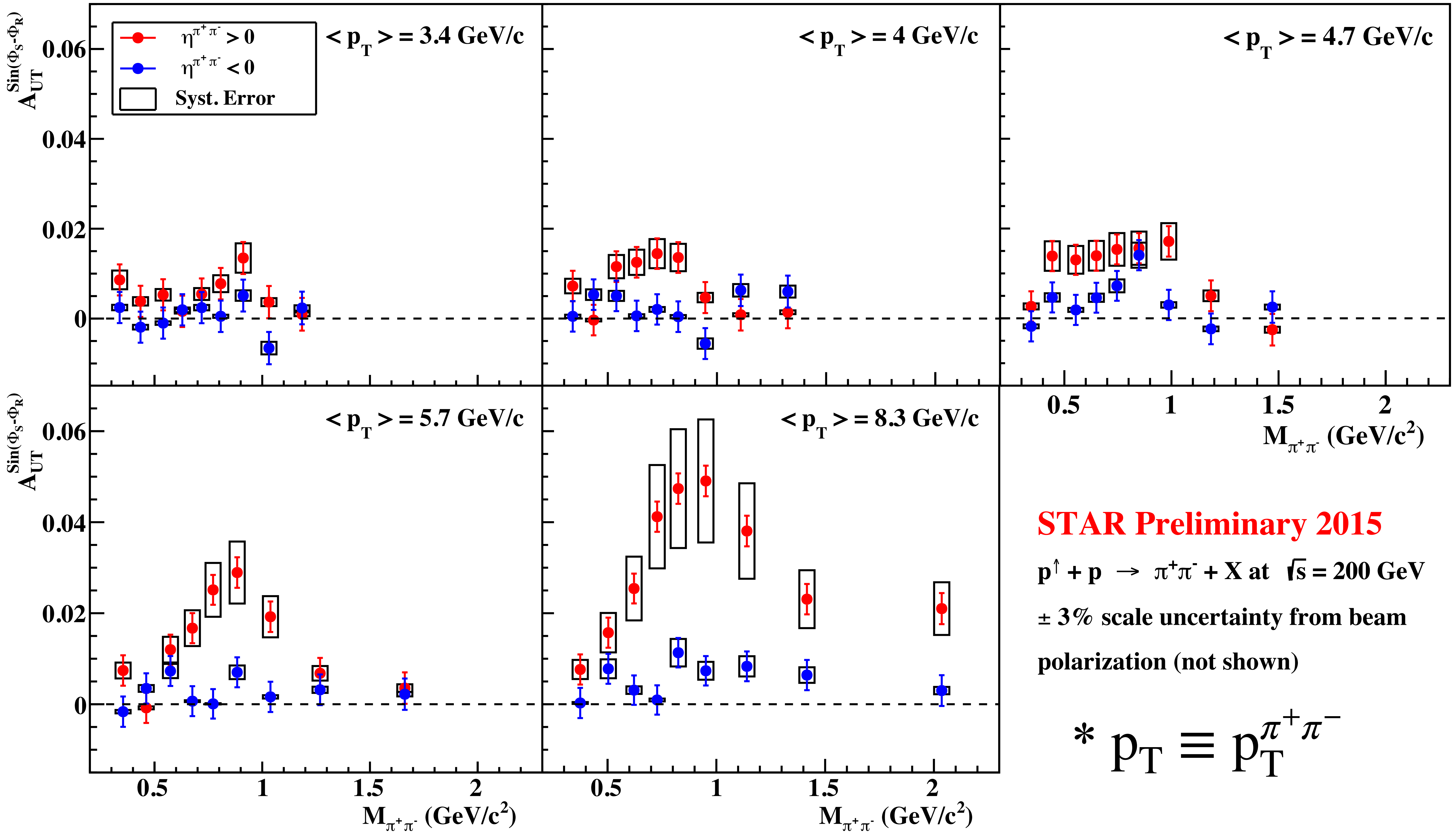
STAR Preliminary: $A_{UT}^{\sin(\phi_{RS})}$ vs $M_{inv}^{\pi^+\pi^-}$

- **$A_{UT}^{\sin(\phi_{RS})}$ vs $M_{inv}^{\pi^+\pi^-}$ in different p_T and $\eta^{\pi^+\pi^-}$ bins.**
- Signal grows stronger at higher p_T in forward $\eta^{\pi^+\pi^-}$ region. Resonance peak around $M_{inv}^{\pi^+\pi^-} \sim 0.8$ GeV/c$^2 \sim M_\rho$.
- Backward $\eta^{\pi^+\pi^-}$ signal is small, mainly from low x quarks from polarized beam.



STAR Preliminary 2015

$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X$ at $\sqrt{s} = 200$ GeV

± 3% scale uncertainty from beam polarization (not shown)

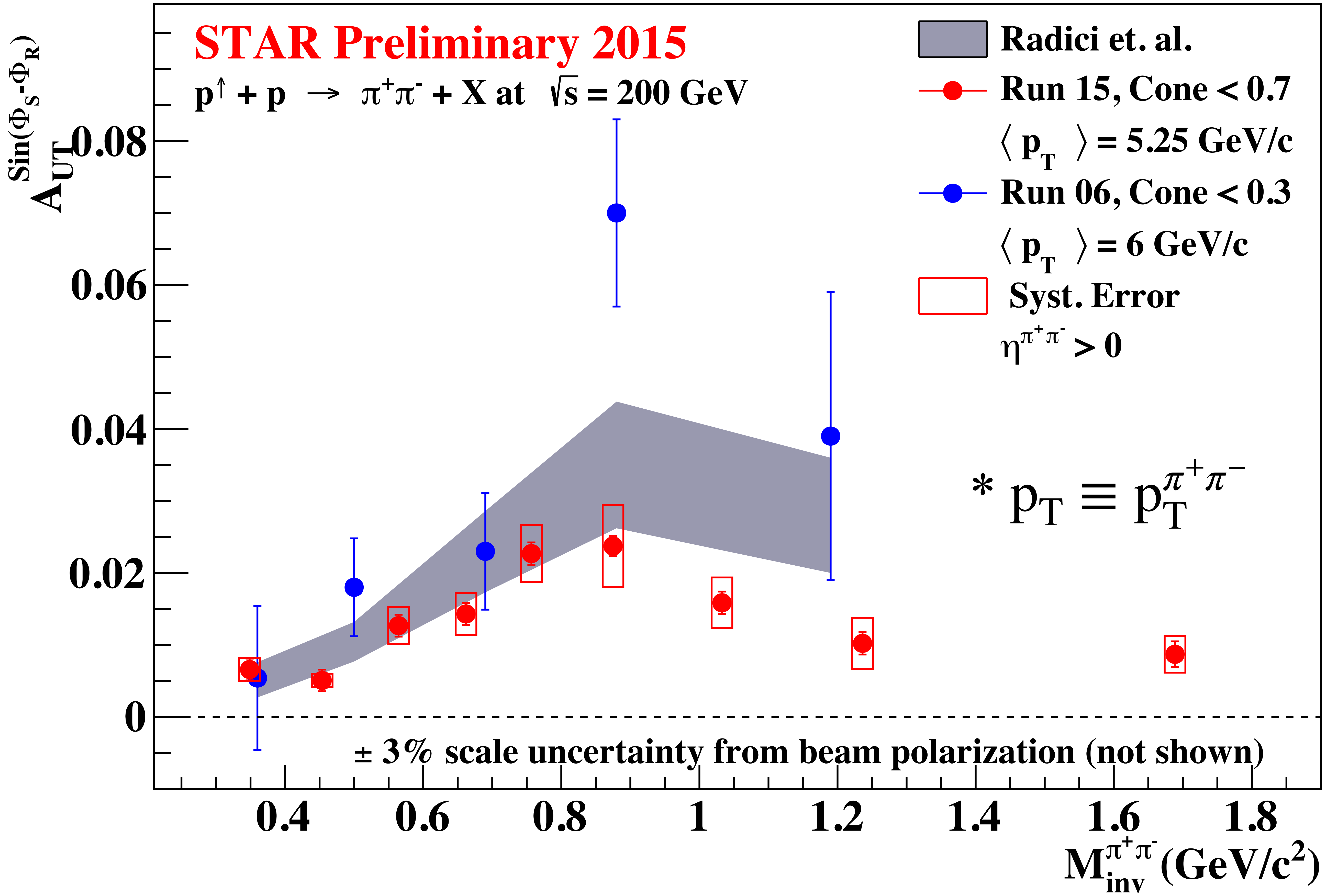
$* \; p_T \equiv p_T^{\pi^+\pi^-}$

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

STAR Preliminary: $A_{UT}^{\sin(\phi_{RS})}$ vs $M_{inv}^{\pi^+\pi^-}$ integrated $p_T^{\pi^+\pi^-}$

- Asymmetry is enhanced around $M_{inv}^{\pi^+\pi^-} \sim 0.8$, consistent with the previous measurement and theory prediction.
- Theory overshoots the new measurement starting around and beyond the resonance peak.
- Statistical precision is significantly improved in the new result.



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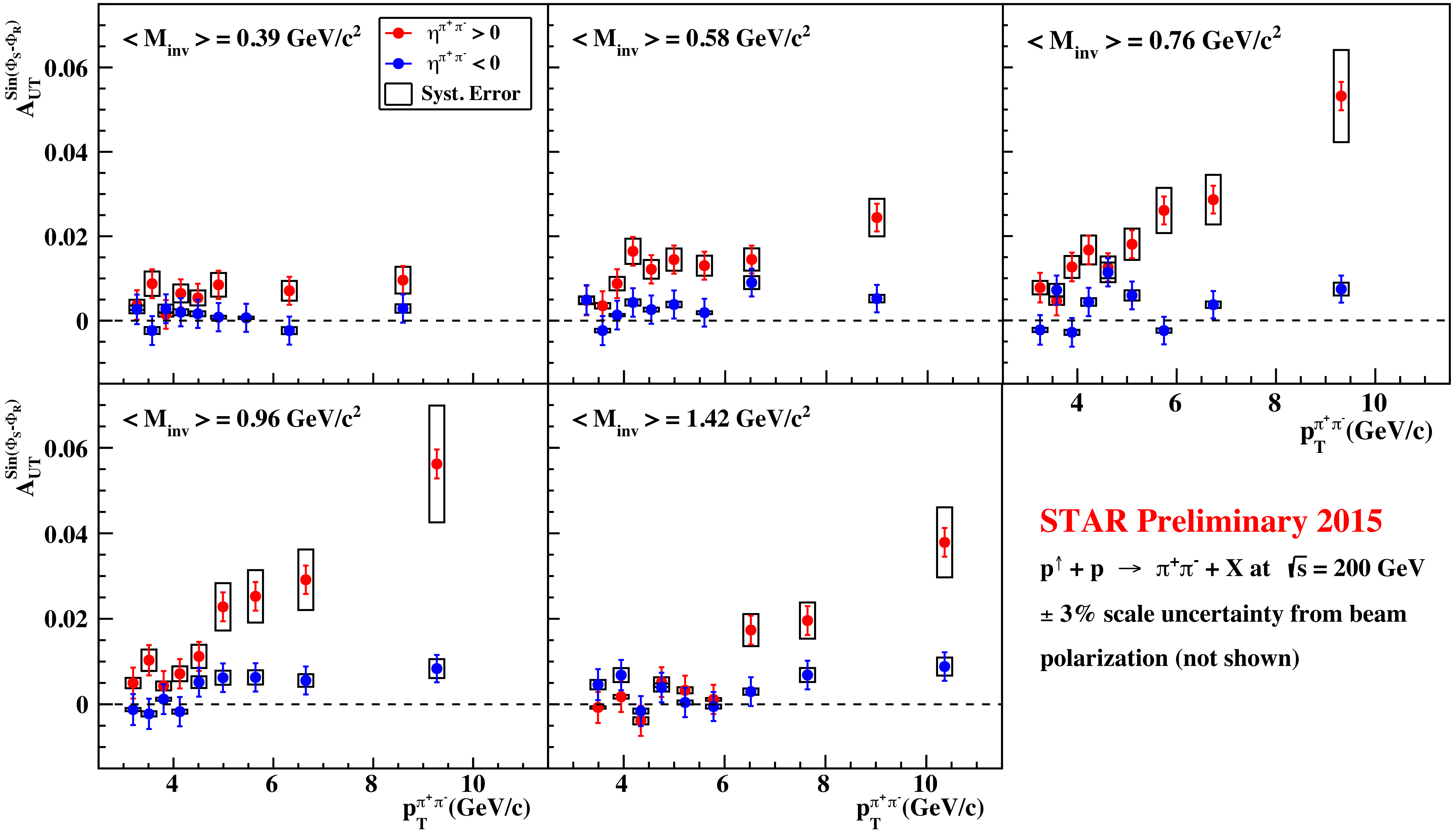
STAR Preliminary: $A_{UT}^{\sin(\phi_{RS})}$ vs $p_T^{\pi^+\pi^-}$

- Large asymmetry signal at higher p_T in forward $\eta^{\pi^+\pi^-}$ region. Stronger signal when $\langle M_{inv}\rangle \sim M_\rho$.
- Backward $\eta^{\pi^+\pi^-}$ signal is small, mainly from low x quarks from polarized beam.



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Particle Identification Including TOF (After Preliminary)

- The PID dominated large systematic uncertainty is concerning.
- Further investigated including TOF.

TPC based PID

- PID: $-1 < n\sigma_\pi < 2$
- $\pi^+\pi^-$ purity: ~ 70 - 80 %

Preliminary Results

TPC and/or TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and/or TOF PID
- $\pi^+\pi^-$ purity: ~ 85 - 90 %

TPC and TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and TOF PID
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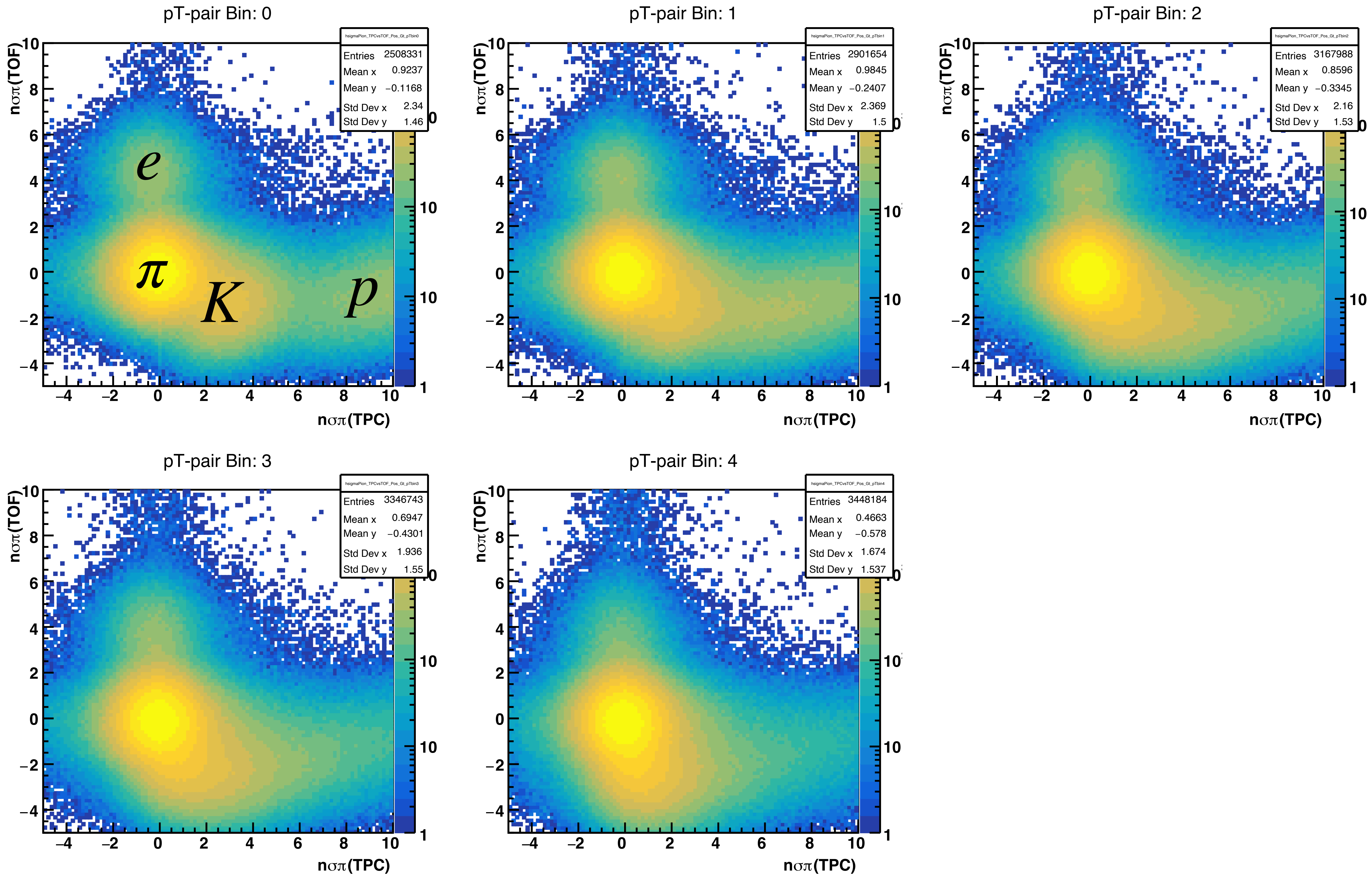

- $n\sigma_\pi^{TPC}$ vs. $n\sigma_\pi^{TOF}$
- Use of the TPC and TOF together improves PID.

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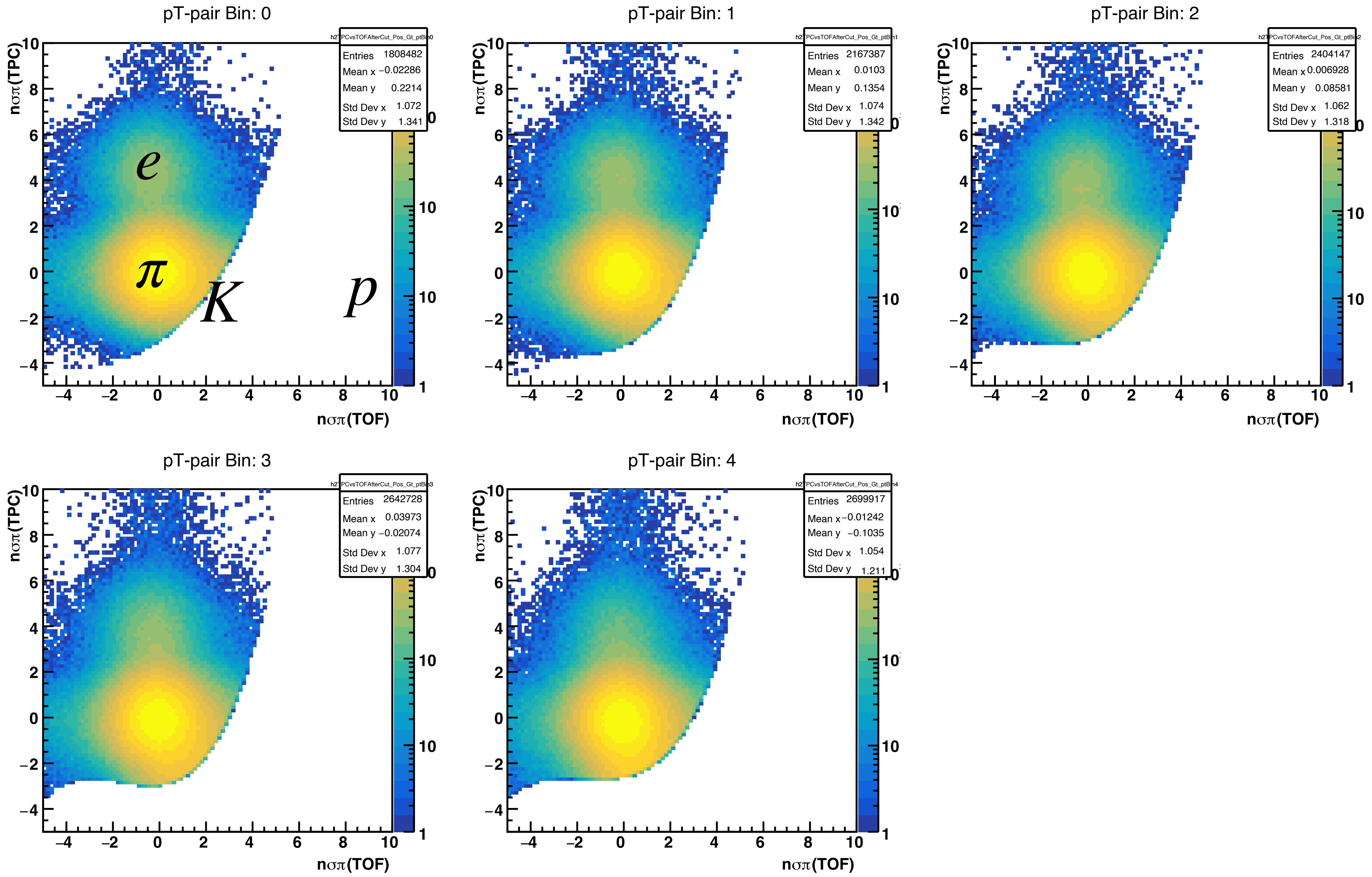

- $n\sigma_\pi^{TPC}$ vs. $n\sigma_\pi^{TOF}$
- Use of the TPC and TOF together improves PID.
- **Polynomial TOF cut** significantly removes p+K backgrounds.

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

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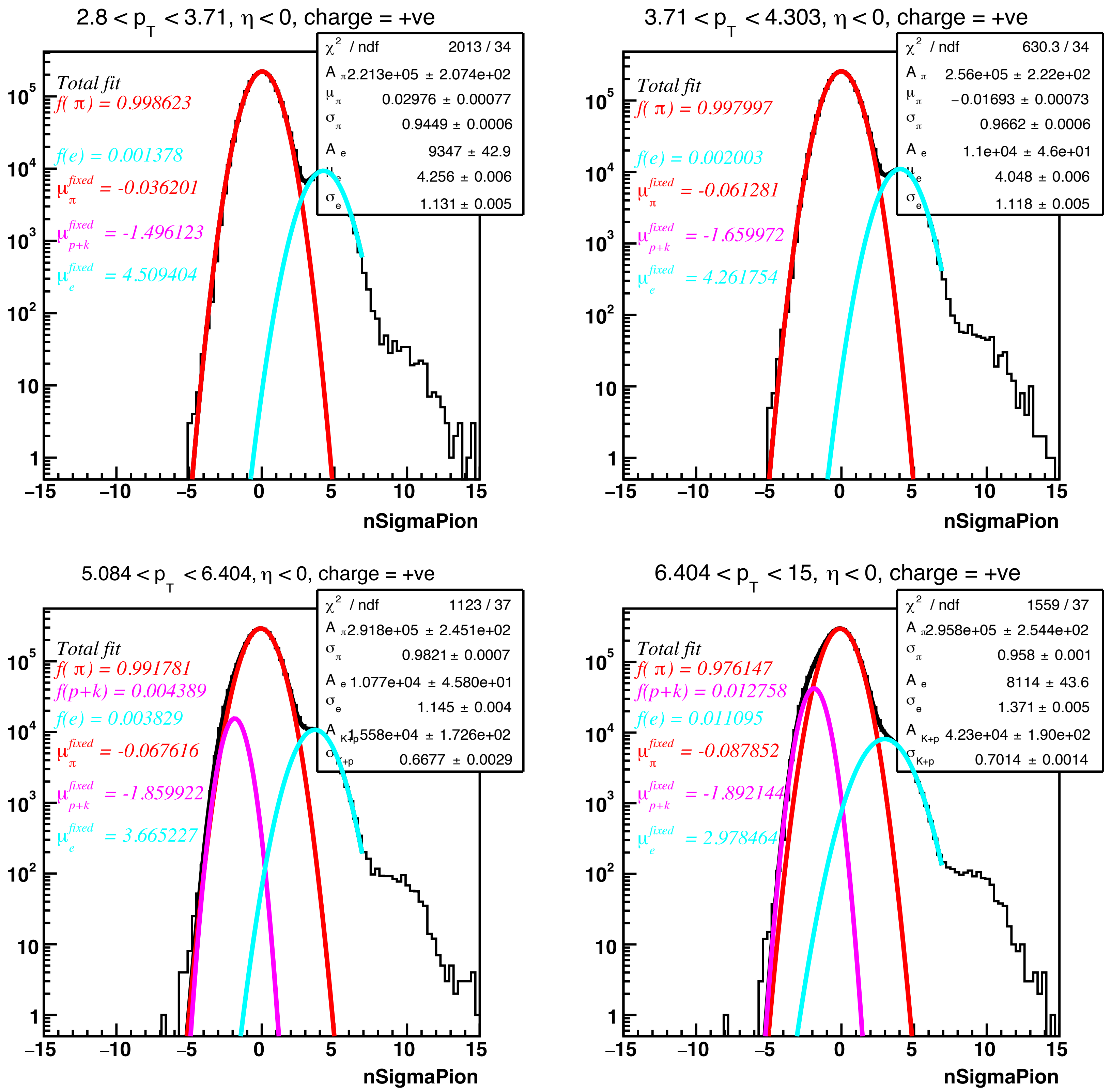
Preliminary Results

TPC and/or TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and/or TOF PID
- $\pi^+\pi^-$ purity: ~ 85 - 90 %

TPC and TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and TOF PID
- $\pi^+\pi^-$ purity: > 97 %



- $n\sigma_\pi^{TPC}$ vs . $n\sigma_\pi^{TOF}$
- Use of the TPC and TOF together improves PID.
- **Polynomial TOF cut** significantly removes p+K backgrounds.

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^\uparrow + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Particle Identification Including TOF (After Preliminary)

- The PID dominated large systematic uncertainty is concerning.
- Further investigated including TOF.

TPC based PID

- PID: $-1 < n\sigma_\pi < 2$
- $\pi^+\pi^-$ purity: ~ 70 - 80 %

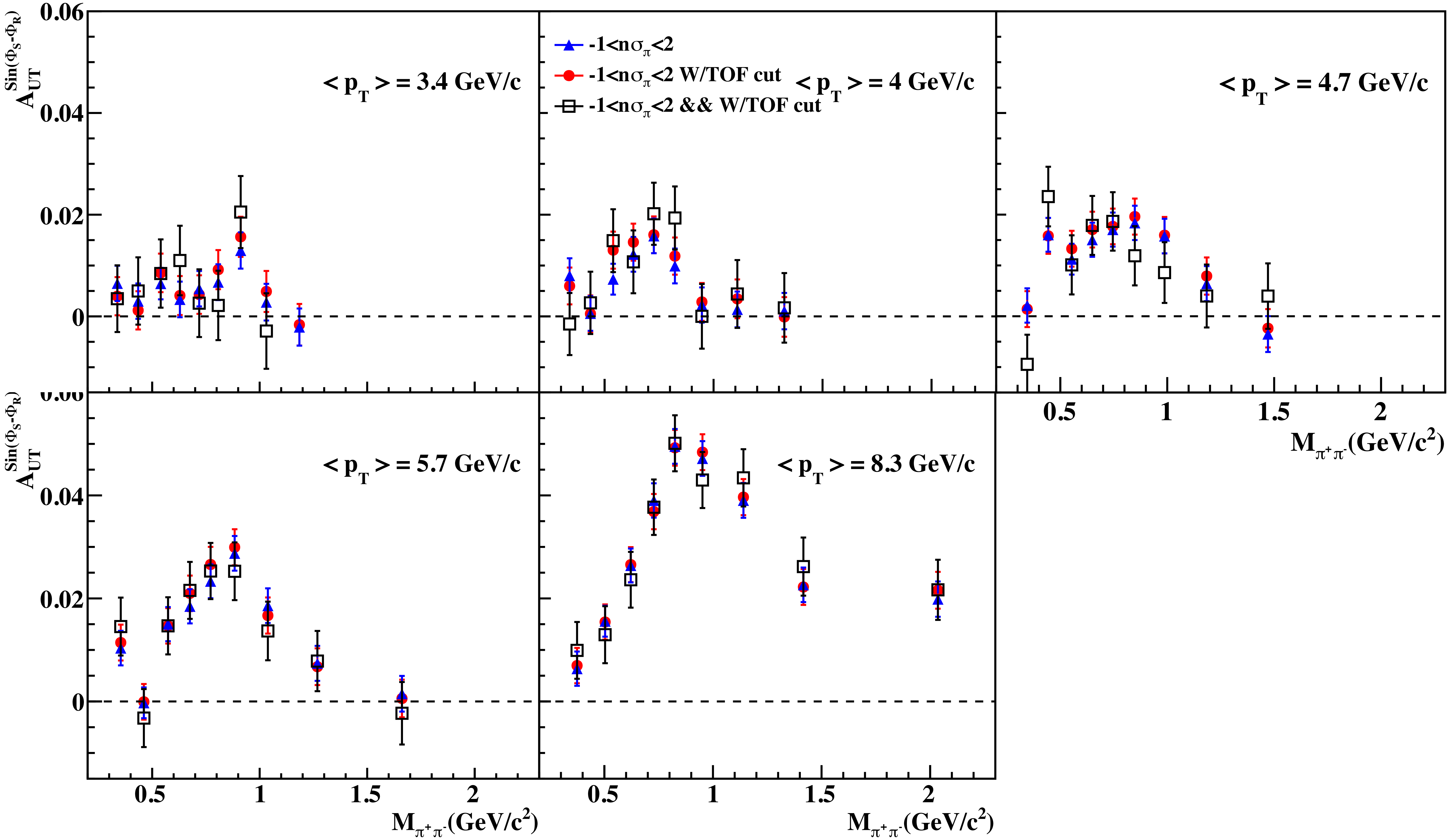
Preliminary Results

TPC and/or TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and/or TOF PID
- $\pi^+\pi^-$ purity: ~ 85 - 90 %

TPC and TOF based PID

- PID: $-1 < n\sigma_\pi < 2$ and TOF PID
- $\pi^+\pi^-$ purity: > 97 %



- **Consistent A_{UT} irrespective of the background sizes suggests negligible background effects in the signal A_{UT}.**

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Particle Identification Including TOF (After Preliminary)

- Background A_{UT}

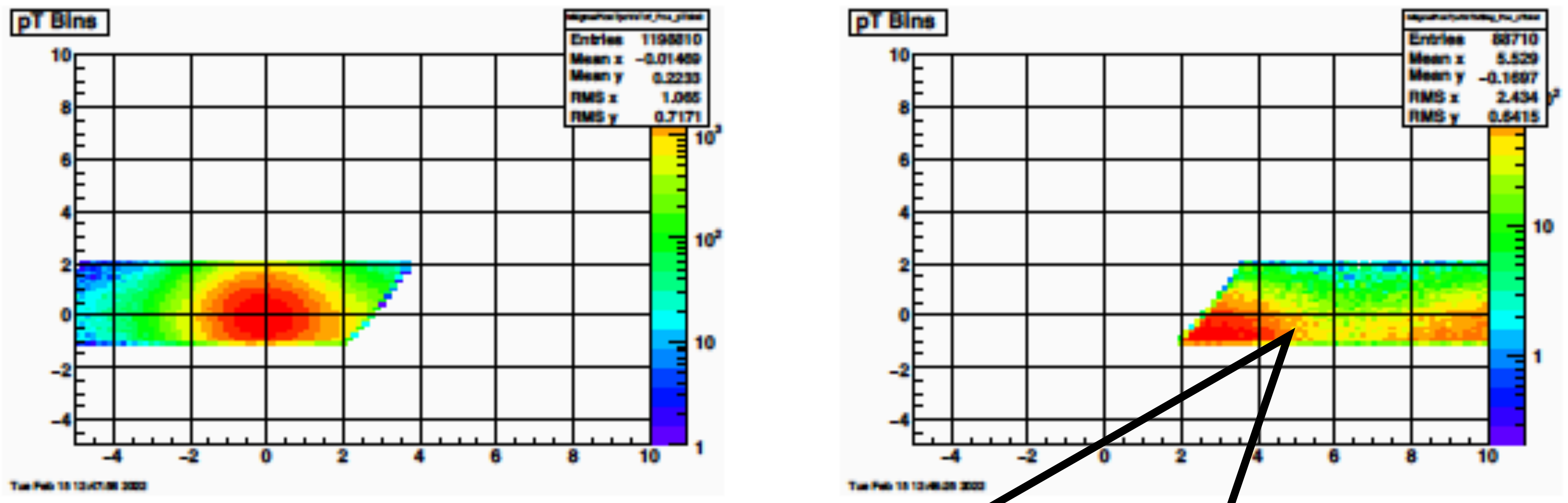


Figure A.11: Left: pion signal Right: Pion background.

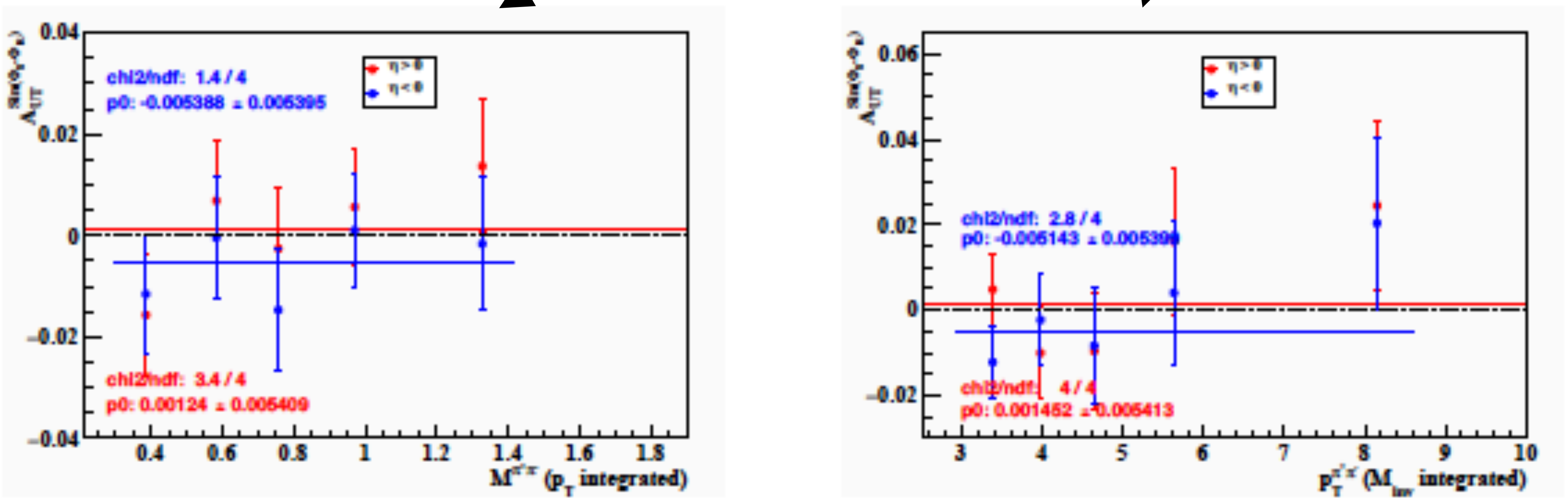


Figure A.12: Left: A_{UT}^{bkg} vs M_{inv}, p_T integrated. Right: A_{UT}^{bkg} vs p_T, M_{inv} integrated.

- Measured background A_{UT} are consistent to zero within statistical errors.

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

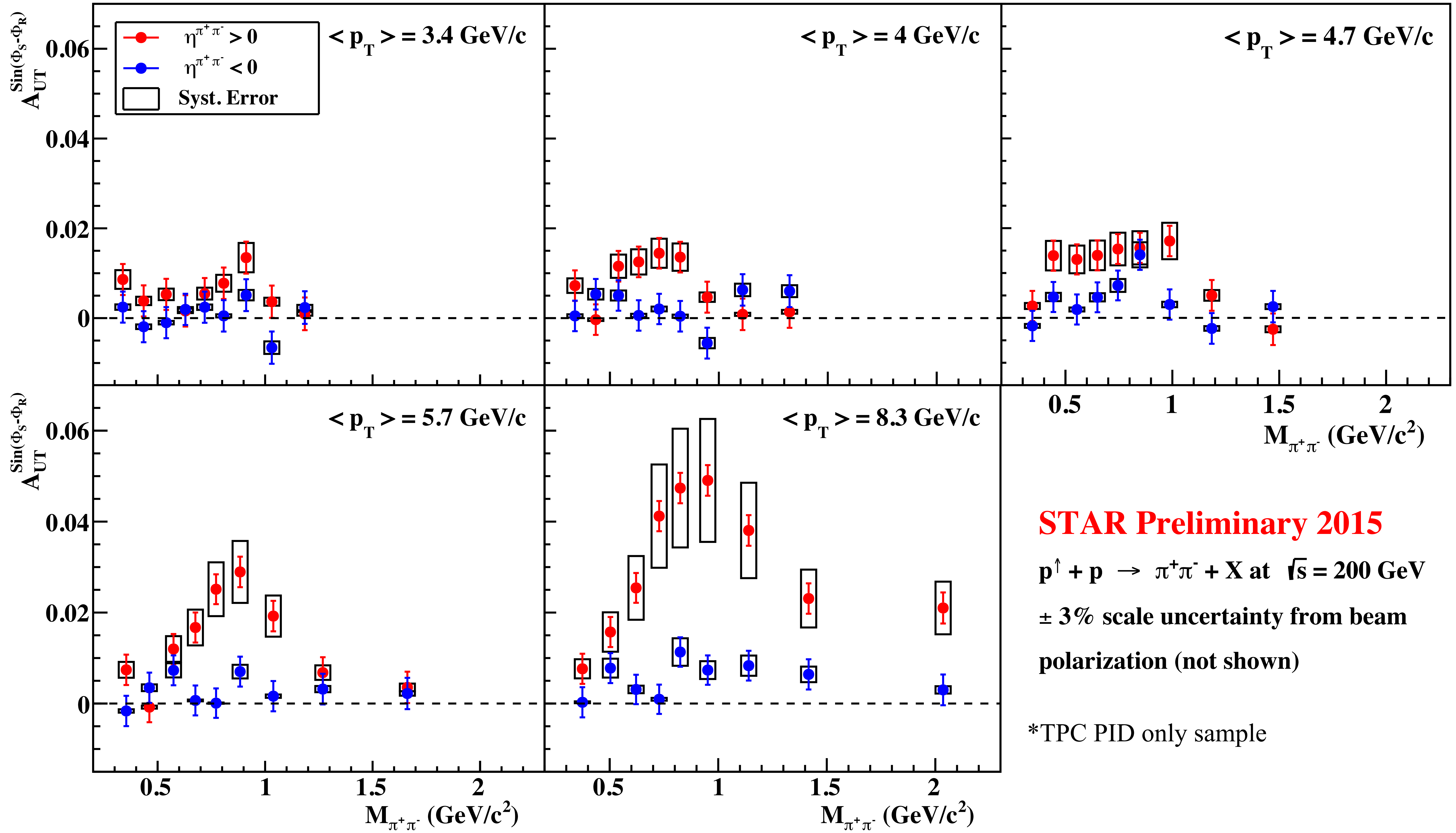
Particle Identification Including TOF (After Preliminary)

- PID systematic uncertainty estimated as a difference between asymmetries of different pair purities:

$$\delta_{\text{pid,sys}} = |A_{UT}^{\text{TPC\&/orTOF}} - A_{UT}^{\text{TPC\&\&TOF}}|$$

- Significant improvement in the systematic uncertainty, comparable to the statistical uncertainty, is expected.

Preliminary Level Systematic Error



STAR Preliminary 2015

$p^{\uparrow} + p \rightarrow \pi^+\pi^- + X$ at $\sqrt{s} = 200$ GeV

± 3% scale uncertainty from beam polarization (not shown)

*TPC PID only sample

STAR Run 2015 $\pi^+\pi^-$ Asymmetry Analysis

$$p^\uparrow + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

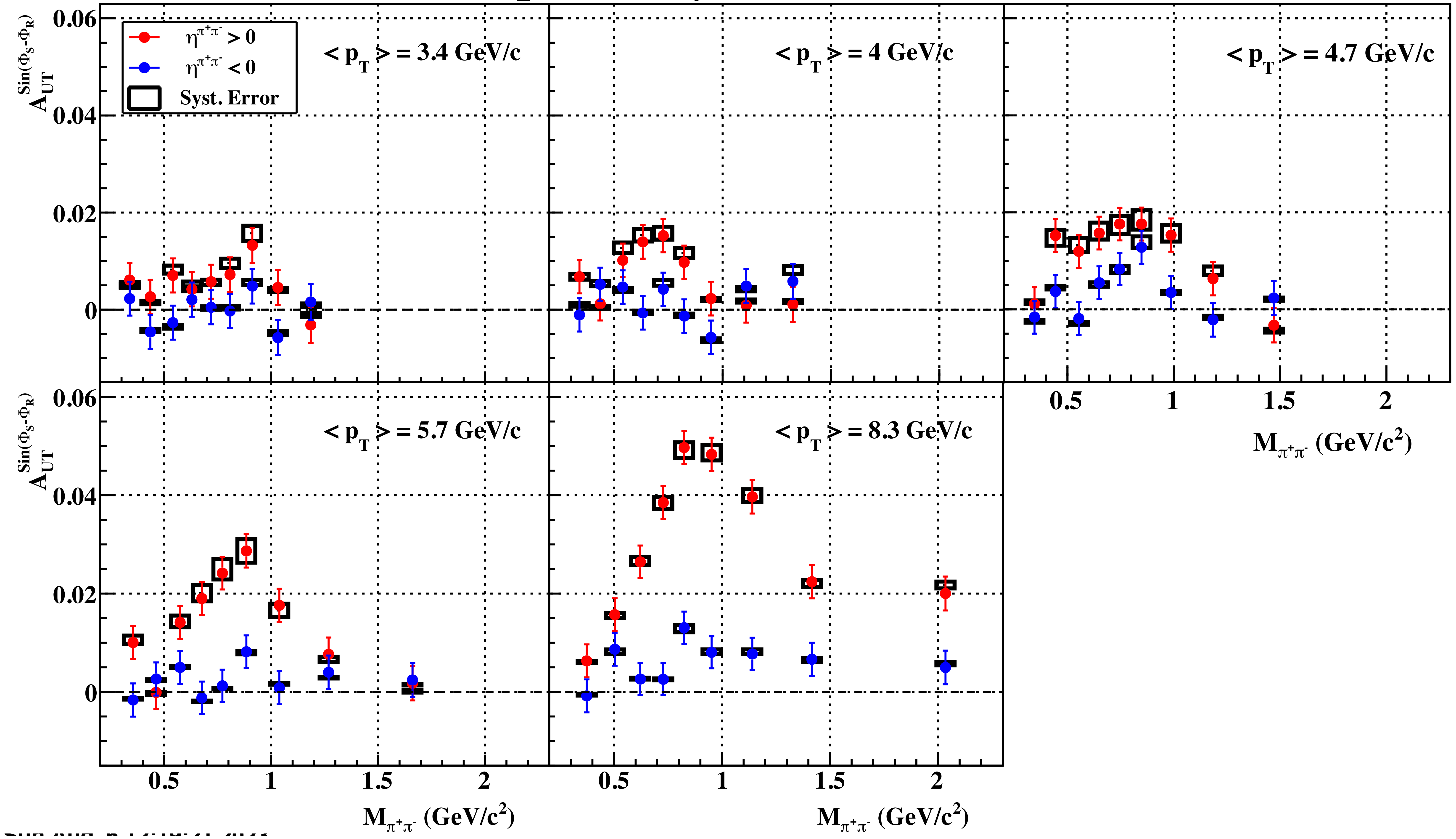
Particle Identification Including TOF (After Preliminary)

- PID systematic uncertainty estimated as a difference between asymmetries of different pair purities:

$$\delta_{\text{pid,sys}} = |A_{UT}^{\text{TPC\&/orTOF}} - A_{UT}^{\text{TPC\&\&TOF}}|$$

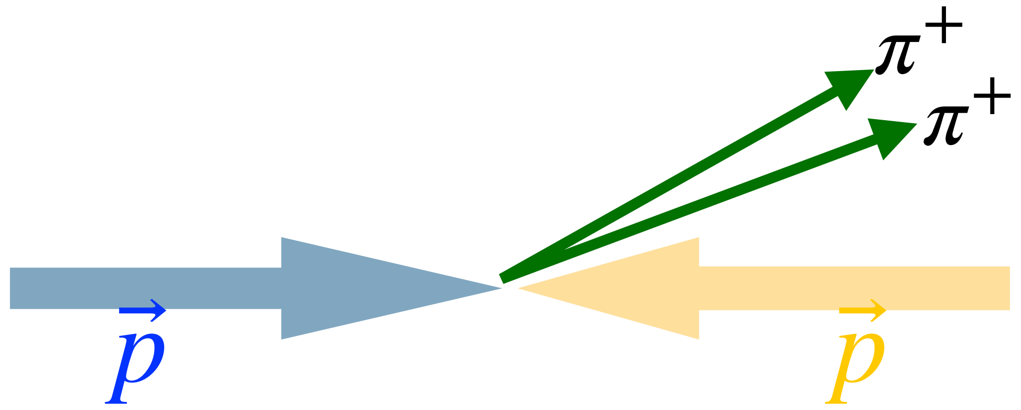
- Significant improvement in the systematic uncertainty, comparable to the statistical uncertainty, is expected.

Expected Systematic Error



STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

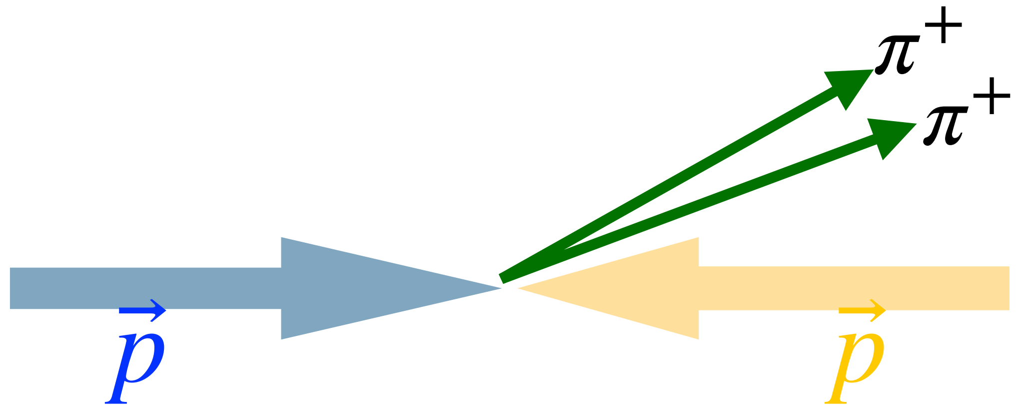
$$p + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$



Polarization integrated ≈ Unpolarized

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$$p + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

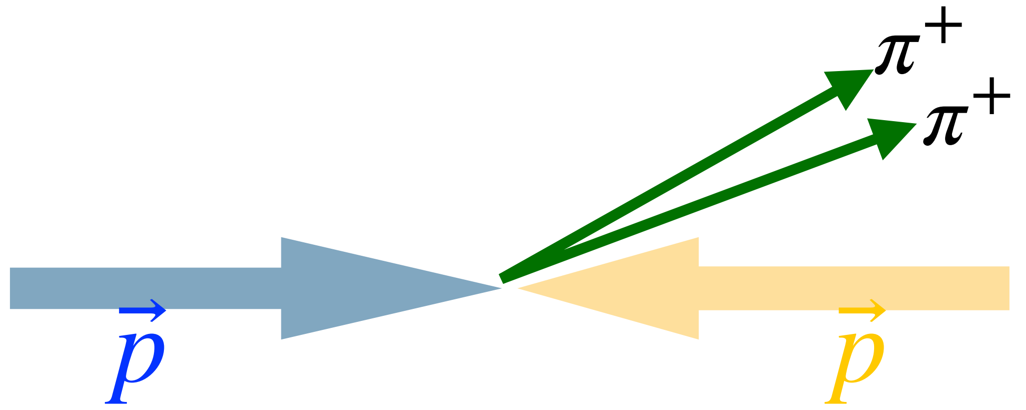


Polarization integrated ≈ Unpolarized

- Inclusive $\pi^+\pi^-$ differential cross section:
  - As a function of invariant mass, $M_{inv}^{\pi^+\pi^-}$, in $|\eta|<1$.
  - Much needed for the $D_1^{h_1h_2}$ extraction.
  - Access to $D_1^{h_1h_2/g}$.

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$$p + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$



Polarization integrated ≈ Unpolarized

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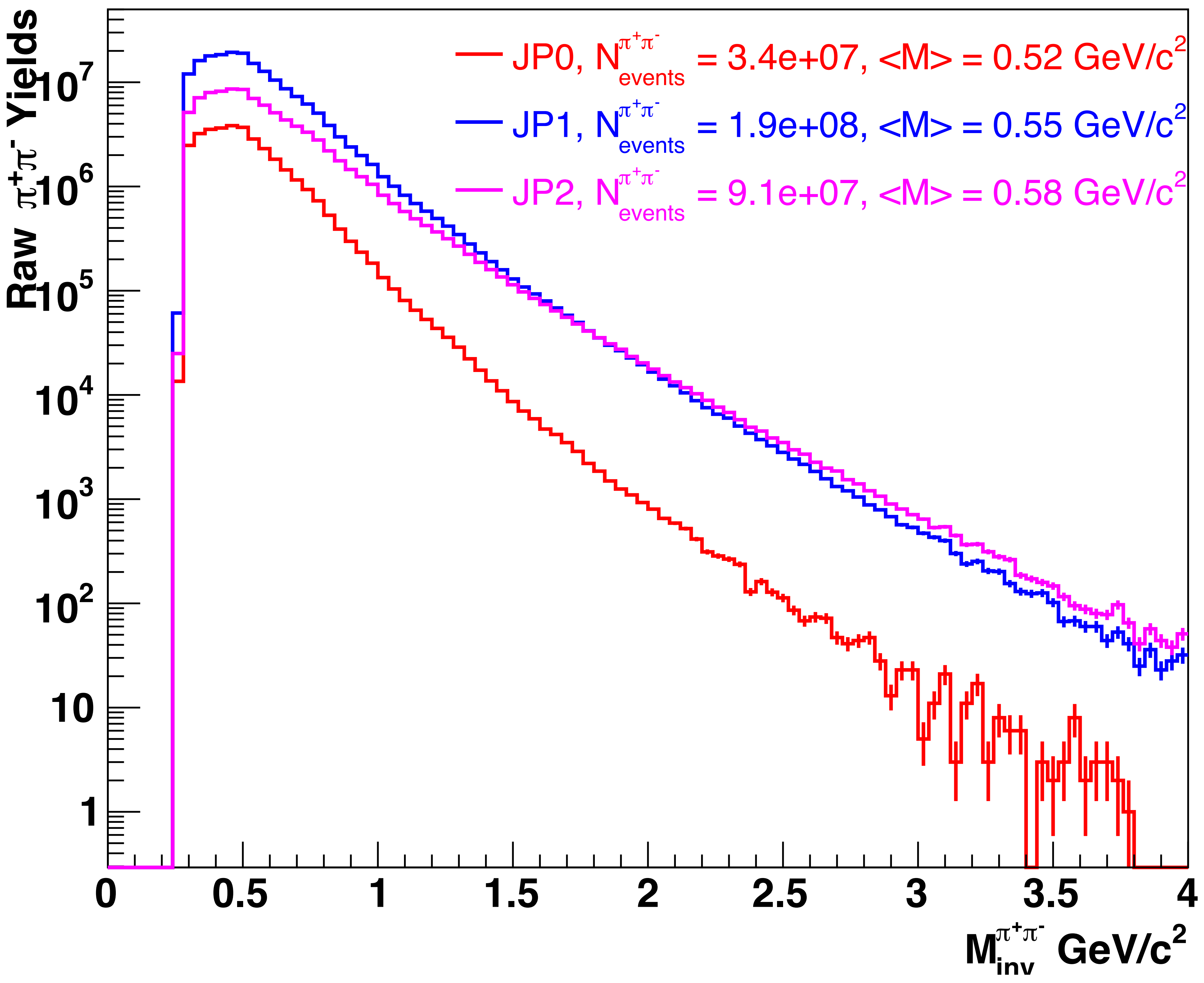
  - As a function of invariant mass, $M_{inv}^{\pi^+\pi^-}$, in $|\eta|<1$.
  - Much needed for the $D_1^{h_1h_2}$ extraction.
  - Access to $D_1^{h_1h_2/g}$.
- STAR Run 2012 dataset @ $\sqrt{s}$ = 200 GeV
- Triggers: JP0, JP1, JP2
- Lower trigger threshold provides better gluon sensitivity than Run 2015.
- $\pi^+\pi^-$ construction is same as in the IFF analysis, except for the track $p_T > 0.5$ GeV/c.



Simulation and Embedding Sample

Simulation and Embedding Sample

- Simulation and Embedding sample needed for:
  - Data unfolding
  - PID corrections
  - Efficiency analysis
  - Systematic studies

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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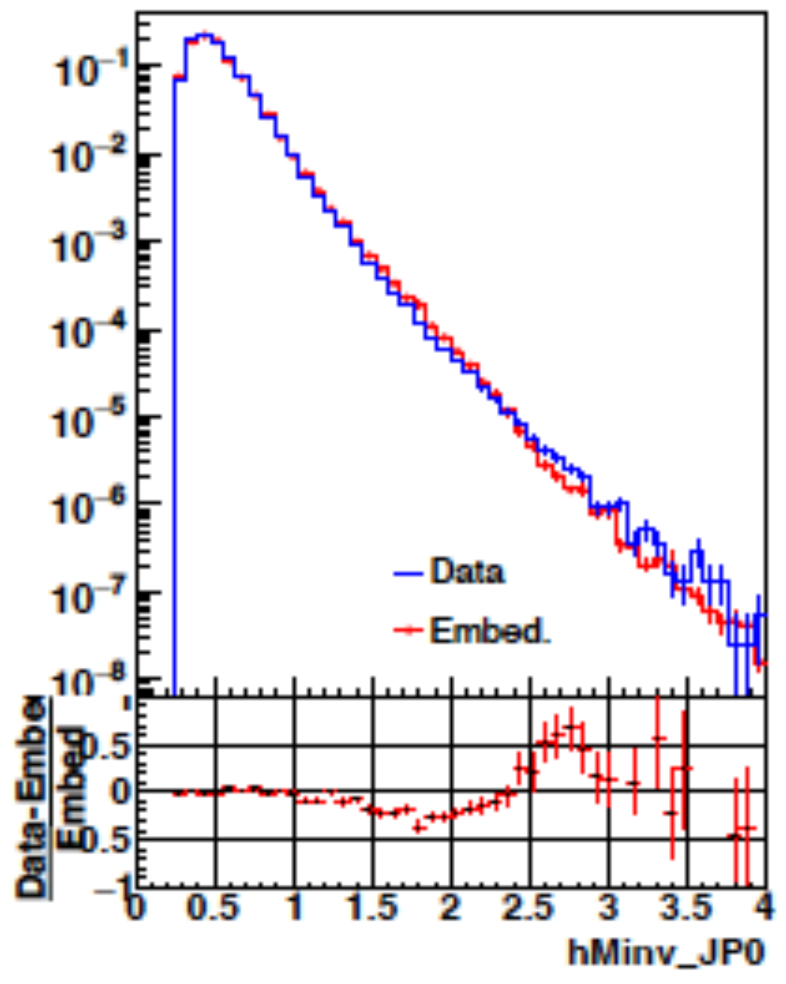
Simulation and Embedding Sample

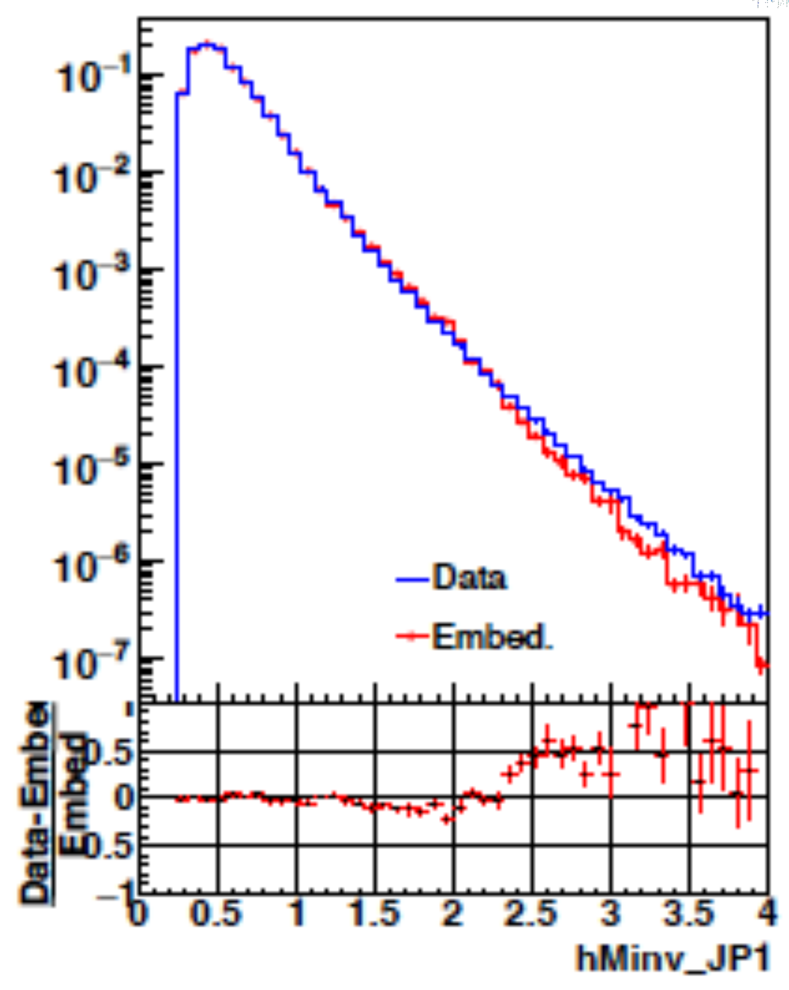
- Simulation and Embedding sample needed for:
  - Data unfolding
  - PID corrections
  - Efficiency analysis
  - Systematic studies

- **PYTHIA** simulated events, reconstructed through the **GEANT 3** package embedded with the real collision events to effectively reconstruct STAR detector responses (Embedding).

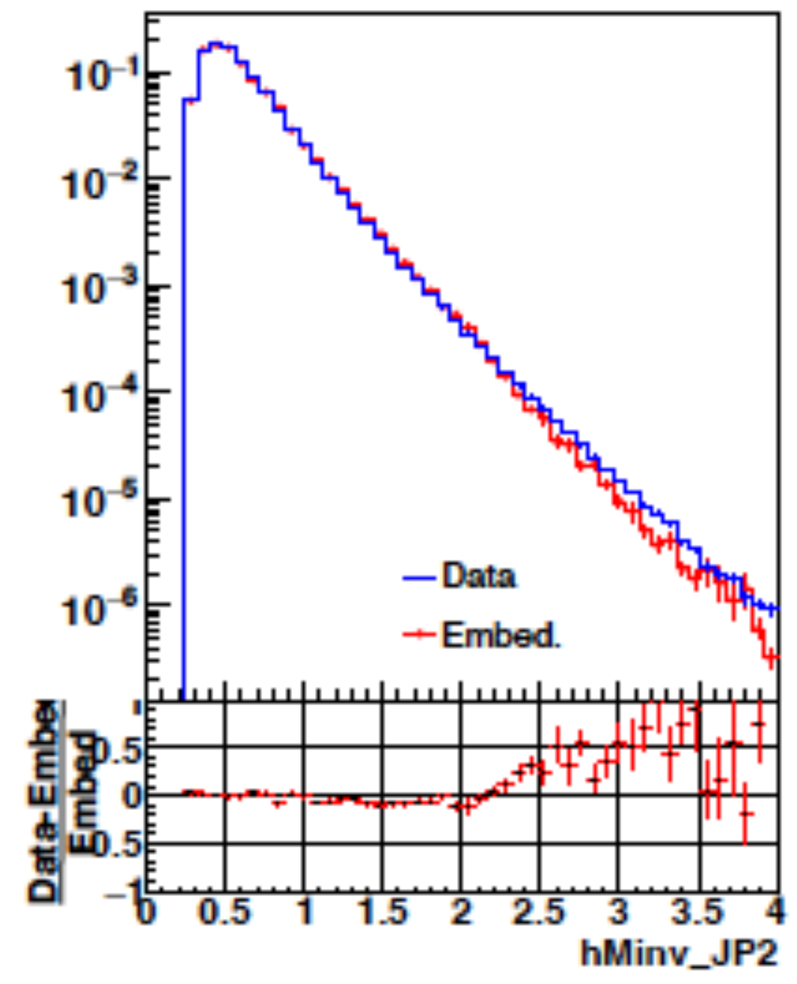
- The embedding sample must have good description of the data to directly apply embedding driven measurement to the data.



(a) $M_{inv}^{\pi^+\pi^-}$ comparison for JP0 trigger.



(b) $M_{inv}^{\pi^+\pi^-}$ comparison for JP1 trigger.



(c) $M_{inv}^{\pi^+\pi^-}$ comparison for JP2 trigger

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$$p + p \rightarrow \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$$p + p \to \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

- Unfolding accounts for the bin migration effect and backgrounds.

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$p + p \rightarrow \pi^+\pi^- + X$ at $\sqrt{s} = 200$ GeV

- Unfolding accounts for the bin migration effect and backgrounds.

Unfolding Using TUnfoldingDensity

- Unfolding solves the inverse problem for ***x***:

$$y_i = \sum_{j=1}^{m} A_{ij} x_i + b_i \ , \ 1 \leq i \leq n, \ n \geq m$$

y = detector level, ***x*** = particle level

A = Migration matrix , ***b*** = background

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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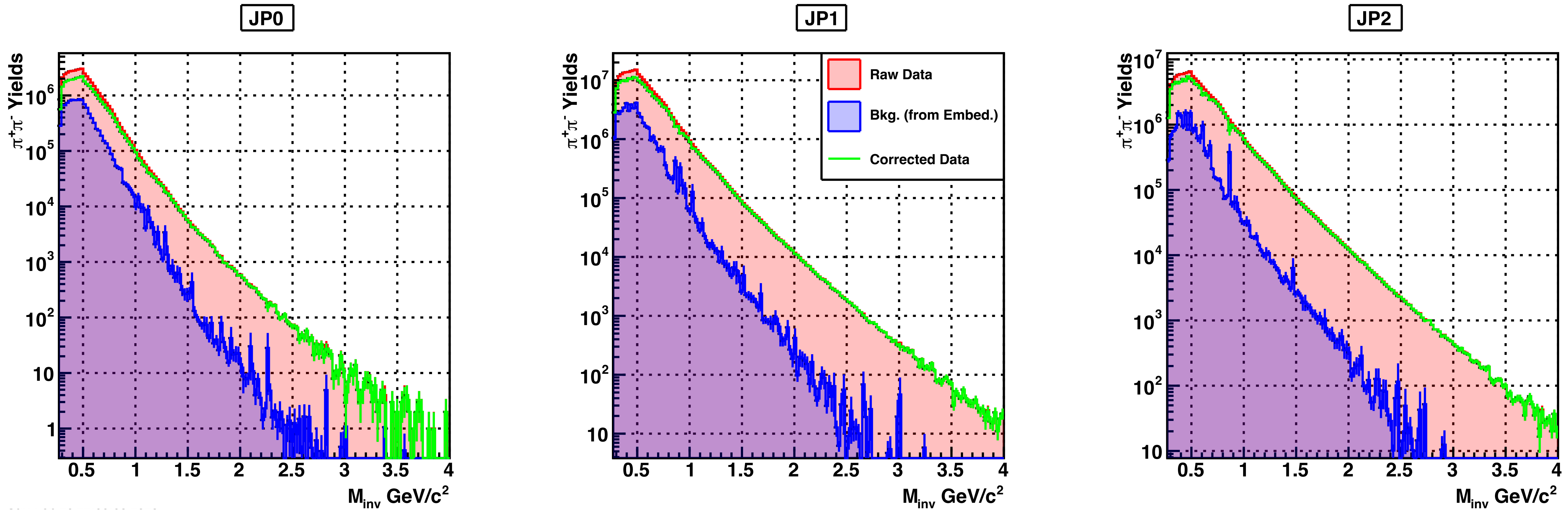
$\boldsymbol{y}$ = detector level, $\boldsymbol{x}$ = particle level

$\boldsymbol{A}$ = Migration matrix , $\boldsymbol{b}$ = background

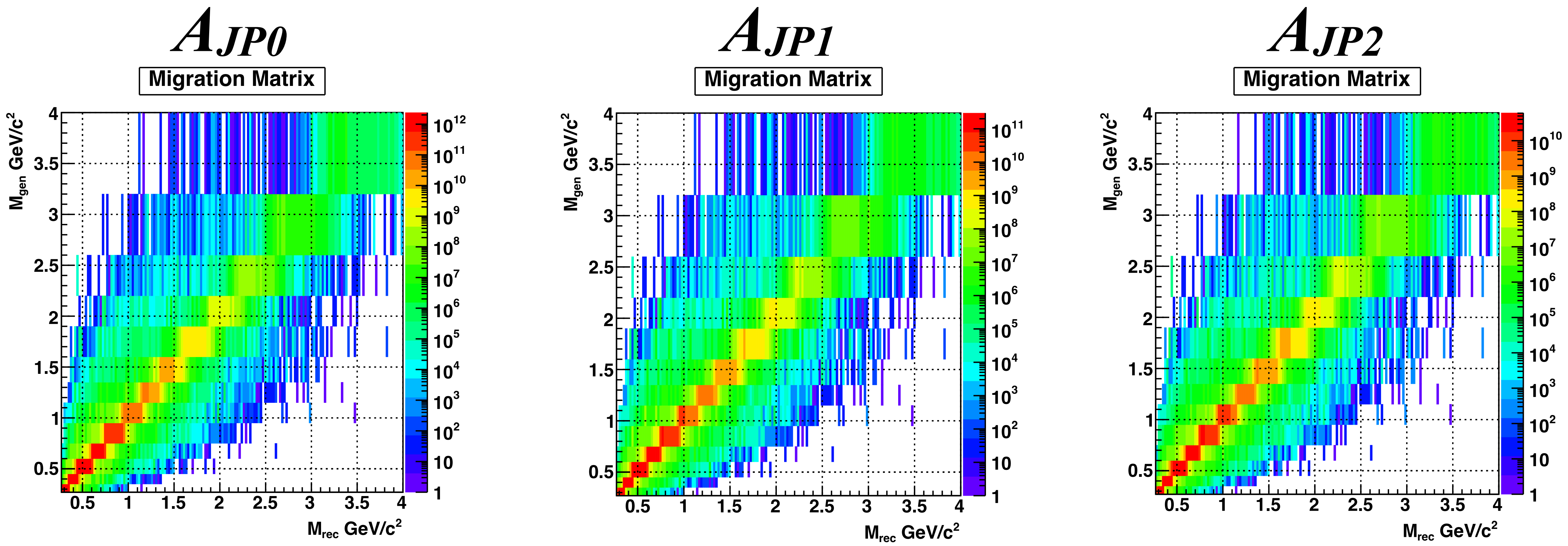
- $A_{n\times m} = A_{160\times 13}$, y ≡ uniform bin widths

  $x \equiv$ variable bin widths (Cross-Section bins)

- Unfolding is performed for each trigger, allowing independent measurement of triggered cross-section.



$b \equiv$ Backgrounds $\quad y \equiv$ Raw yields $\quad y - b \equiv$ Raw yields – Backgrounds



STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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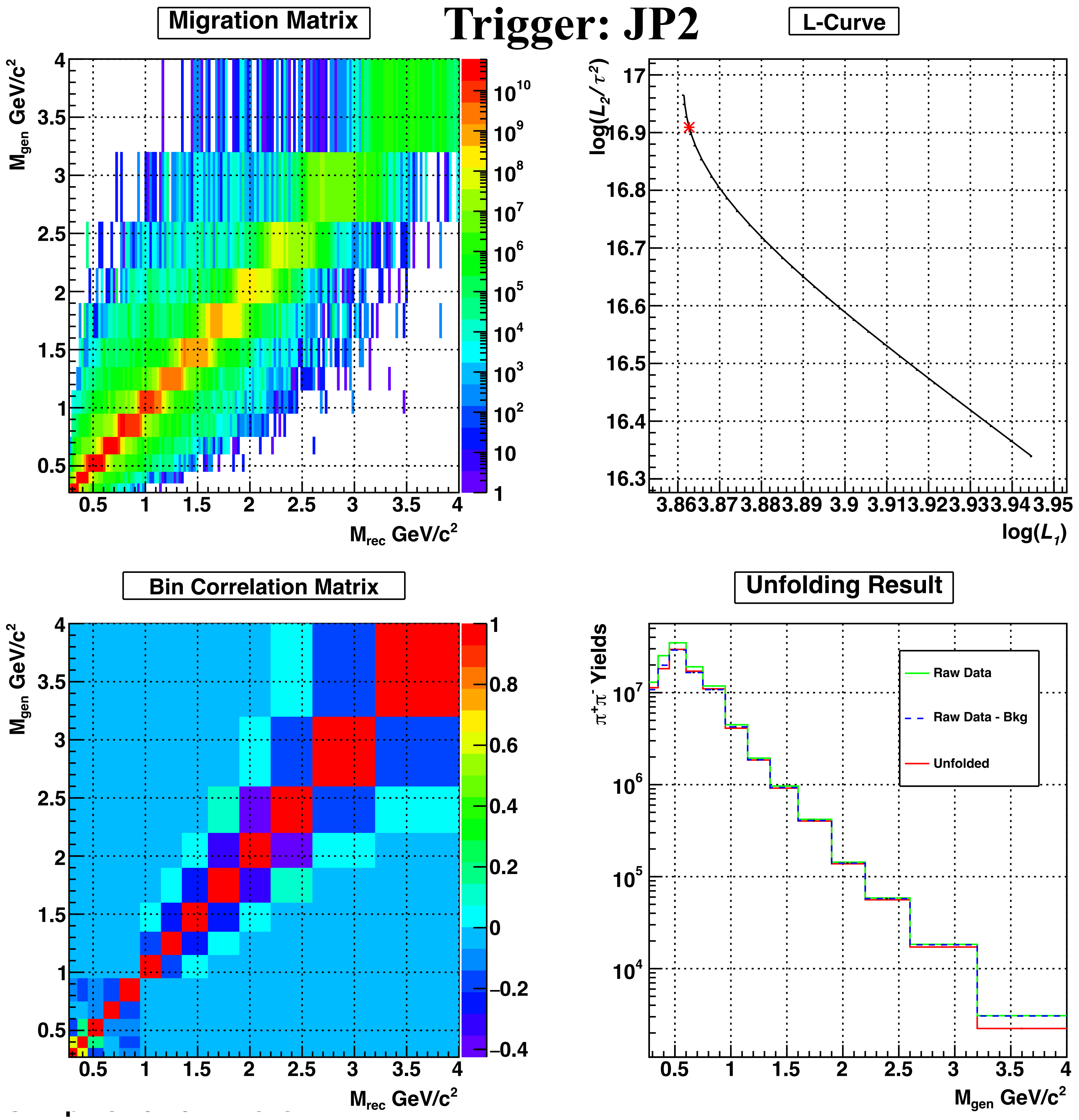

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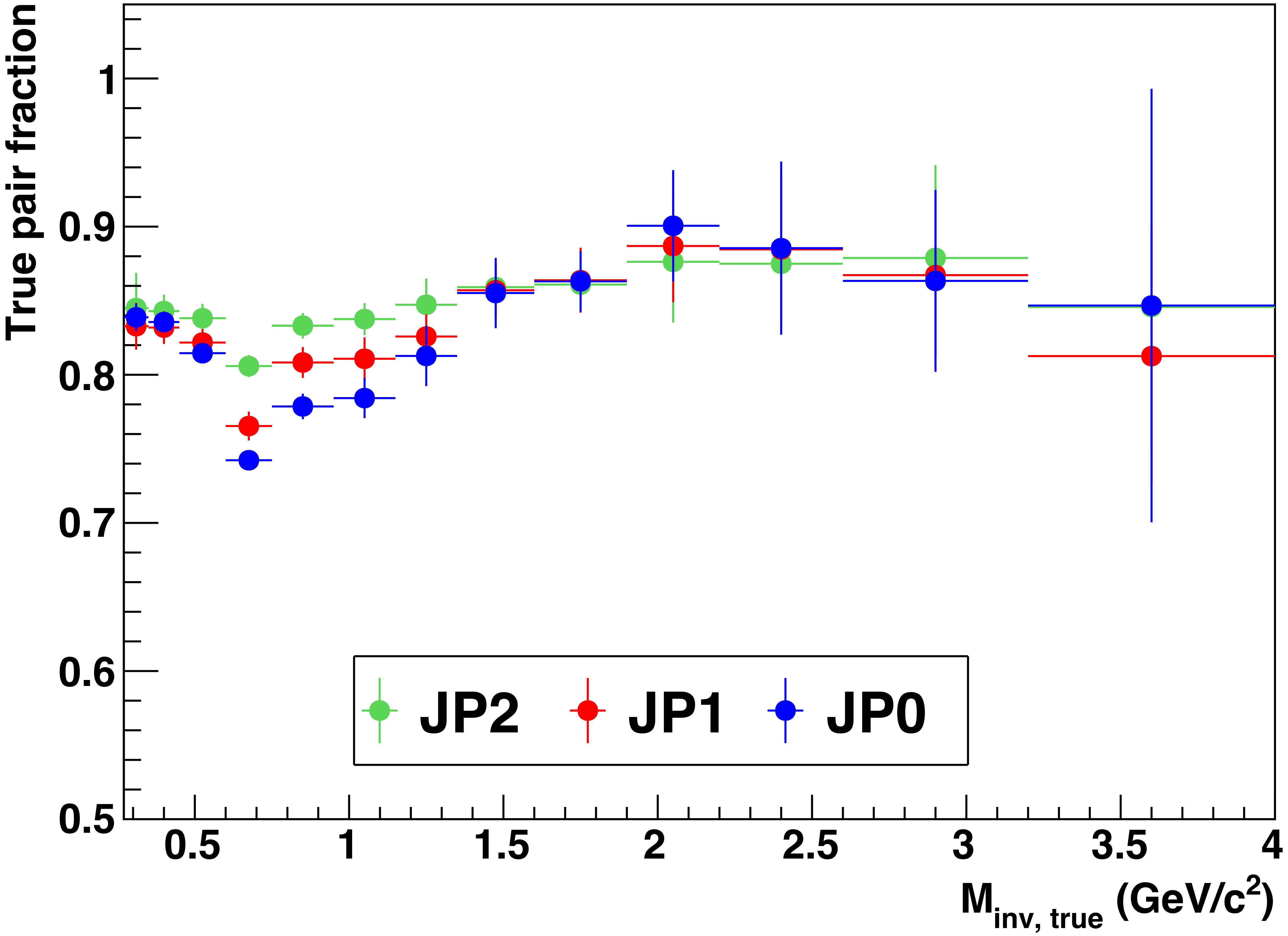
$$p + p \to \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Corrections (Bin by bin)

1. $\pi^+\pi^-$ Purity Fraction (f_{fake})
   - Embedding-driven PID
   - Pion selection: $-1 < n\sigma_\pi < 2$
   - $f_{fake} = \dfrac{N_{true \&\& -1<n\sigma_\pi<2}}{N_{-1<n\sigma_\pi<2}}$



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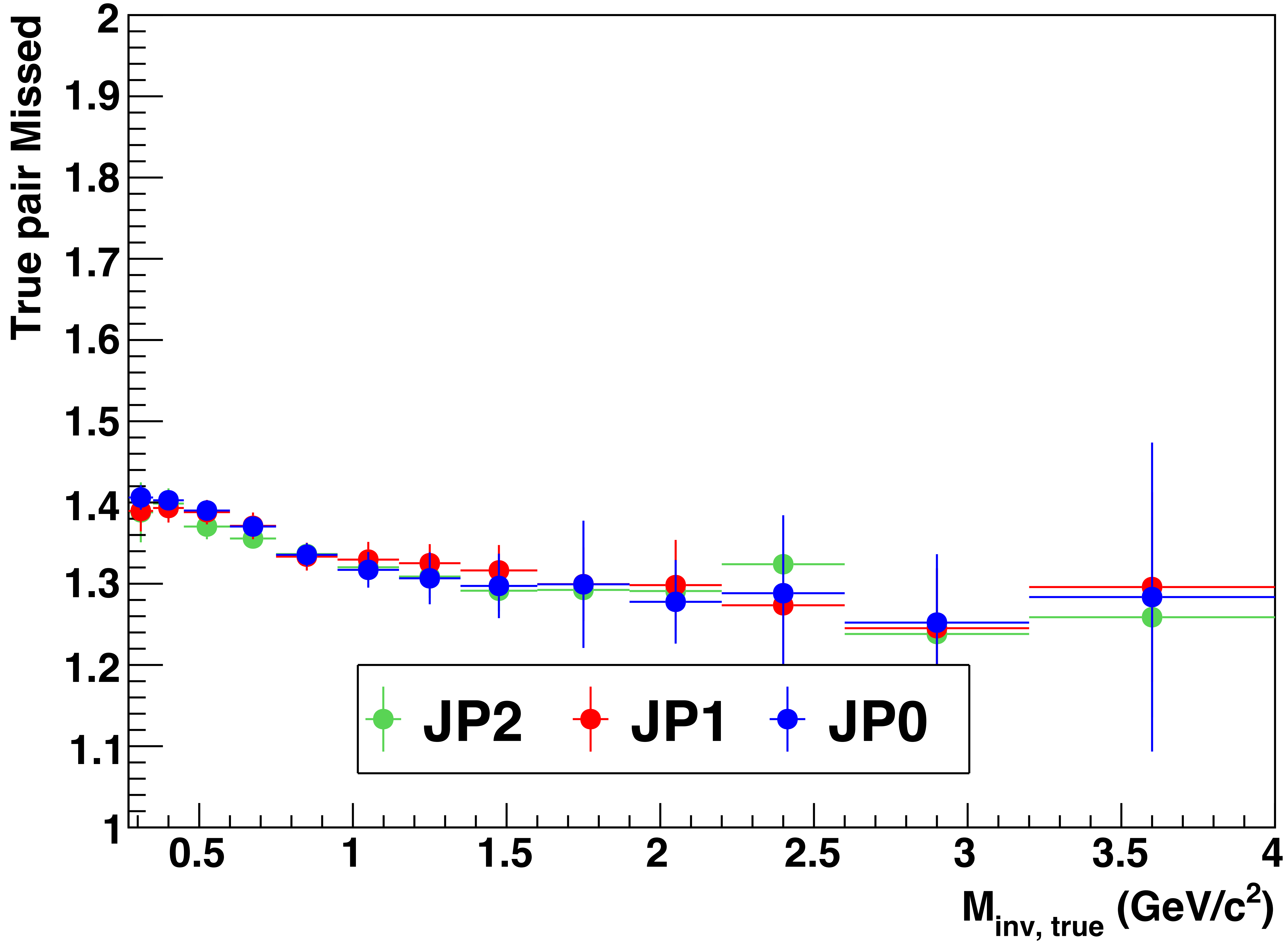
$$p + p \to \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Corrections (Bin by bin)

1. $\pi^+\pi^-$ Purity Fraction (f_{fake})
2. $\pi^+\pi^-$ Loss Fraction (f_{loss})

- $f_{loss} = \dfrac{N_{true}}{N_{-1<n\sigma_\pi<2\&\&true}}$



STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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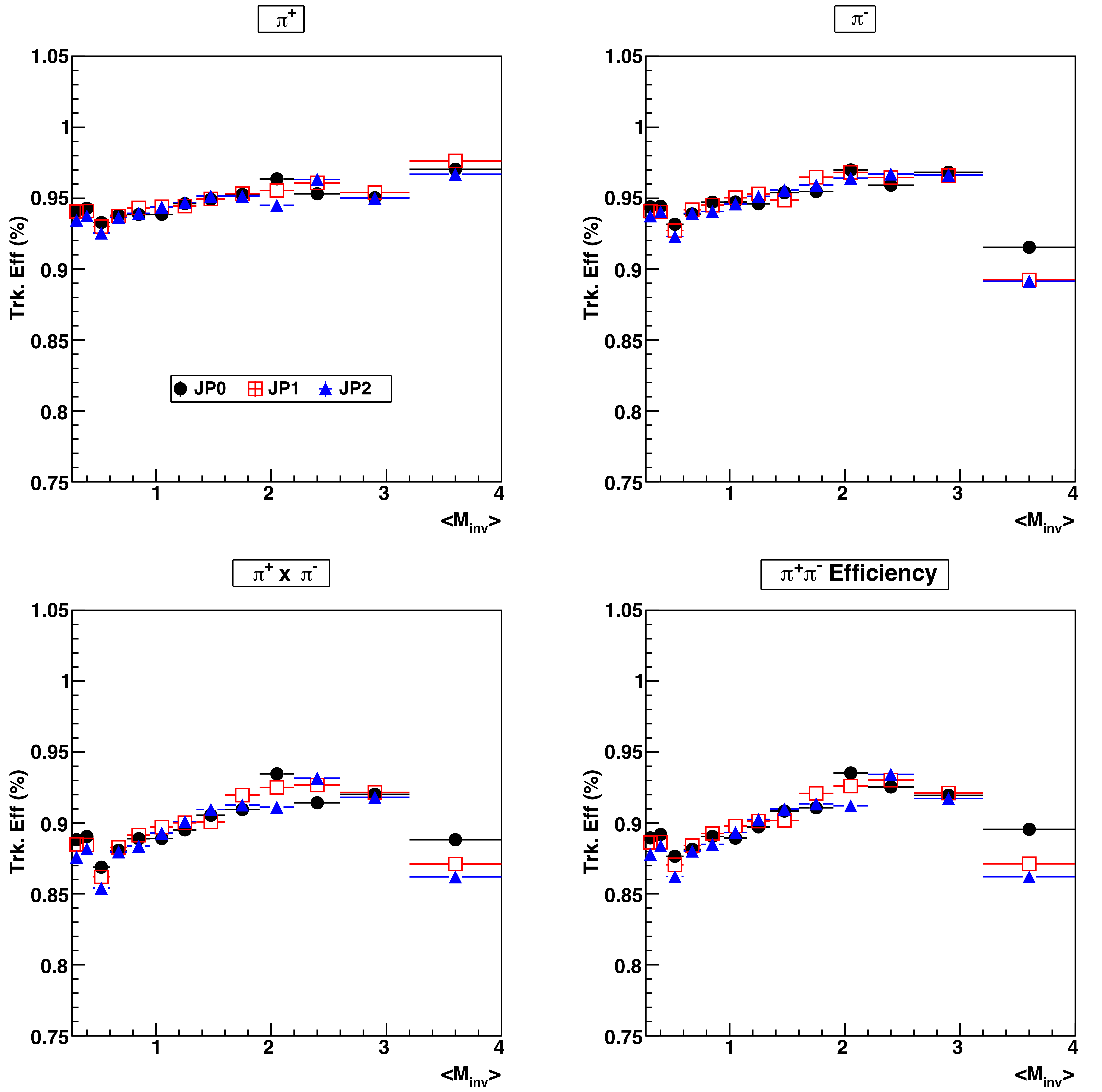
Corrections (Bin by bin)

1. $\pi^+\pi^-$ Purity Fraction (f_{fake})
2. $\pi^+\pi^-$ Loss Fraction (f_{loss})
3. Tracking Efficiency (ϵ^{π}_{trk})

$$\epsilon^{\pi}_{trk} = \frac{N^{\pi}_{gen,rec}}{N^{\pi}_{gen}}$$



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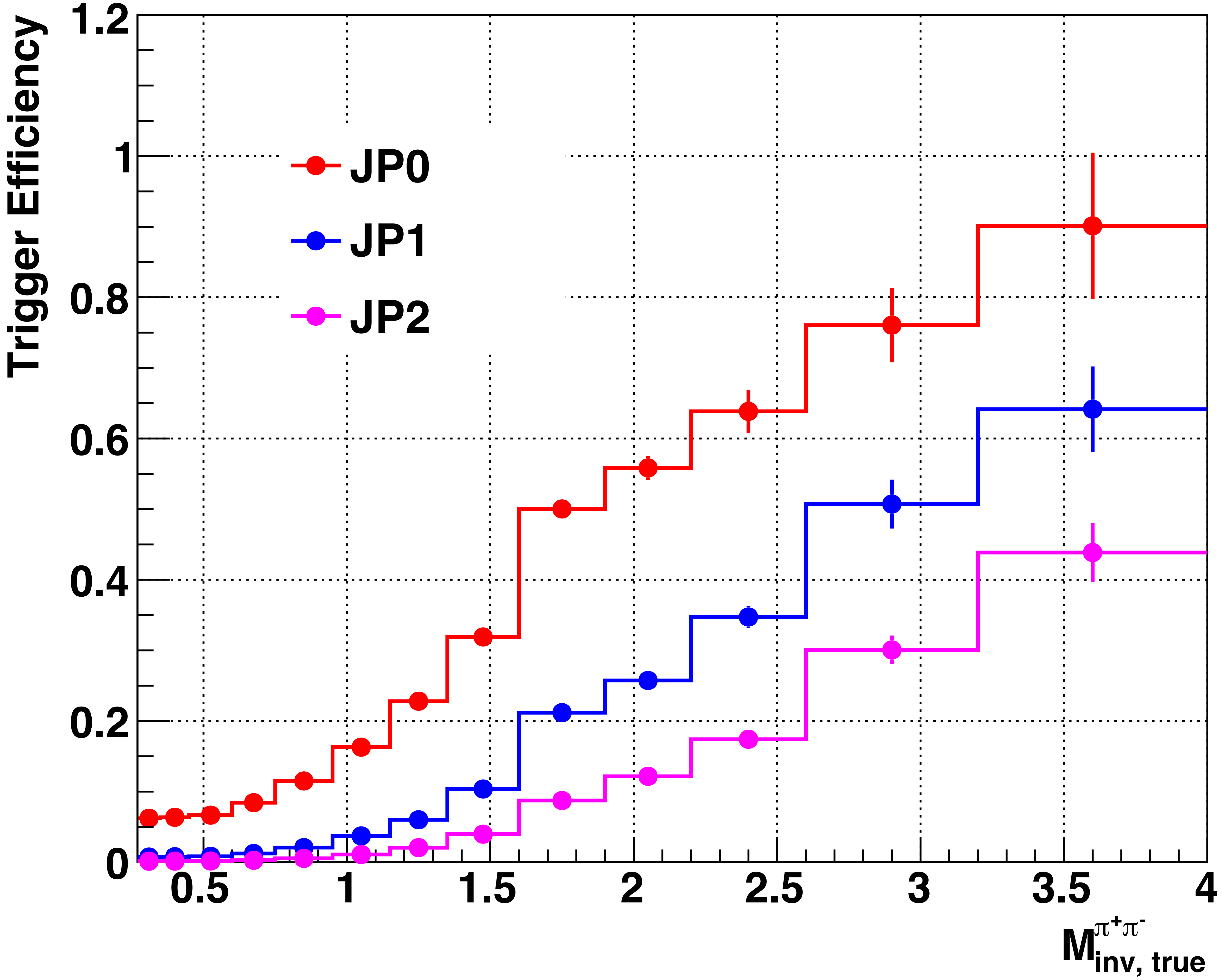
1. $\pi^+\pi^-$ Purity Fraction (f_{fake})
2. $\pi^+\pi^-$ Loss Fraction (f_{loss})
3. Tracking Efficiency (ϵ_{trk}^{π})
4. Trigger Efficiency ($\epsilon_{trg}^{\pi^+\pi^-}$)

$$\epsilon_{trg}^{\pi^+\pi^-} = \frac{N_{triggered}^{\pi^+\pi^+}}{N_{unbiased}^{\pi^+\pi^+}}$$



STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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STAR

Corrections (Bin by bin)

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3. Tracking Efficiency (ϵ_{trk}^{π})
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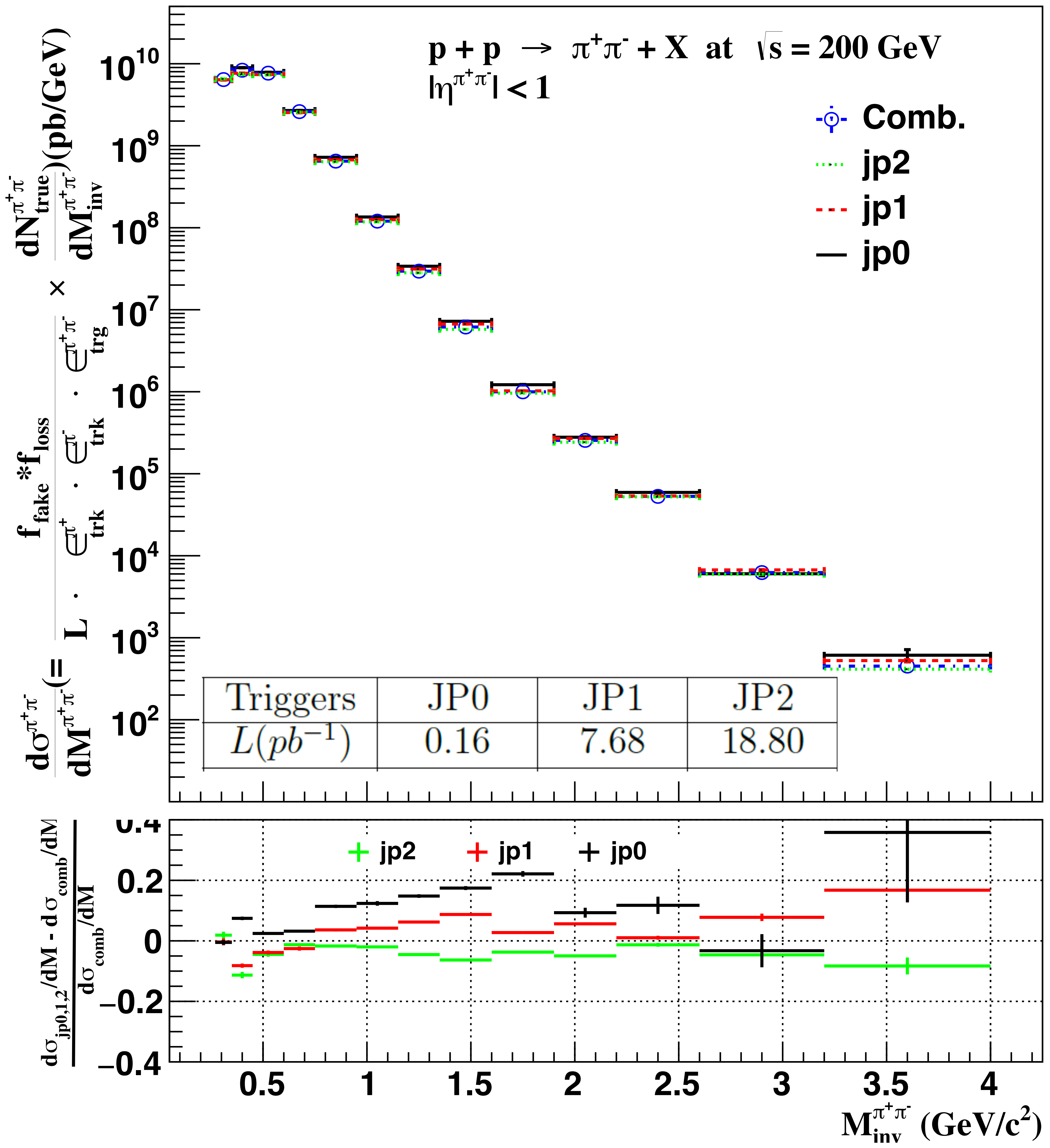
Triggered Cross Sections

$$\frac{d\sigma^{pp\rightarrow\pi^+\pi^-}}{dM^{\pi^+\pi^-}} = \frac{f_{fake} \cdot f_{loss}}{L \cdot \epsilon_{trk}^{\pi^+} \cdot \epsilon_{trk}^{\pi^-} \cdot \epsilon_{trg}^{\pi^+\pi^-}} \cdot \frac{dN_{true}^{\pi^+\pi^-}}{dM^{\pi^+\pi^-}}$$

- Good agreement between triggered cross-sections; disagreement is considered as “Trigger Inefficiency”.
- Final cross-section (“Comb.” in the figure) is the weighted average of triggered cross-sections.



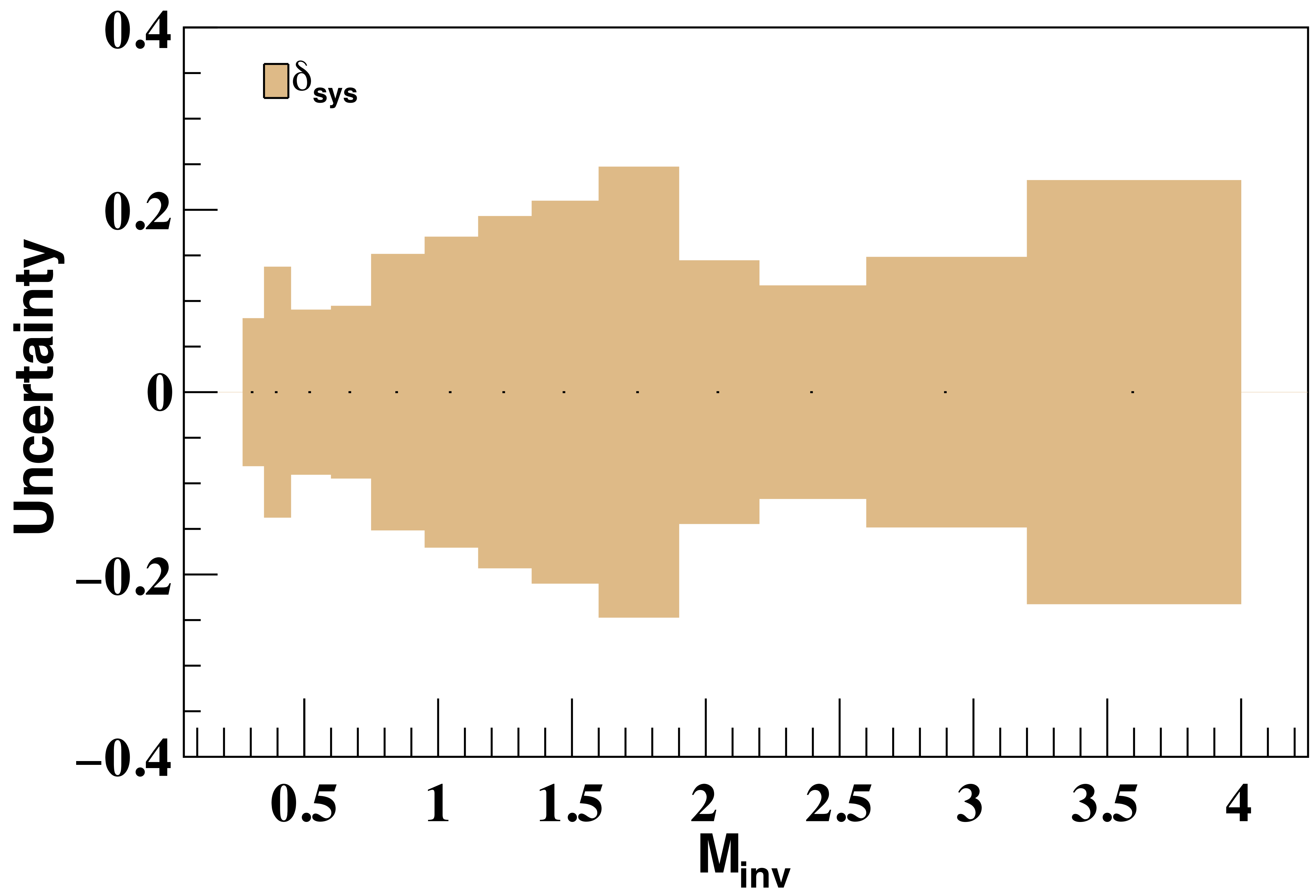
TEMPLE UNIVERSITY

STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

$$p + p \to \pi^+\pi^- + X \text{ at } \sqrt{s} = 200 \text{ GeV}$$

Systematic uncertainties

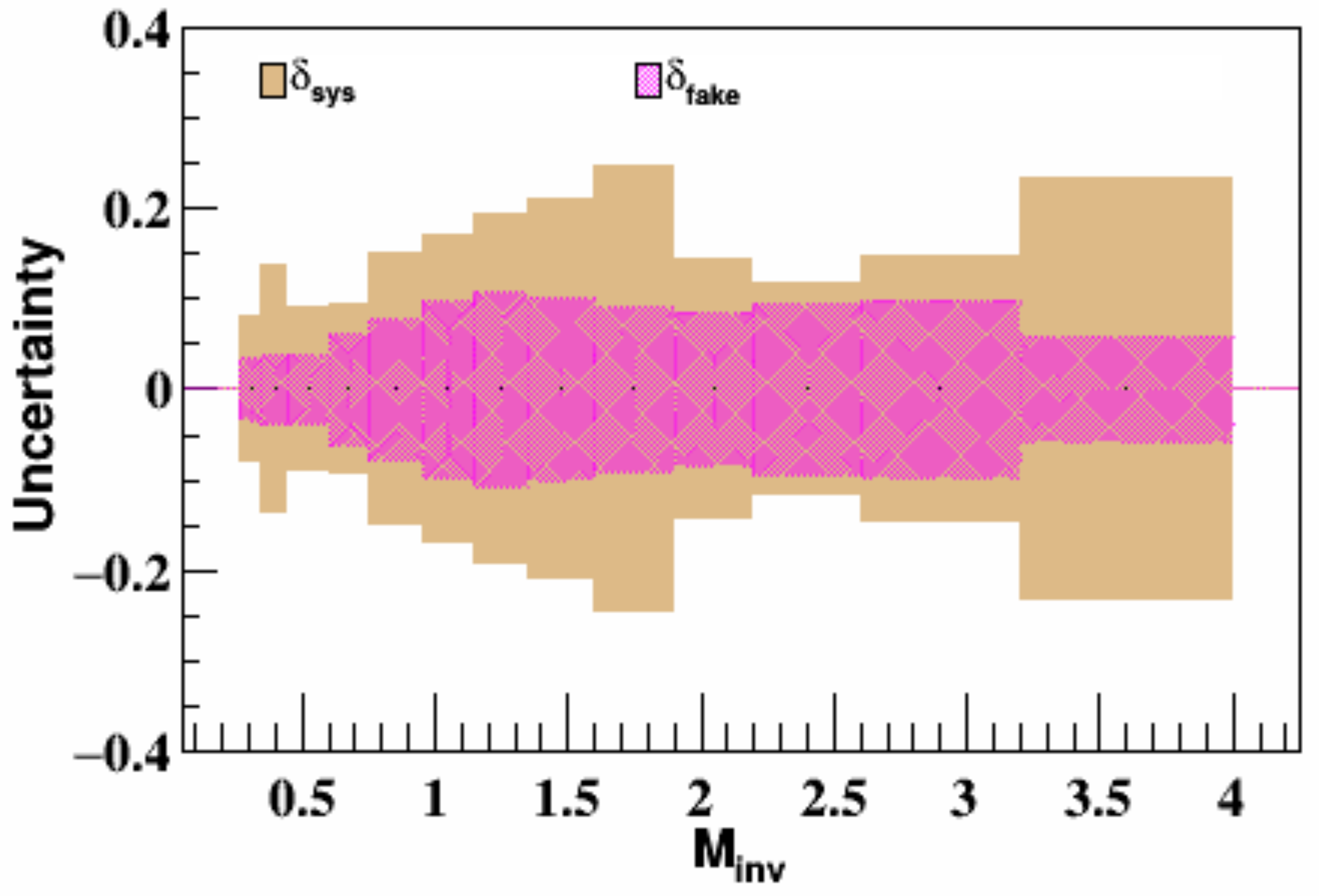
$$\delta_{sys} = \sqrt{\delta_{fake}^2 + \delta_{loss}^2 + \delta_{trg}^2 + \delta_{bias}^2 + \delta_{embstat}^2}$$



Systematic uncertainties

$$\delta_{\rm sys} = \sqrt{\delta^2_{\rm fake} + \delta^2_{\rm loss} + \delta^2_{\rm trg} + \delta^2_{\rm bias} + \delta^2_{\rm embstat}}$$

1. $\pi^+\pi^-$ Purity Fraction ($\delta_{\rm fake}$)



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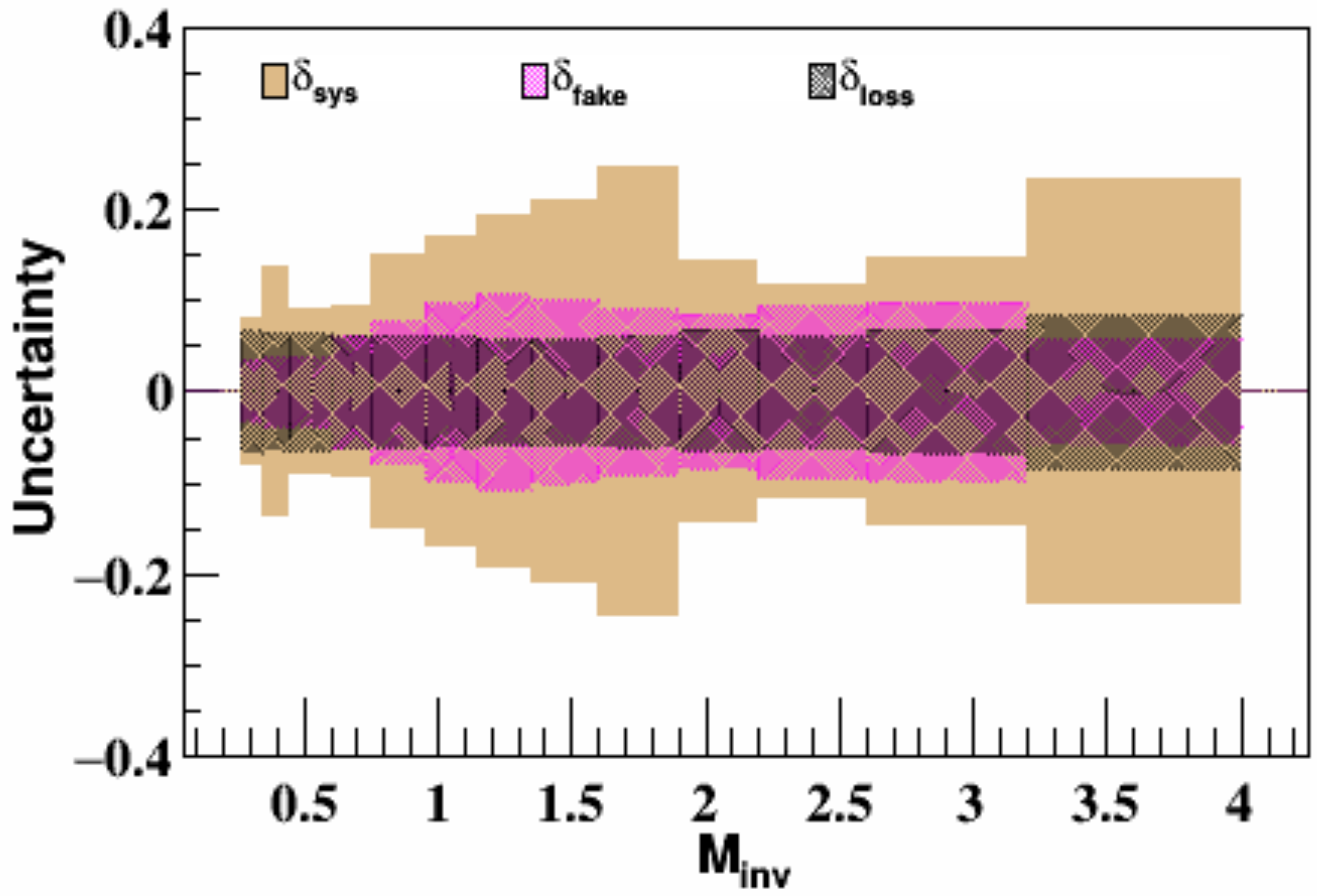

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1. $\pi^+\pi^-$ Purity Fraction (δ_{fake})
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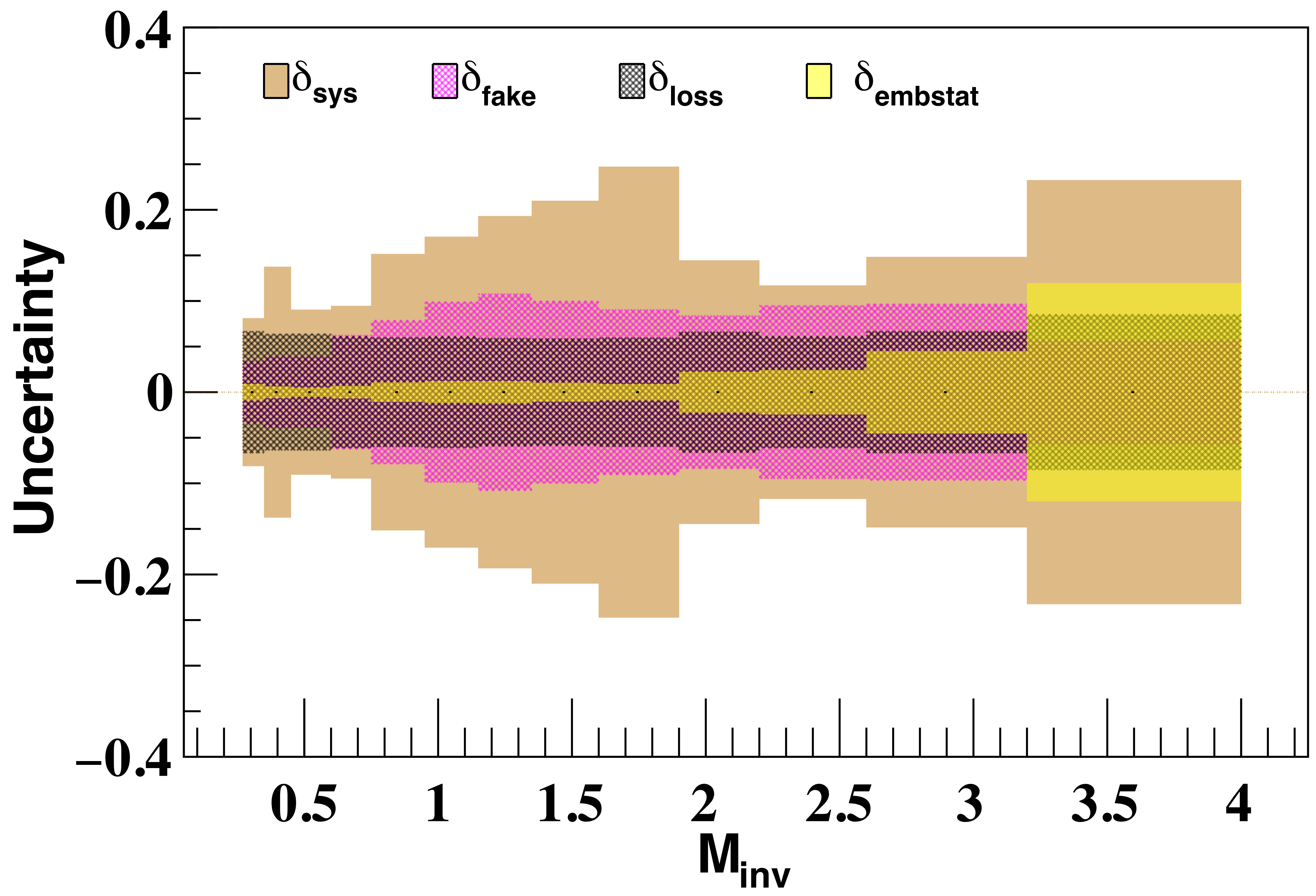

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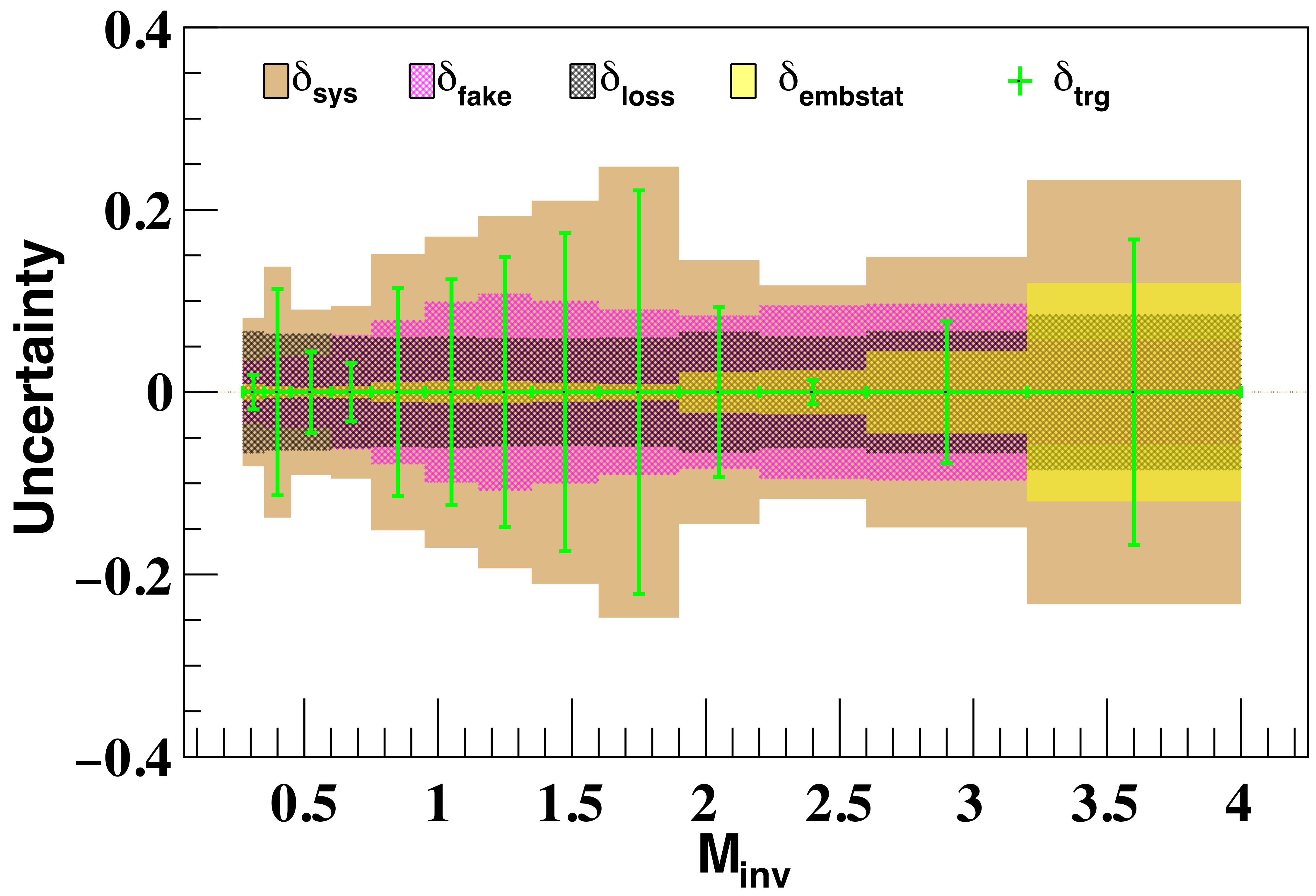

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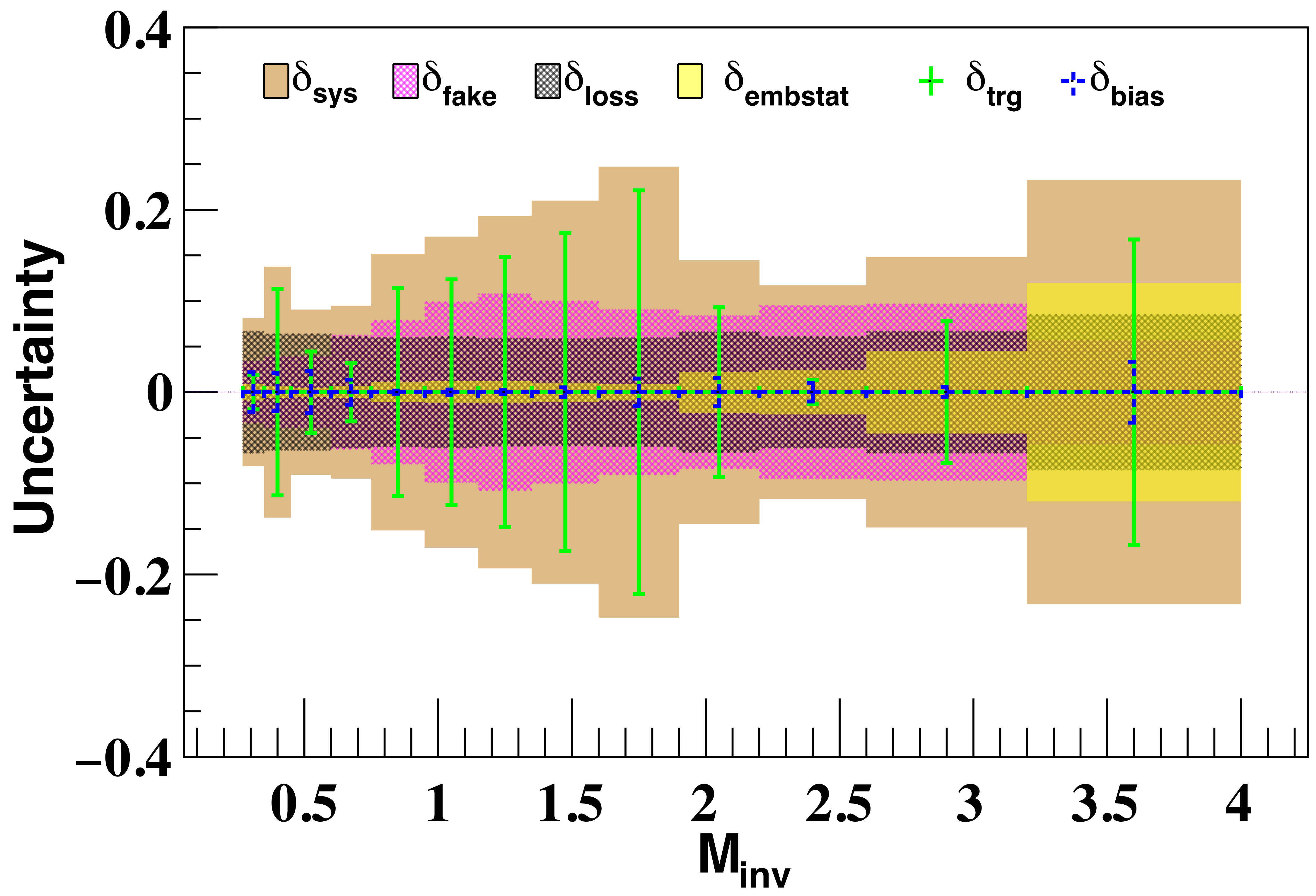

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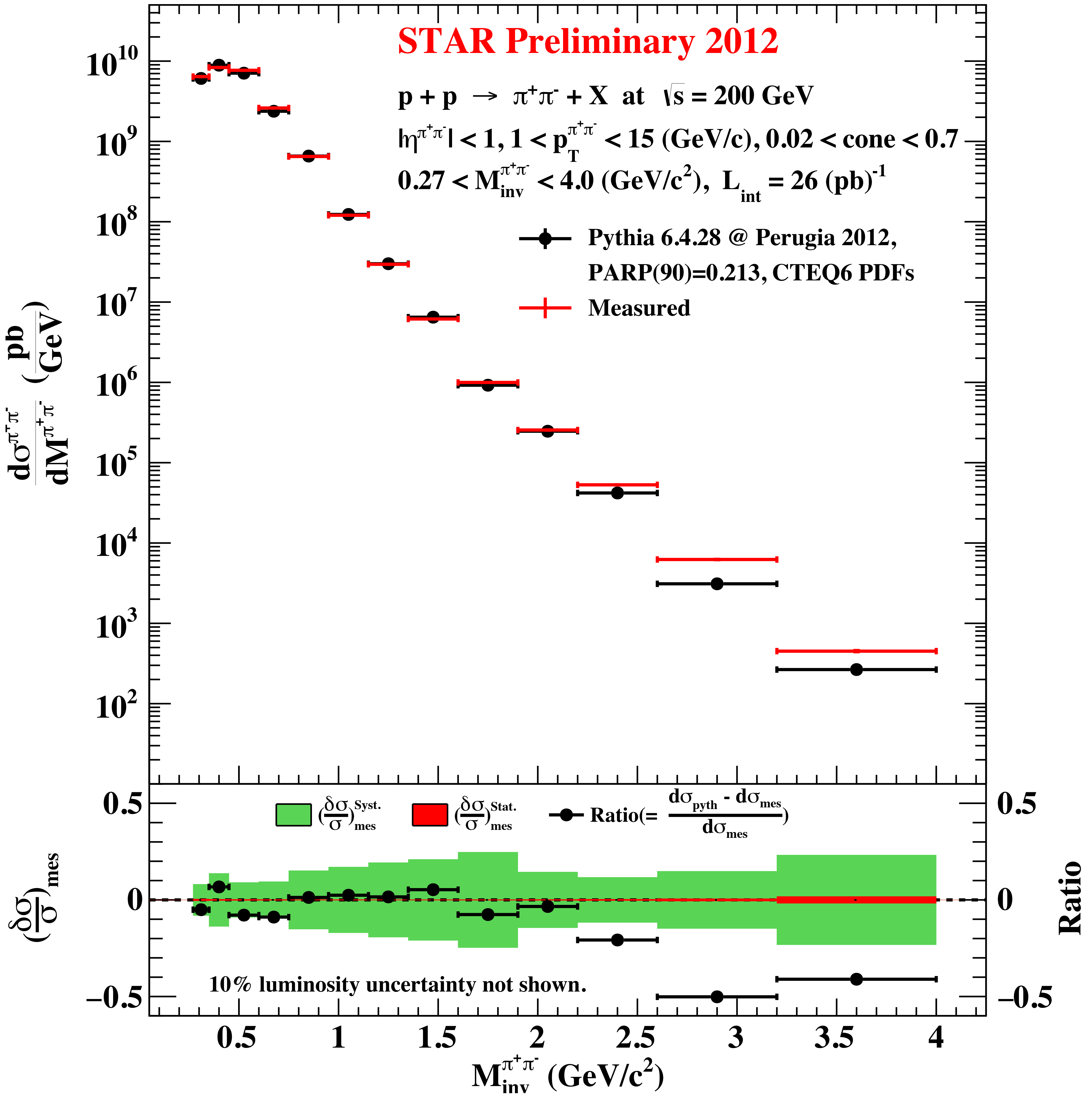
4. Trigger Bias (δ_{bias})
5. Simulation Statistics ($\delta_{embstat}$)

$$\delta_{sys} = \sqrt{\delta_{fake}^2 + \delta_{loss}^2 + \delta_{trg}^2 + \delta_{bias}^2 + \delta_{embstat}^2}$$

Final Result with PYTHIA Cross Section



STAR Run 2012 Unpolarized $\pi^+\pi^-$ Cross-Section ($d\sigma_{UU}^{\pi^+\pi^-}$) Measurement

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Systematic uncertainties

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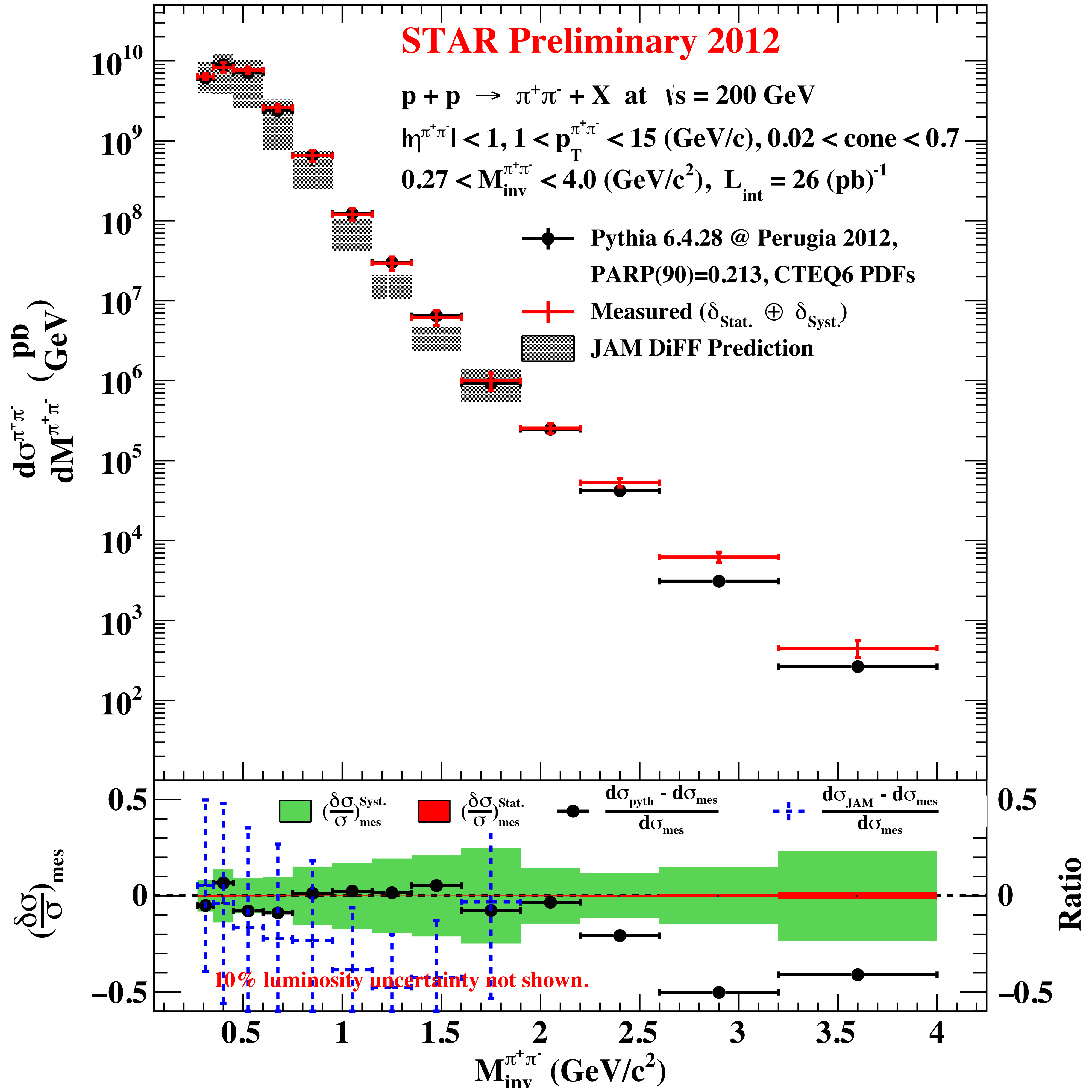
4. Trigger Bias (δ_{bias})
5. Simulation Statistics ($\delta_{embstat}$)

$$\delta_{sys} = \sqrt{\delta_{fake}^2 + \delta_{loss}^2 + \delta_{trg}^2 + \delta_{bias}^2 + \delta_{embstat}^2}$$

Final Result with PYTHIA and JAM DiFF Cross-Section



Summary and Outlook

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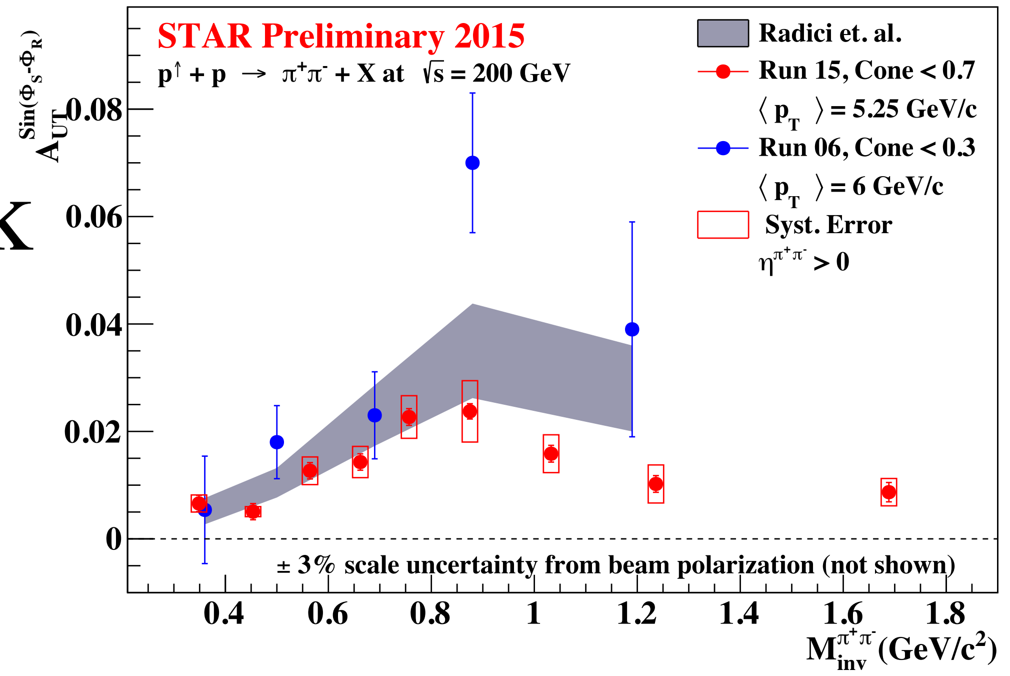
- **Precision $\pi^+\pi^-$ IFF asymmetry measurement**
  - Probes valence quarks (u and d) transversity.
  - Dominant PID systematic uncertainty expected to shrink comparable to the statistical uncertainty, including TOF.



$$A_{UT} \propto \sum_{i,j,k,l} \frac{f_1^{j/b}(x_b)h_1^{i/a}(x_a)H_1^{\sphericalangle h_1h_2/k}(z_h,M_h^2)}{f_1^{j/b}(x_b)f_1^{i/a}(x_a)\underbrace{D_1^{h_1h_2/l}(z_h,M_h^2)}}$$

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- **Precision $\pi^+\pi^-$ IFF asymmetry measurement**
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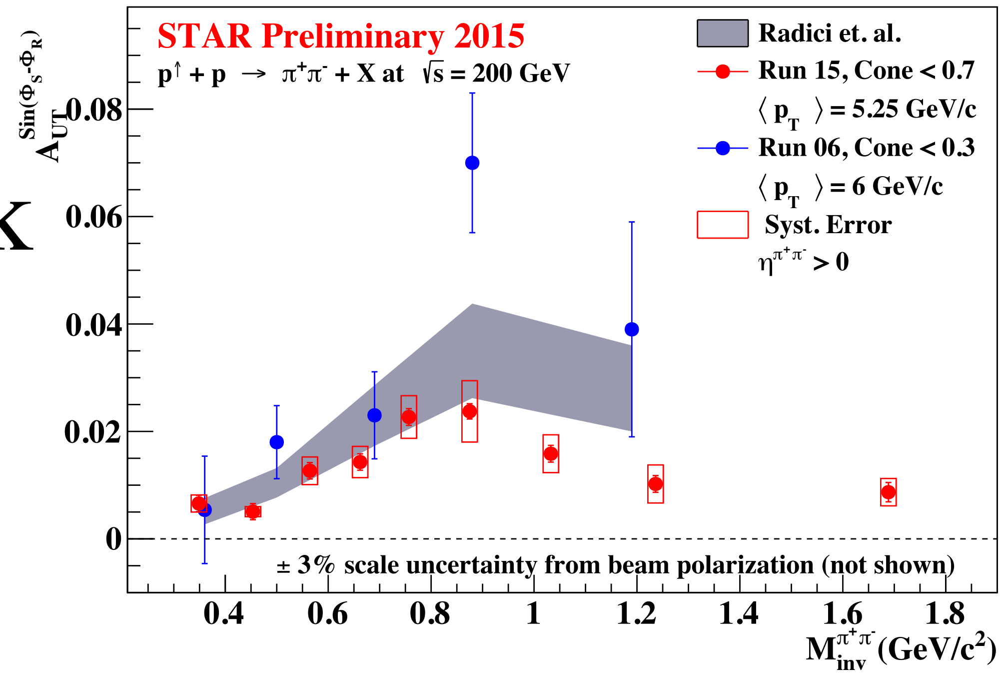

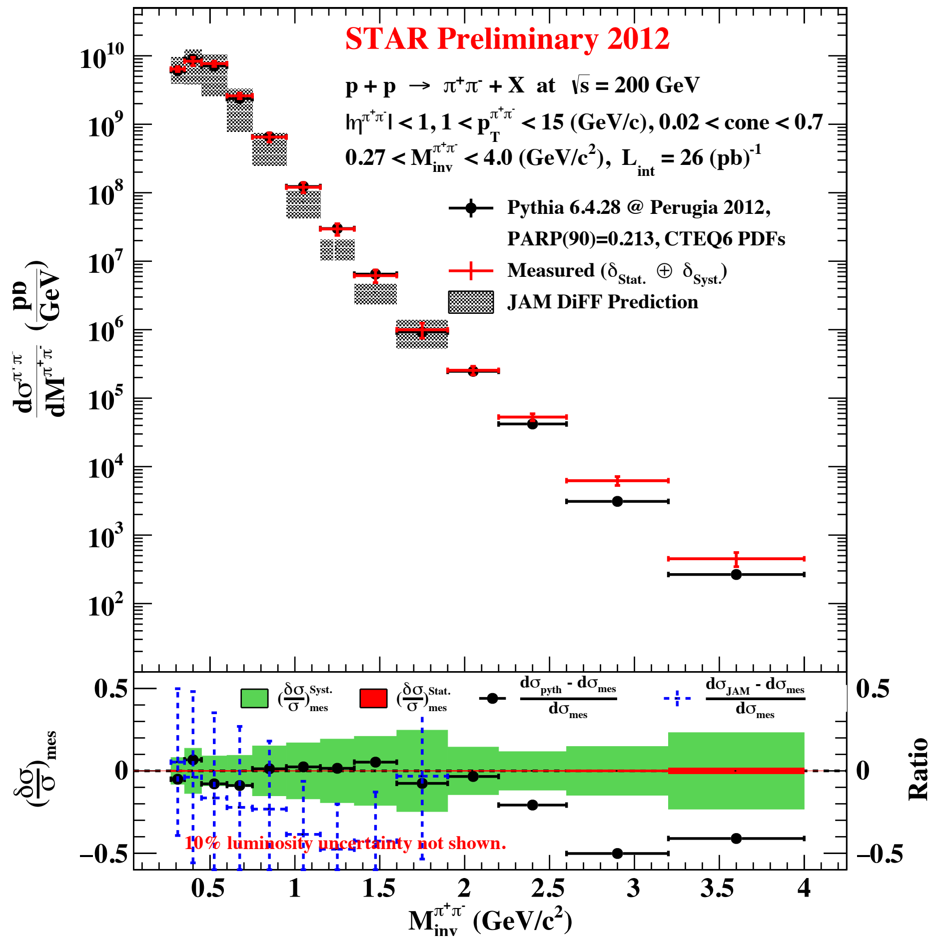

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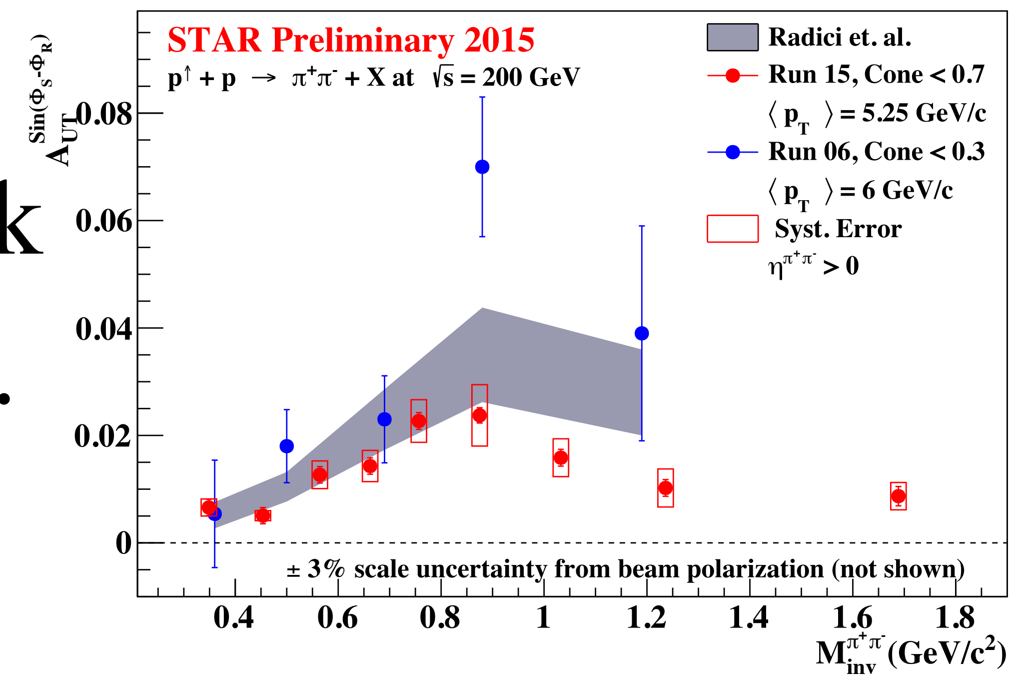

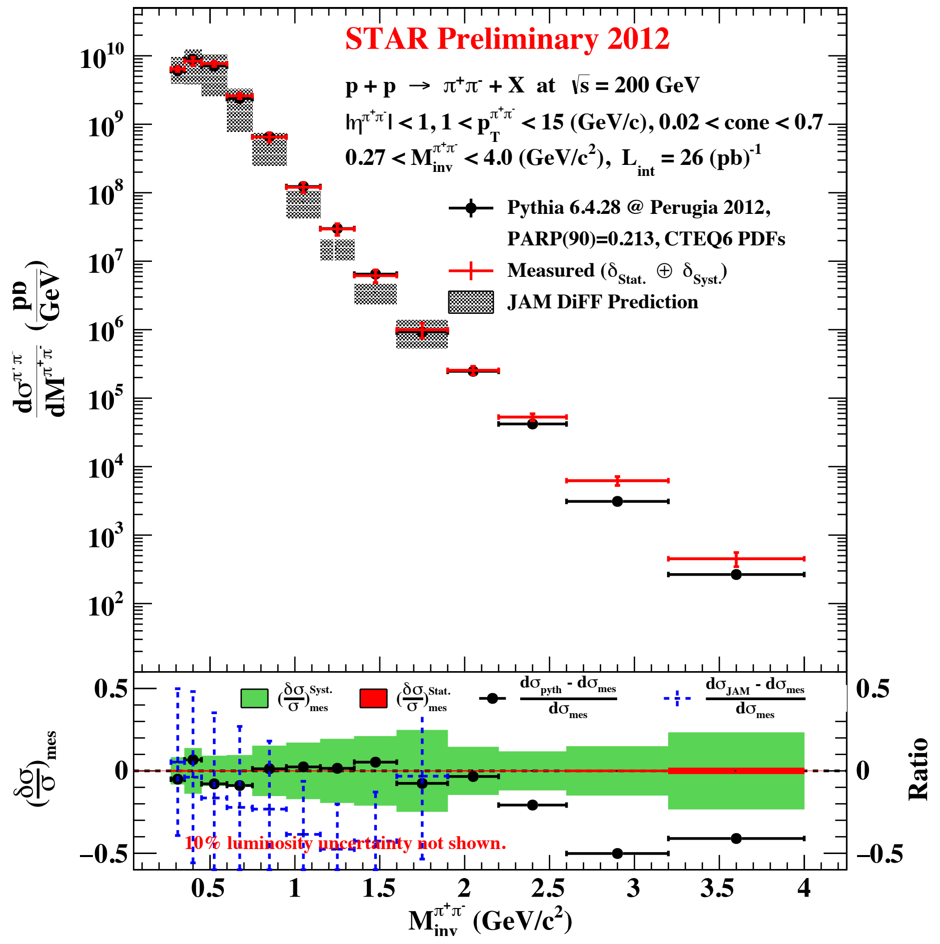

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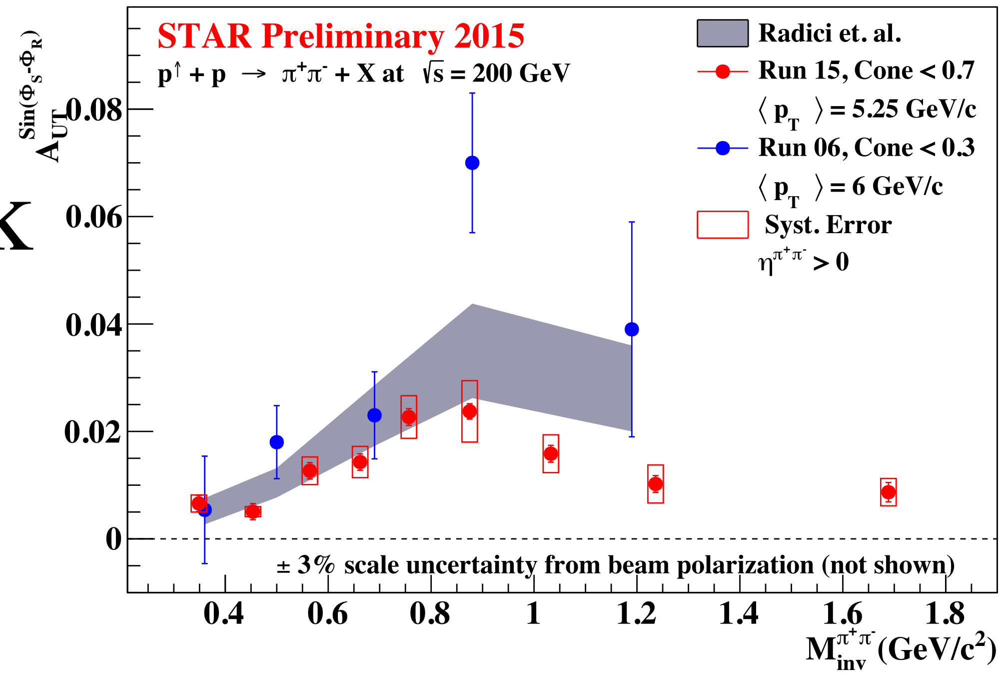

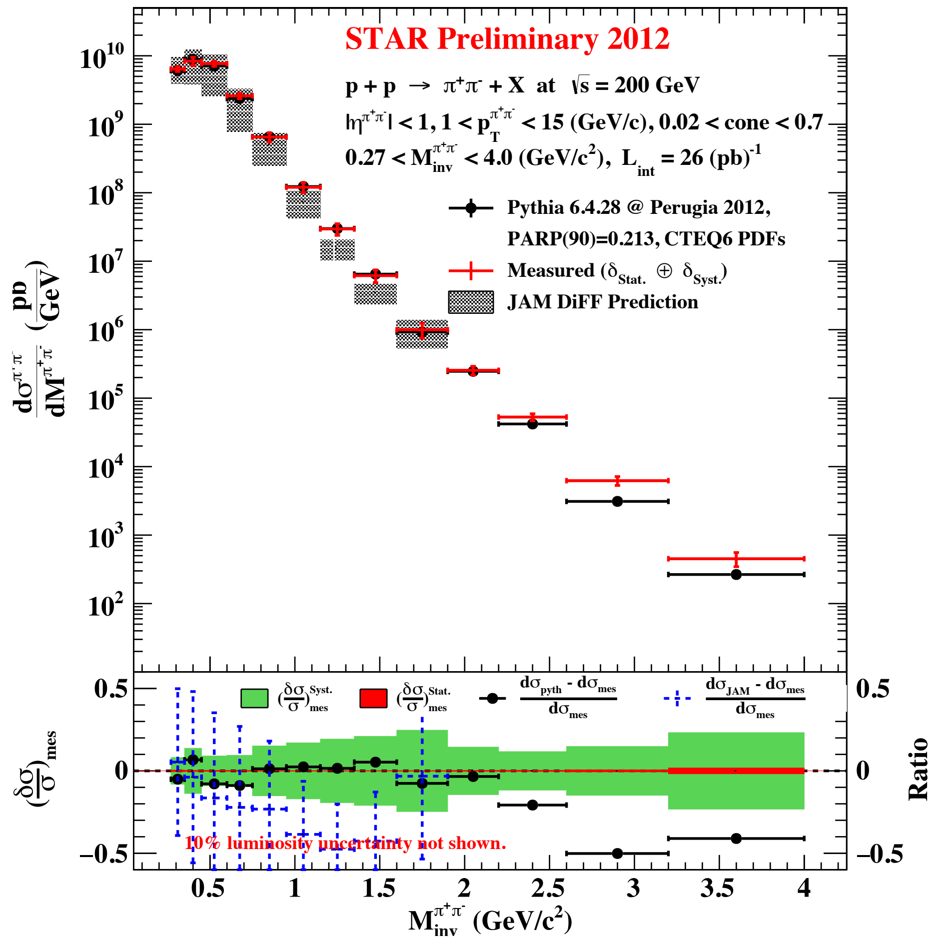

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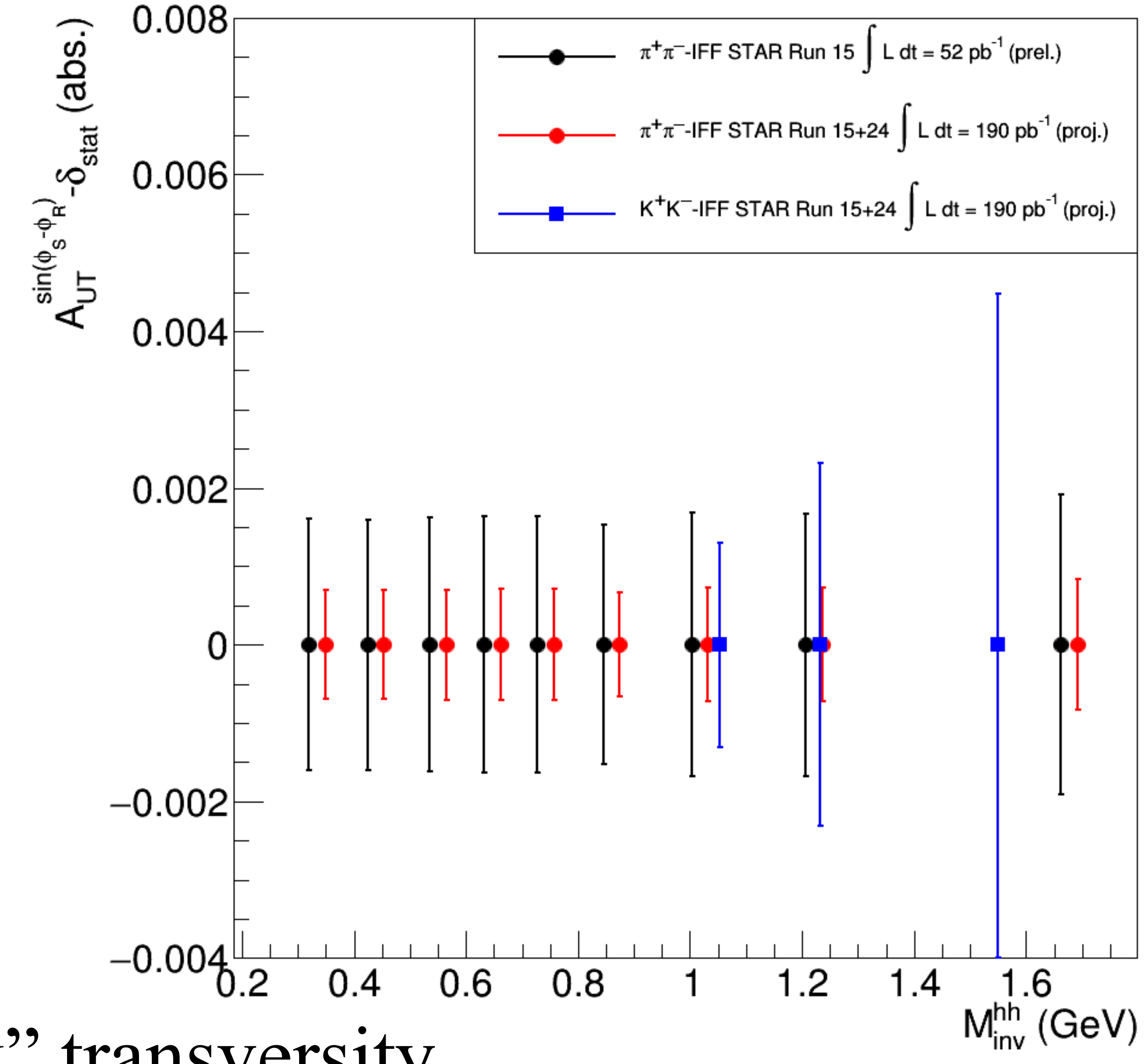

- **IFF and cross-section results at $\sqrt{s} = 200$ GeV will undergo publication together.**
- **Precision IFF asymmetries @ $\sqrt{s} = 200$ GeV probing transversity beyond valence quarks**
  - Precision $\pi^+\pi^-$ **IFF asymmetry** from **Run 2015+2024**
  - Proposed K^+K^- **IFF asymmetry**, sensitive to the "strange quark" transversity.

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Moral Support:	My wife, Sujata My family Friends

Back Up

Transversity Properties

- In the context of transversity, the term **"chiral-odd"** refers to the transformation properties of the transversity distribution under **chiral transformations**.

- **Chiral transformations** involve **changing the handedness (chirality)** of particles while keeping their **momentum unchanged**. Chirality is associated with the helicity of a particle, which is the projection of its spin onto its momentum direction.

- The **transversity distribution** is considered **chiral-odd because it flips sign under chiral transformations**. In other words, if you **swap a particle with its antiparticle and simultaneously change its helicity (chirality), the transversity distribution changes sign**.

- Mathematically, for a quark distribution function q(x) representing the probability of finding a quark with momentum fraction x inside a nucleon, and a corresponding antiquark distribution function Δq(x) representing the probability of finding an antiquark with momentum fraction x inside a transversely polarized nucleon, the chiral transformation can be denoted as:
- $q(x) + \Delta q(x) \rightarrow -q(x) + \Delta q(x)$

- This sign change under chiral transformations is what makes the transversity distribution "chiral-odd."

- Transversity and its chiral properties provide insights into the more intricate details of the internal structure of nucleons and the fundamental symmetries of the strong interaction.

Transversity Properties

The reason transversity should appear with another chiral-odd function in certain physical processes is due to the conservation of angular momentum and parity in strong interactions, as described by Quantum Chromodynamics (QCD). Let's break down the key concepts:

Conservation of Angular Momentum: In QCD processes, the total angular momentum of the initial and final states should be conserved. This includes both the intrinsic angular momentum (spin) and the orbital angular momentum. The transversity distribution involves the intrinsic angular momentum of quarks with respect to their transverse momentum.

Chirality and Angular Momentum Conservation: Chiral-odd functions like transversity are associated with the helicity of quarks, which is related to their intrinsic angular momentum aligned with their momentum direction. When quarks undergo interactions, their helicity can change. To conserve angular momentum, a change in helicity should be accompanied by an opposite change in the angular momentum of the particle's orbital motion, often involving gluon exchange. This exchange requires another chiral-odd function to maintain angular momentum conservation.

Parity Conservation: Parity is another important symmetry in particle physics. Chiral-odd functions are related to parity-odd observables. When you consider processes where parity is conserved, chiral-odd functions must appear with other chiral-odd functions to ensure that the parity of the overall process remains unchanged.

In summary, the appearance of transversity with another chiral-odd function in certain processes is a consequence of the conservation of angular momentum and parity in strong interactions. This combination of chiral-odd functions is necessary to ensure that angular momentum and parity are conserved throughout the interaction, in accordance with the principles of Quantum Chromodynamics.

$\pi^+\pi^-$ Asymmetry Analysis

$p^\uparrow + p \rightarrow \pi^+\pi^- + X$ at $\sqrt{s} = 200$ GeV

Event-by-event polarization correction

Imperfection Resonances

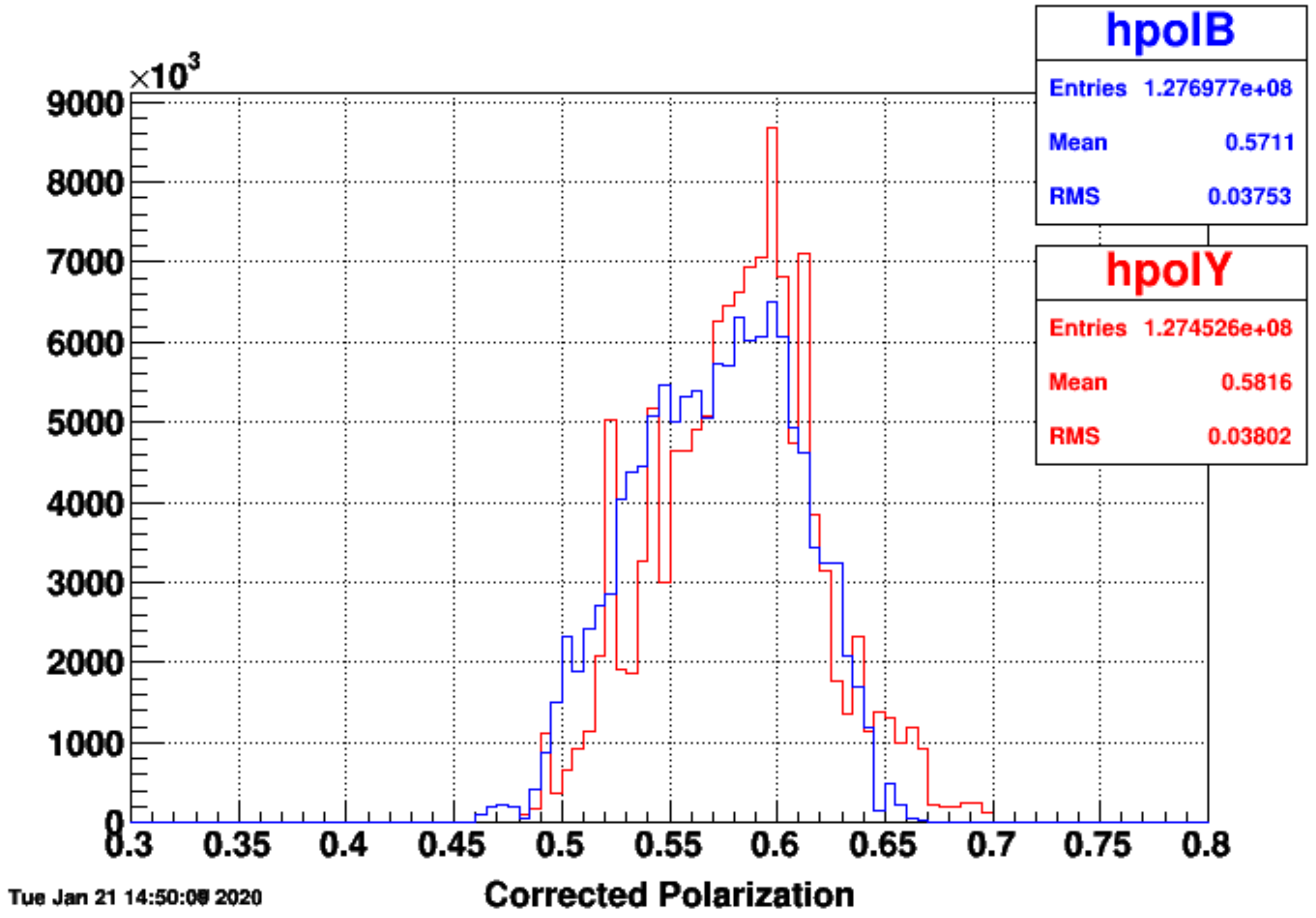
(due to misalignment and magnet errors)

Intrinsic Resonances

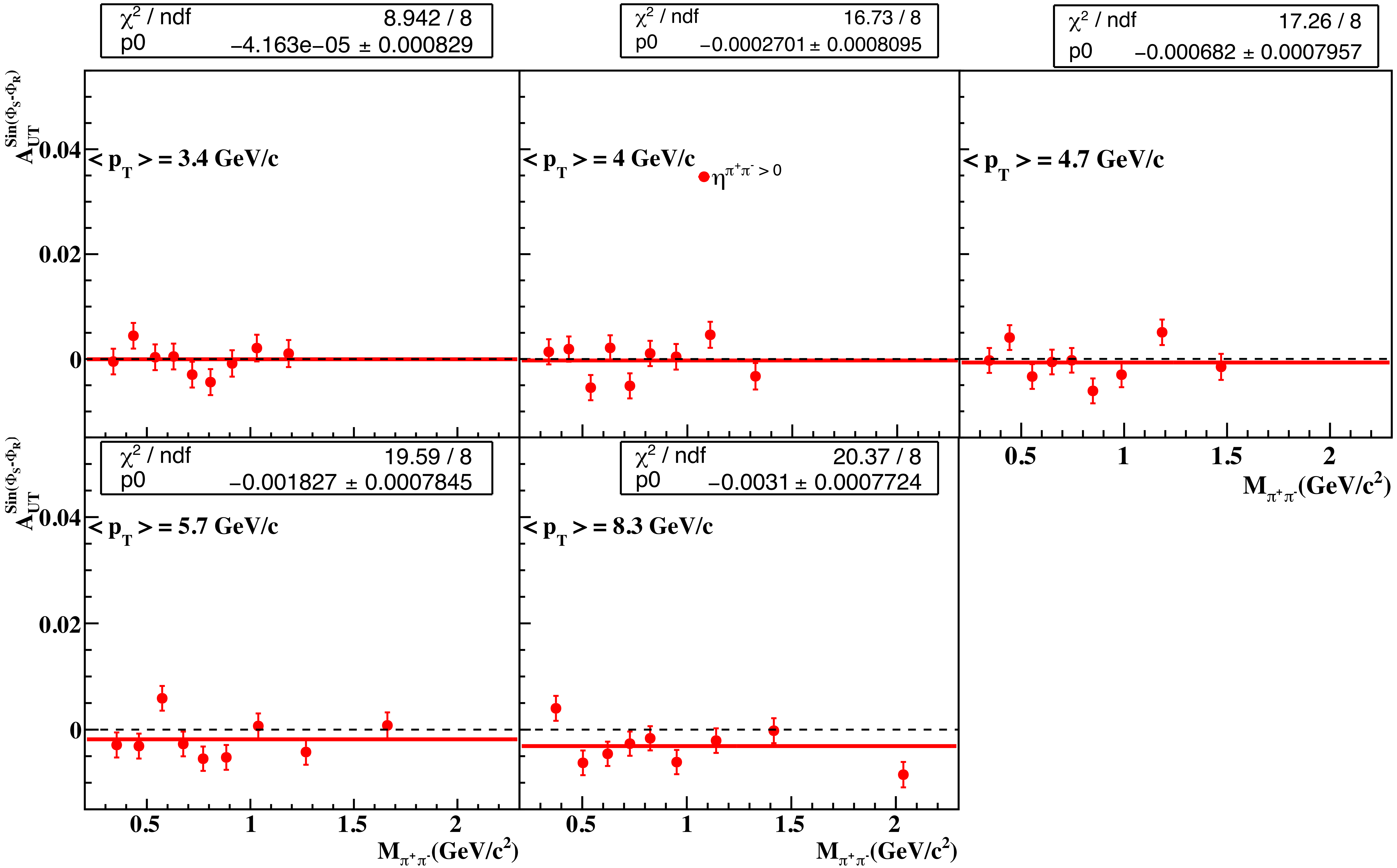
(imperfection in the focusing magnetic field)

Depolarization

Corrected Polarization $P_t = P_0 - \frac{dP}{dt}\Delta t$, $\Delta t = t_{event} - t_0$



Random Asymmetry



Jet Patch Trigger Definition

- BEMC towers split up into 18 $\Delta\eta \times \Delta\phi = 1 \times 1$ “Jet Patches” each containing 400 towers.
- ADC values from each patches are summed and passed to DSM
- ADC to E_T conversion: $E_T = 0.236 \times (ADC - 5)$ in GeV.

		Thresholds		
Name	Offline ID	DSM ADC	E_T	Avg. Prescale
JP0	370601	20	3.5 GeV	141.35
JP1	370611	28	5.4 GeV	2.5
JP2	370621	36	7.3 GeV	1.0

Run 2012 Trigger Parameters